

PacT Series

ComPacT NSX – DCM Modbus SL Communication

User Guide

PacT Series offers world-class circuit breakers and switches.

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12/2024



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Safety Information

Important Information

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result in** minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

Please Note

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

Cybersecurity Safety Notice

⚠ WARNING

POTENTIAL COMPROMISE OF SYSTEM AVAILABILITY, INTEGRITY, AND CONFIDENTIALITY

- Change default passwords at first use to help prevent unauthorized access to device settings, controls, and information.
- Disable unused ports/services and default accounts to help minimize pathways for malicious attackers.
- Place networked devices behind multiple layers of cyber defenses (such as firewalls, network segmentation, and network intrusion detection and protection).
- Use cybersecurity best practices (for example, least privilege, separation of duties) to help prevent unauthorized exposure, loss, modification of data and logs, or interruption of services.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

For more information, refer to DOCA0122•• *MasterPacT, ComPacT, PowerPacT - Cybersecurity Guide*, page 8.

About the Book

Document Scope

The aim of this guide is to provide users, installers, and maintenance personnel with the technical information needed to operate the Modbus protocol on:

- ComPacT™ NSX100-630 circuit breakers and switch-disconnectors.
- ComPacT™ NSX100-1200 DC circuit breakers and switch-disconnectors.

Validity Note

This document is valid for:

- ComPacT NSX100-630 circuit breakers
 - with the DCM Modbus SL module
 - and/or with MicroLogic™ 5, 6, or 7 trip unit
 - and/or with communication motor mechanism
- ComPacT NSX100-630 switch-disconnectors with the DCM Modbus SL module
- ComPacT NSX100-1200 DC circuit breakers and switch-disconnectors with the DCM Modbus SL module and communication motor mechanism.

and connected to an RS485 serial line Modbus network using a DCM Modbus SL module.

This document describes the registers and commands available for the IMU modules with the following firmware version:

IMU module	Part number	Firmware version
MicroLogic 5A/E, and 6A/E trip units	–	≥ V001.003.000
MicroLogic 7E trip unit	–	≥ V002.001.001
DCM Modbus SL module	LV434311	≥ V001.003.000
PAS700 gateway	PAS700	≥ V1.20.05

Online Information

The information contained in this guide is likely to be updated at any time. Schneider Electric strongly recommends that you have the most recent and up-to-date version available on www.se.com/ww/en/download.

The technical characteristics of the devices described in this guide also appear online. To access the information online, go to the Schneider Electric home page at www.se.com.

Related Documents

Title of Documentation	Reference Number
ComPacT NSX Circuit Breakers and Switch-Disconnectors - User Guide	DOCA0187EN DOCA0187ES DOCA0187FR DOCA0187ZH
ComPacT NSX MicroLogic 5/6/7 Electronic Trip Units - User Guide	DOCA0188EN DOCA0188ES DOCA0188FR DOCA0188ZH
ComPacT NSX DC Circuit Breakers and Switch-Disconnectors - User Guide	DOCA0186EN DOCA0186ES DOCA0186FR DOCA0186ZH
<i>MasterPacT, ComPacT, PowerPacT - Cybersecurity Guide</i>	DOCA0122EN DOCA0122ES DOCA0122FR DOCA0122ZH
DCM Modbus SL Module - Instruction Sheet	PKR7481102

To find documents online, visit the Schneider Electric download center (www.se.com/ww/en/download/).

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Modbus Communication with ComPacT NSX Circuit Breakers

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Introduction

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PacT Series Master Range

Future-proof your installation with Schneider Electric’s low-voltage and medium-voltage PacT Series. Built on legendary Schneider Electric innovation, the PacT Series comprises world-class circuit breakers, switches, residual current devices and fuses, for all standard and specific applications. Experience robust performance with PacT Series within the EcoStruxure-ready switchgear, from 16 to 6300 A in low-voltage and up to 40.5 kV in medium-voltage.

Description

Modbus Communication

The Modbus communication option enables Schneider Electric low voltage circuit breakers to be connected to a supervisor or to any other device with a Modbus client communication channel.

The circuit breakers can be connected to:

- An RS-485 serial line network with Modbus protocol using dedicated interfaces such as:
 - The DCM Modbus SL module for one circuit breaker.
 - The IFM Modbus SL communication interface for one circuit breaker.
- An Ethernet network with Modbus TCP/IP protocol using dedicated interfaces such as:
 - The IFE Ethernet communication interface for one circuit breaker.
 - The IFE Ethernet switchboard server.

Access to Functions

The Modbus communication option provides access to many functions, including:

- read metering and diagnostic data
- read status conditions and remote operations
- transfer of time-stamped events
- display protection settings
- read the circuit breakers identification and configuration data
- remote control of the circuit breaker
- time-setting and synchronization

This list depends on the composition of the intelligent modular unit (IMU) (type of circuit breaker, of MicroLogic trip unit, and so on) and the enabled functions.

Intelligent Modular Unit

Definition

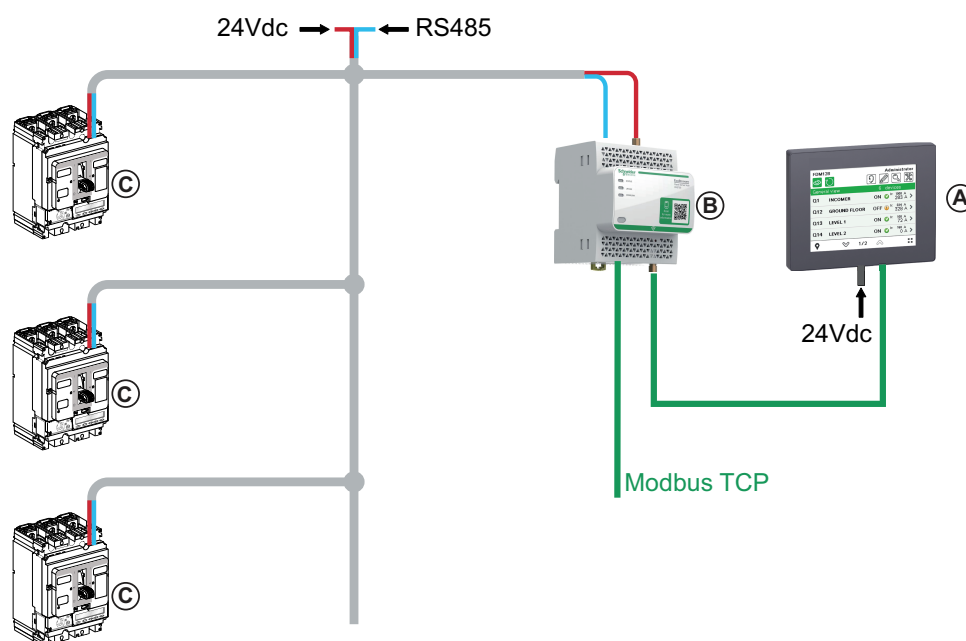
A modular unit is a mechanical and electrical assembly containing one or more products to perform a function in a switchboard (incoming protection, motor command, and control).

The circuit breaker with its internal communicating components (MicroLogic control unit or MicroLogic trip unit) and DCM Modbus SL module connected to one communication interface is called an intelligent modular unit (IMU).

An IMU is composed around a circuit breaker from the following ranges:

- ComPacT NSX circuit breakers

Communication Architecture



- A. HMI
- B. PAS700 Panel Server
- C. ComPacT NSX with DCM Modbus SL module

NOTE:

- Do not connect using the ring topology.
- A 120 Ω parallel resistor must be connected when the DCM Modbus SL module is connected at the end of Modbus network.
- Maximum 12 devices are allowed to connect to a PAS700 Panel Server.

Remote Controller

A remote controller is a device that is able to communicate with an IMU using a communication interface, such as the PAS700 Ethernet communication interface. For example, FDM128 Ethernet display for eight devices, supervisor, PLC, BMS, SCADA system, and so on, are remote controllers.

For the description of Modbus registers and commands, refer to the *Modbus Communication Guides*.

EcoStruxure Power Commission Software

Overview

EcoStruxure Power Commission software helps you to manage a project as part of testing, commissioning, and maintenance phases of the project life cycle. The innovative features in it provide simple ways to configure, test, and commission the smart electrical devices.

EcoStruxure Power Commission software automatically discovers the smart devices and allows you to add the devices for an easy configuration. You can generate comprehensive reports as part of Factory Acceptance Test and Site Acceptance Test to replace your heavy manual work. Additionally, when the panels are under operation, any change of settings made can be easily identified by a yellow highlighter. This indicates the difference between the project and device values, and hence provides a system consistency during the operation and maintenance phase.

EcoStruxure Power Commission software enables the configuration of the following circuit breakers, modules, and accessories:

Circuit breaker ranges	Modules	Accessories
- ComPacT NSX circuit breakers - PowerPacT H-, J- and L-frame circuit breakers	- MicroLogic trip units - Communication interface modules: DCM Modbus SL module	SDTAM and SDx output modules

For more information, refer to the *EcoStruxure Power Commission Online Help*.

Click [here](#) to download the latest version of EcoStruxure Power Commission software.

Key Features

EcoStruxure Power Commission software performs the following actions for the supported devices and modules:

- Create projects by device discovery
- Save the project in the EcoStruxure Power Commission cloud for reference
- Upload settings to the device and download settings from the device
- Compare the settings between the project and the device
- Perform control actions in a secured way
- Generate and print the device settings report
- Perform a communication wiring test on the entire project and generate and print test report
- View the communication architecture between the devices in a graphical representation
- View the measurements, logs, and maintenance information
- Export Waveform Capture on Trip Event (WFC)
- View the status of device and IO module
- View the alarm details
- Buy, install, remove, or retrieve the Digital Modules
- Check the system firmware compatibility status
- Update to the latest device firmware
- Perform force trip and automatic trip curve tests

DCM Mobus SL Module

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Introduction

Overview

DCM Modbus SL module supports Modbus communication and control function. It is used to connect the ComPacT NSX circuit breaker to Modbus Serial Line network.

DCM Modbus SL module is installed inside the circuit breaker. Two indication LEDs on the front side indicate the communicating and working status. It can obtain the circuit breakers status and data from MicroLogic 5A/E, 6A/E, and 7E trip units in real time. It also can be used to control the motor mechanism. DCM module is a digital input and output center for auxiliary contacts (OF, SD or SDE), terminal thermal sensor and MicroLogic 5A/E, 6A/E, and 7E trip units. The commissioning parameters under 24VDC, trip and mechanical operation counts are stored in DCM Modbus SL module.

DCM module connects to Modbus Client (e.g. Modbus gateway or PLC) through RS485 port. You can do commissioning for multiple DCM modules at one time and update its firmware with EcoStruxure Power Commission software.

DCM Type

DCM Modbus SL commercial reference: LV434311.

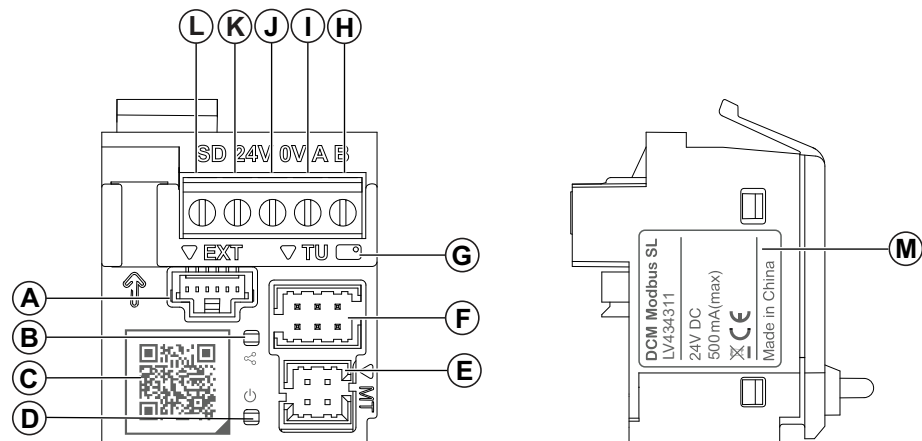
DCM Features

Main features of the DCM Modbus SL module include:

- Real time communication: obtain circuit breakers status (OF, SDE and SD) and data from MicroLogic trip units in real time, and control motor mechanism MTc with communication function remotely.
- Efficient commissioning: commissioning through PAS700 Panel Server webpage, serial number, or Modbus commands.
- Compact structure: fully embedded design to save space.
- Standard connection: connect to Modbus Client through RS485 port.
- Rich extension: support different types of indication contacts and terminal thermal sensor.
- Data security: trip and mechanical operation counts, and communication parameters are stored in nonvolatile memory.
- Easy check: two LEDs at front side indicate communication and operation status.
- Enhanced security: built-in remote control lock to block intrusive commands.

Hardware Description

General Description



- A. Extension port
- B. Communication LED
- C. Go2SE QR code
- D. Operation LED
- E. Port for motor mechanism
- F. Port for MicroLogic trip units
- G. G&I red cable position indication
- H. RS485-B port
- I. RS485-A port
- J. 0V terminal
- K. +24V power supply terminal
- L. SD signal terminal
- M. Rating plate

Installation

DCM Modbus SL module can be installed in OF and SDE slots on the ComPacT NSX circuit breaker. For more information, consult the instruction sheet available on Schneider Electric website: [PKR7481102](#).

If you need to install the motor mechanism with communication function, insert the premade cord available in the motor mechanism package to the port for motor mechanism on DCM module. For more information on installation, consult the instruction sheet available on Schneider Electric website: [GHD16272AA](#), [GHD16318AA](#).

24 Vdc Power Supply

The DCM Modbus SL module must always be supplied with 24 Vdc.

It is recommended to use a UL listed/UL recognized limited voltage/limited current or a class 2 power supply with a 24 Vdc, 3 A maximum.

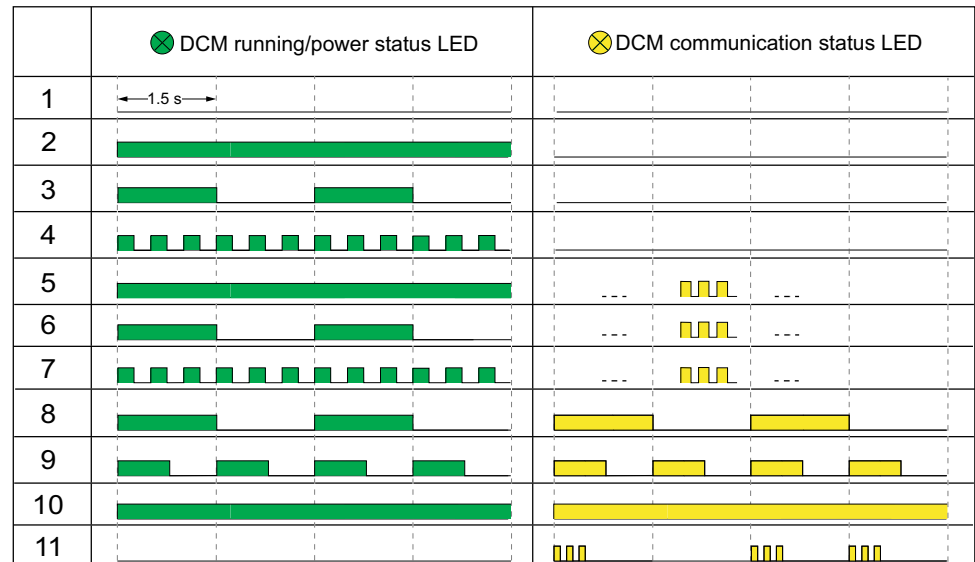
NOTE: For 24 Vdc power supply connection, use copper conductors only.

DCM Status LED

The DCM status LED indication types:

- DCM running/power status LED color is green.
- DCM communication status LED color is yellow.

DCM communication status yellow LED indicates that the traffic is being transmitted or received over the Modbus serial line network through IMU. The LED is ON during the transmission and reception of the messages. The LED is OFF otherwise.



Status	DCM status description	Action
1	Power off	-
2	DCM is running. No Modbus frame received.	Check and confirm Modbus RTU connection and configuration is correct.
3	DCM running in remote control command blocked mode. No Modbus frame is received.	Check and confirm Modbus RTU connection and configuration is correct.
4	DCM is in alarm status, and no Modbus frame is received.	• Check the connection between DCM and MicroLogic trip units. • Confirm MicroLogic trip unit versions are compatible.
5	DCM is running. Modbus frames are received.	-
6	DCM running in remote control command blocked mode. Modbus frames are received.	If necessary, send Modbus command to unlock intrusive command.
7	DCM is in alarm status, and Modbus frames are received.	• Check the connection between DCM and MicroLogic trip units. • Confirm MicroLogic trip unit versions are compatible.
8	DCM is locating.	-
9	DCM is allocating Modbus addresses.	-
10	DCM restores the factory setting.	-
11	DCM internal fault.	Replace the DCM module.

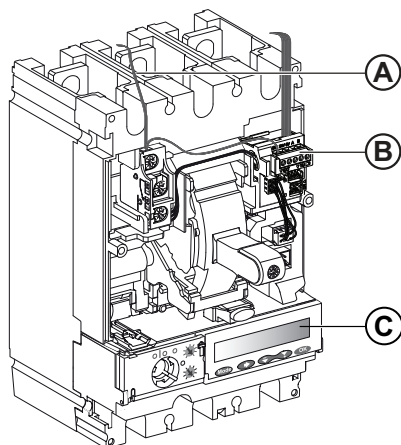
Schematics with ComPacT NSX Circuit Breakers

General Description

Depending on the DCM Modbus SL module mode used, the user must connect the DCM Modbus SL module using one of the following configurations:

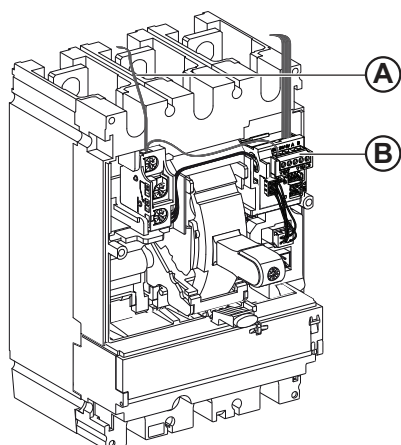
- Connection of the DCM Modbus SL module to the MicroLogic trip unit.
- Connection of the DCM Modbus SL module only.

Connection of the DCM to MicroLogic Trip Unit



- A. 24 Vdc DCM Modbus SL cord
- B. DCM Modbus SL modbus
- C. MicroLogic trip unit

Connection of the DCM Only



- A. 24 Vdc DCM Modbus SL cord
- B. DCM Modbus SL modbus

Configuration

Overview

Two configurations are available for DCM Modbus SL modules:

- PAS700 Panel Server configuration
- SN configuration

Default Parameters

The DCM Modbus SL module has following default parameters of communication network:

- Modbus Master address: 247
- Baud rate: 19200 bps
- 8 data bits
- 1 stop bit
- Parity: even

PAS700 Panel Server Configuration

Modbus baud rate and Modbus address can be configured with PAS700 Panel Server webpage.

Perform the following steps to configure:

Step	Action
1	Connect DCM modules to PAS700 Panel Server by using cables. Connect PAS700 to PC with a network cable.
2	Login in PAS700 Panel Server webpage. Select Network Parameter Configuration tab from the function bar on the left to configure Modbus serial port parameters.
3	Select Configure RTU Devices tab to enter baud rate and Modbus Server address configuration page.
4	Click Baud Rate Synchronization button to synchronize the baud rate of network gateway and DCM modules. If baud rates are already synchronized, skip this step.
5	Click Import Template button to import the preset DCM circuit setting and Modbus address. You can also enter the DCM circuit setting and Modbus addresses manually in PAS700 webpage.
6	Click Start to apply the configuration automatically.
7	Open or close the circuit breaker in the specified time according to the pop-up message. If the circuit breaker supports motor mechanism with communication function, set the motor mechanism to manual mode.
8	When finished, close the pop-up window. If some Modbus addresses configuration fails, repeat step 5-7.

For more detailed information, refer to Modbus RTU chapter in PAS700 Panel Server User Guide.

NOTE: If the baud rate or parity is changed after the DCM Modbus SL module has automatically detected these settings, the DCM module must be restarted (power off/power on) in order to detect the new network parameters.

SN Configuration

DCM Modbus SL modules support Auto-Go baud rate self adaptation function. When the DCM module connects to Modbus serial line network, it automatically detects baud rate and parity.

The Modbus Client must send at least 25 frames on the Modbus Serial Line communication network in order to allow the auto sensing algorithm to work. The transmission format is binary with one start bit, eight data bits, and one stop bit. For example, sent a **Read Multiple Register** (function code 0x03) request to Modbus Server 1.

NOTE: If the baud rate or parity is changed after the DCM Modbus SL module has automatically detected these settings, the DCM module must be restarted (power off/power on) in order to detect the new network parameters.

If you already have the baud rate of DCM modules, refer to [Set DCM Modbus Baud Rate](#), page 193 to modify the DCM Modbus baud rate.

Perform the following steps to modify the Modbus address using DCM product serial number:

Step	Action
1	Find the serial number from the SN tag of DCM module, or scan the QR code on DCM module to get the serial number.
2	Modify Modbus address of each device. For more information, refer to Configure Modbus Address Using DCM Serial Number , page xxx. This example command configures with broadcast mode.

Communication Test

Test with PAS700

Suggest testing the serial line network communication of each circuit breaker using PAS700 Panel Server webpage.

Step	Action
1	Connect DCM modules to PAS700 Panel Server by using cables. Connect PAS700 to PC with a network cable.
2	Finish the configuration described in PAS700 Panel Server Configuration, page 19
3	Click Locate button.
4	Two LEDs on the corresponding DCM module blink for 15 seconds, and the display on MicroLogic 5A/E, 6A/E, and 7E trip unit flashes for 3 seconds.

Test by Sending Locating Frame

To confirm the location of the DCM with specified Modbus address, you can send a DCM device locating request. After sending the request, the LEDs on the corresponding DSM module blink for 15 seconds, and the display on MicroLogic 5A/E, 6A/E, and 7E trip unit flashes for 3 seconds. For more information, refer to Read DCM Modbus SL Module Location, page 194.

Test with EcoStruxure Power Commission Software

To test the serial line communication on various circuit breakers, it is recommended to use EcoStruxure Power Commission software, page 13.

If a PC running EcoStruxure Power Commission (EPC) software and connected on the Modbus network is able to read data from the IMU, the communication is established. Refer to the EcoStruxure Power Commission Online Help.

Modbus Protocol with ComPacT NSX Circuit Breakers

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Modbus Client-Server Principle

Overview

The Modbus protocol exchanges information using a request-reply mechanism between a client and a server. The client-server principle is a model for a communication protocol in which one device (the client) controls one or more other devices (the servers). In a standard Modbus network, there is 1 client and up to 31 servers.

A detailed description of the Modbus protocol is available at www.modbus.org.

Characteristics of the Client-Server Principle

The client-server principle is characterized as follows:

- Only 1 client is connected to the network at a time.
- Only the client can initiate communication and send requests to the servers.
- The client can address each server individually using its specific address or all servers simultaneously using address 0.
- The servers can only send replies to the client.
- The servers cannot initiate communication, either to the client or to other servers.

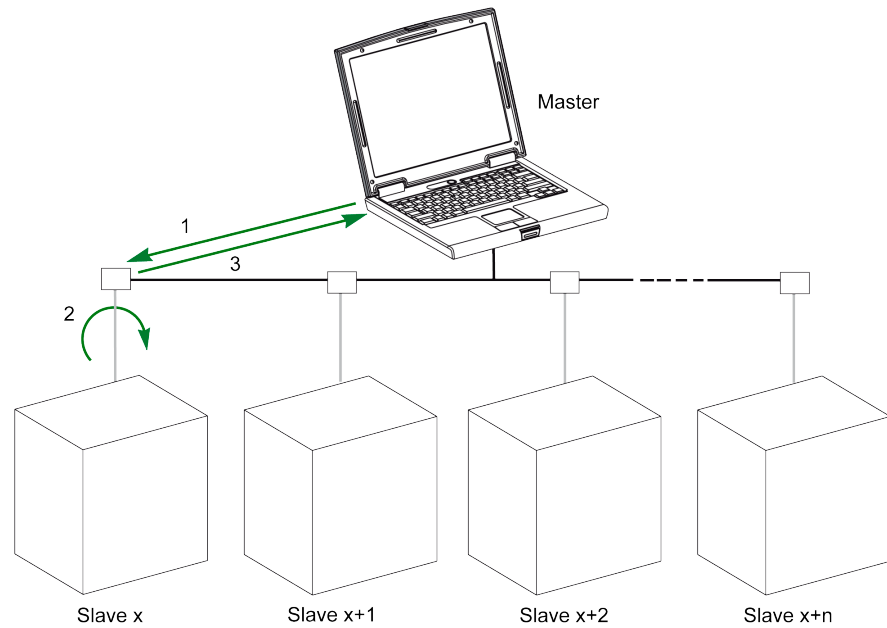
Client-Server Communication Modes

The Modbus protocol can exchange information using 2 communication modes:

- unicast mode
- broadcast mode

Unicast Mode

In unicast mode, the client addresses a server using the specific address of the server. The server processes the request then replies to the client.



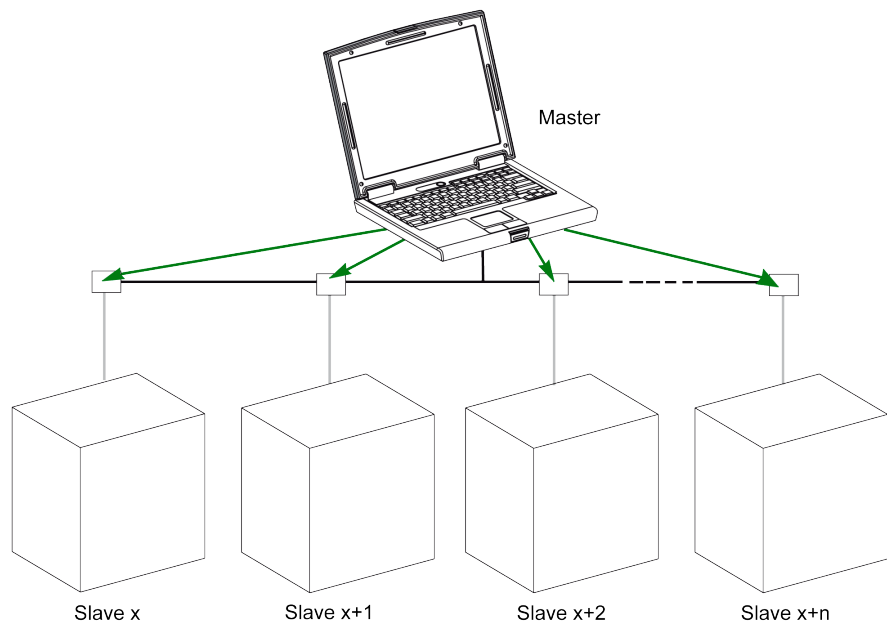
1 Request

2 Process

3 Reply

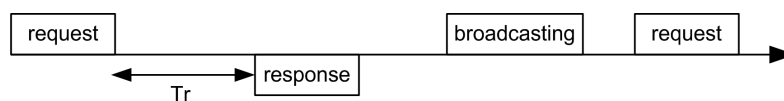
Broadcast Mode

The client can also address all servers using address 0. This type of exchange is called broadcasting. The servers do not reply to broadcasting messages.



Response Time

The response time T_r is the time needed by a server to respond to a request sent by the client:



Values with the Modbus protocol:

- Typical value < 10 ms for 90% of the exchanges
- Maximum value is around 700 ms, so it is recommended to implement a 1 second time out after sending a Modbus request.

Data Exchange

The Modbus protocol uses 2 types of data:

- Single bit
- Register (16 bits)

MasterPacT MTZ, MasterPacT NT/NW, ComPacT NS, and ComPacT NSX circuit breakers support registers only.

Each register has a register number. Each type of data (bit or register) has a 16-bit address.

The messages exchanged with the Modbus protocol contain the address of the data to be processed.

Registers and Addresses

The address of register number n is $n-1$. The tables detailed in the following parts of this document provide both register numbers (in decimal format) and corresponding addresses (in hexadecimal format). For example, the address of register number 12000 is 0x2EDF (11999).

Frames

All the frames exchanged with the Modbus protocol have a maximum size of 256 bytes and are composed of 4 fields:

Field	Definition	Size	Description
1	Server number	1 byte	Destination of the request • 0: broadcasting (all servers concerned) • 1–247: unique destination
2	Function codes	1 byte or 2 bytes	Refer to function codes description, page 28
3	Data	n registers	Request or reply data NOTE: Number of registers n is limited to 52 with MasterPacT MicroLogic E trip unit.
4	Check	2 bytes	CRC16 (to check transmission errors)

Modbus Programming Recommendations

Read Register Recommendations

The registers of the IMU modules are available through Modbus communication in:

- dataset registers (standard and/or legacy datasets)
- device registers:
 - MicroLogic registers
 - DCM registers

To read the registers:

- First read the registers that are available in datasets.
 - Standard dataset is recommended because it contains more data in a data format that allows a better accuracy.
 - Legacy dataset is used only for legacy equipments.
- Then read the data that is not available in datasets in the device registers.

The benefit of datasets is that the most useful information of each IMU module is collected in one table that can be read with two or three read requests. Each module updates the values in the dataset registers on a regular basis.

The response time of requests to dataset registers is shorter than the response time of requests to device registers. Therefore, it is recommended to read the dataset registers instead of device registers, to improve the overall performance of the communication system.

Register Update

The values in the registers are updated in two different ways:

- The measurement values are updated periodically, with a fixed refreshment rate.
- The other values are updated when the value is changed.

Type of Registers	Register Update
Identification	Triggered by device replacement
Settings	Triggered by setting change
Measurement	Periodically, with fixed refreshment rate
• Real-time measurements	Every 1 s
• Demand values of real-time measurements	Every 1 s
• Harmonic values	Every 3 s
• Energy measurements	Every 5 s
• Peak values of demand values of real-time measurements	Every 5 s
• Minimum and maximum values of real-time measurements	Every 5 s
Maintenance and diagnostic	Triggered by data change
Events	Triggered by event detection
IO status	Triggered by status change

The refreshment rate of values is the same for the dataset registers and the device registers.

Use the refreshment rate to optimize the performance of the communication between the remote controller and the IMU modules.

Modbus Functions

General Description

The Modbus protocol offers a number of functions that are used to read or write data over the Modbus network. The Modbus protocol also offers diagnostic and network-management functions.

Only the Modbus functions handled by the circuit breaker are described here.

Read Functions

The following read functions are available:

Function Code	Subfunction Code	Name	Description
3 (0x03)	–	Read holding registers	Read n output or internal registers
4 (0x04)	–	Read input registers	Read n input registers
43 (0x2B)	14 (0x0E)	Read device identification	Read the identification data of the server
43 (0x2B)	15 (0x0F)	Get date and time	Read the date and time of the server

NOTE: Number of registers n is limited to 52 with MasterPacT MicroLogic E trip unit.

Read Register Example

The following table shows how to read the rms current on phase 1 (I1) in register 1016. The address of register 1016 is $1016 - 1 = 1015 = 0x03F7$. The Modbus address of the Modbus server is $47 = 0x2F$.

Client Request		Server Reply	
Field Name	Example	Field Name	Example
Modbus server address	0x2F	Modbus server address	0x2F
Function code	0x03	Function code	0x03
Address of the register to read (MSB)	0x03	Data length in bytes	0x02
Address of the register to read (LSB)	0xF7	Register value (MSB)	0x02
Number of registers (MSB)	0x00	Register value (LSB)	0x2B
Number of registers (LSB)	0x01	CRC (MSB)	0xFF
CRC (MSB)	0xFF	CRC (LSB)	0xFF
CRC (LSB)	0xFF	–	

The content of register 1016 (address 0x03F7) is 0x022B = 555. Therefore, the rms current on phase 1 (I1) is 555 A.

Get Date and Time Example

The following table shows how to get the date and time of a Modbus server. The Modbus address of the Modbus server is 47 = 0x2F.

Client Request		Server Reply	
Field Name	Example	Field Name	Example
Modbus server address	0x2F	Modbus server address	0x2F
Function code	0x2B	Function code	0x2B
Subfunction code	0x0F	Subfunction code	0x0F
Reserved	0x00	Reserved	0x00
–	–	Date and time	Refer to the DATETIME data type

Set Date and Time Example

The following table shows how to set date and time of a Modbus server. The Modbus address of the Modbus server is 47 = 0x2F, the new date is October 2, 2014, and the new time is 2:32:03:500 p.m.

NOTE: Use the broadcast mode (with Modbus server address = 0) to set the date and time of all Modbus servers.

Client Request		Server Reply	
Field Name	Example	Field Name	Example
Modbus server address	0x2F	Modbus server address	0x2F
Function code	0x2B	Function code	0x2B
Subfunction code	0x10	Subfunction code	0x10
Reserved1	0x00	Reserved1	0x00
Not used	0x00	Not used	0x00
Year = 2014	0x0E	Year = 2014	0x0E
Month = October	0x0A	Month = October	0x0A
Day Of Month = 2	0x02	Day Of Month = 2	0x02
Hour = 14	0x0E	Hour = 14	0x0E
Minutes = 32	0x20	Minutes = 32	0x20
3 sec. 500 ms	0x0DAC	3 sec. 502 ms	0x0DAE

The normal response is an echo of the request, returned after the date-time has been updated in the remote device. If the date-time structure content is not consistent with a true date-time (that is, an invalid date-time), the value returned in the Date-Time field is set to 0 by the device.

In case of 24 Vdc power loss, the date and time of the Modbus servers without battery is not refreshed anymore. It is therefore mandatory to set date and time for all Modbus servers after recovering the 24 Vdc power supply.

Furthermore, due to the clock drift of each Modbus server, it is mandatory to set date and time for all Modbus servers periodically. Recommended period is at least every 15 minutes.

Scattered Holding Register Read Function

The scattered holding register read function is available:

Function Code	Subfunction Code	Name	Description
100 (0x64)	4 (0x04)	Read scattered holding register	Read n non-contiguous registers

The maximum value for n is 100 but when using a MasterPacT MicroLogic A or E trip unit, it is recommended to have n lower or equal to 21.

The scattered holding register read function enables the user to:

- avoid reading a large block of contiguous registers when only few registers are needed
- avoid multiple use of functions 3 and 4 in order to read non-contiguous registers

Read Scattered Holding Register Example

The following table shows how to read the addresses of the register 664 (address 0x0297) and register 666 (address 0x0299) of a Modbus server. The Modbus address of the Modbus server is 47 = 0x2F.

Client Request		Server Reply	
Field Name	Example	Field Name	Example
Modbus server address	0x2F	Modbus server address	0x2F
Function code	0x64	Function code	0x64
Data length in bytes	0x06	Data length in bytes	0x06
Subfunction code	0x04	Subfunction code	0x04
Transmission number ⁽¹⁾	0xXX	Transmission number ⁽¹⁾	0xXX
Address of first register to read (MSB)	0x02	Value of the first register read (MSB)	0x12
Address of first register to read (LSB)	0x97	Value of the first register read (LSB)	0x0A
Address of second register to read (MSB)	0x02	Value of the second register read (MSB)	0x74
Address of second register to read (LSB)	0x99	Value of the second register read (LSB)	0x0C
CRC (MSB)	0xXX	CRC (MSB)	0xXX
CRC (LSB)	0xXX	CRC (LSB)	0xXX
(1) The client gives the transmission number in the request. The server returns the same number in the reply.			

Write Functions

The following write functions are available:

Function Code	Subfunction Code	Name	Description
6 (0x06)	–	Preset single register	Write 1 register
16 (0x10)	–	Preset multiple registers	Write n registers
43 (0x2B)	16 (0x10)	Set date and time	Write the date and time of the server

NOTE: Number of registers n is limited to 52 with MasterPacT MicroLogic E trip units.

Diagnostic Functions

The following diagnostic functions are available:

Function Code	Subfunction Code	Name	Description
8 (0x08)	–	Diagnostic	Manage diagnostic counters
8 (0x08)	10 (0x0A)	Clear counters and diagnostic register	Reset all diagnostic counters
8 (0x08)	11 (0x0B)	Return bus message counter	Read the counter of correct bus messages managed by the server
8 (0x08)	12 (0x0C)	Return bus communication error counter	Read the counter of incorrect bus messages managed by the server
8 (0x08)	13 (0x0D)	Return bus exception error counter	Read the counter of exception responses managed by the server
8 (0x08)	14 (0x0E)	Return server message counter	Read the counter of messages sent to the server
8 (0x08)	15 (0x0F)	Return server no response counter	Read the counter of broadcast messages
8 (0x08)	16 (0x10)	Return server negative acknowledge counter	Read the counter of messages sent to the server but not answered because of the Negative Acknowledge exception code 07
8 (0x08)	17 (0x11)	Return server busy counter	Read the counter of messages sent to the server but not answered because of the Server Device Busy exception code 06
8 (0x08)	18 (0x12)	Return bus overrun counter	Read the counter of incorrect bus messages due to overrun errors
11 (0x0B)	–	Get communication event counter	Read Modbus event counter

Diagnostic Counters

Modbus uses diagnostic counters to enable performance and error management. The counters are accessible using the Modbus diagnostic functions (function codes 8 and 11). The Modbus diagnostic counters and the Modbus event counter are described in the following table:

Counter Number	Counter Name	Description
1	Bus message counter	Counter of correct bus messages managed by the server
2	Bus communication error counter	Counter of incorrect bus messages managed by the server
3	Server exception error counter	Counter of exception responses managed by the server and incorrect broadcast messages
4	Server message counter	Counter of messages sent to the server
5	Server no response counter	Counter of broadcast messages
6	Server negative acknowledge counter	Counter of messages sent to the server but not answered because of the Negative Acknowledge exception code 07
7	Server busy count	Counter of messages sent to the server but not answered because of the Server Device Busy exception code 06
8	Bus character overrun counter	Counter of incorrect bus messages due to overrun errors
9	Comm. event counter	Modbus event counter (this counter is read with function code 11)

Counters Reset

The diagnostic counters are reset to 0 when:

- the maximum value 65535 is reached,
- they are reset by a Modbus command (function code 8, sub-function code 10),
- the power supply is lost,
- the communication parameters are modified.

Modbus Exception Codes

Exception Responses

Exception responses from either the client or a server can result from data processing errors. One of the following events can occur after a request from the client:

- If the server receives the request from the client without a communication error and can handle the request correctly, it returns a normal response.
- If the server does not receive the request from the client due to a communication error, it does not return a response. The client program eventually processes a timeout condition for the request.
- If the server receives the request from the client but detects a communication error, it does not return a response. The client program eventually processes a timeout condition for the request.
- If the server receives the request from the client without a communication error, but cannot handle it (for example, the request is to read a register that does not exist), the server returns an exception response to inform the client of the nature of the error.

Exception Frame

The server sends an exception frame to the client to report an exception response. An exception frame is composed of 4 fields:

Field	Definition	Size	Description
1	Server number	1 byte	Destination of the request • 1–247: unique destination
2	Exception function code	1 byte	Request function code + 128 (0x80)
3	Exception code	n bytes	See next paragraph
4	Check	2 bytes	CRC16 (to check transmission errors)

Exception Codes

The exception response frame has two fields that differentiate it from a normal response frame:

- The exception function code of the exception response is equal to the function code of the original request plus 128 (0x80).
- The exception code depends on the communication error that the server encounters.

The following table describes the exception codes handled by the circuit breaker:

Exception Code	Name	Description
01 (0x01)	Illegal function	The function code received in the request is not an authorized action for the server. The server may be in the wrong state to process a specific request.
02 (0x02)	Illegal data address	The data address received by the server is not an authorized address for the server.
03 (0x03)	Illegal data value	The value in the request data field is not an authorized value for the server.
04 (0x04)	Server device failure	The server fails to perform a requested action because of an unrecoverable error.
05 (0x05)	Acknowledge	The server accepts the request but needs a long time to process it.

Exception Code	Name	Description
06 (0x06)	Server device busy	The server is busy processing another command. The client must send the request once the server is available.
07 (0x07)	Negative acknowledgment	The server cannot perform the programming request sent by the client.
08 (0x08)	Memory parity error	The server detects a parity error in the memory when attempting to read extended memory.
10 (0x0A)	Gateway path unavailable	The gateway is overloaded or not correctly configured.
11 (0x0B)	Gateway target device failed to respond	The server is not present on the network.

Illegal Data Address

This guide describes the registers available for each IMU module with the latest firmware revision. When a register described in the guide is not implemented in an IMU module that has an old firmware revision, an exception response is returned with the exception code 02 (0x02), illegal data address.

You can upgrade the firmware of the IMU modules by using EcoStruxure Power Commission software.

Write Protection

General Description

WARNING

HAZARD OF NUISANCE TRIPPING OR FAILURE TO TRIP

Protection setting adjustments must be done by qualified electrical personnel.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Remote modifications of Modbus registers can either be dangerous to personnel near the circuit breaker or can cause equipment damage if the protection settings are altered. Therefore, remote control commands are software protected.

Software Protection

To prevent an inadvertent change to the MicroLogic configuration, remote modifications of the Modbus registers are protected by both of the following:

- a robust data structure and a set of dedicated Modbus registers
- a user profile password scheme

This combination is called the command interface. Failure to conform to these results in an error code and the operation is not performed.

Password Management

General Description

Remote access to data on MicroLogic trip units and the ULP modules of the IMU is protected by password. Remote access includes:

- The communication network
- EcoStruxure Power Commission software

The following four profiles are defined for remote access. Each IMU has a different password for each user profile.

- Administrator
- Services
- Engineer
- Operator

The Administrator password is required to write the settings to the MicroLogic trip unit and the ULP modules of the IMU using EcoStruxure Power Commission software, page 13.

Each intrusive command via the command interface is assigned to one or several user profiles, and protected by the corresponding user profile password. The password for each intrusive command is indicated in the description of the command.

No password is required for non-intrusive commands through the command interface.

Default Passwords

WARNING

POTENTIAL COMPROMISE OF SYSTEM AVAILABILITY, INTEGRITY, AND CONFIDENTIALITY

Change default passwords at first use to help prevent unauthorized access to device settings, controls, and information.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

The default password for each user profile is as follows:

User profile	Default password
Administrator	'0000' = 0x30303030
Services	'1111' = 0x31313131
Engineer	'2222' = 0x32323232
Operator	'3333' = 0x33333333

Changing a Password

A password can be changed with [EcoStruxure Power Commission software](#), page 13.

Entering the current password for a given user profile is required to change the password of this user profile. Entering the Administrator password enables you to change the password of any user profile.

A password is composed of exactly 4 ASCII characters. It is case-sensitive and the allowed characters are:

- Digits from 0 to 9
- Letters from a to z
- Letters from A to Z

Passwords of the IMU

The MicroLogic trip unit and the ULP modules of the IMU must be protected by the same passwords for each user profile.

When using [EcoStruxure Power Commission software](#) to modify a password, the password gets modified in the MicroLogic trip unit and the ULP modules of the IMU.

It is compulsory to assign the current IMU passwords to the new module in the IMU, in case of:

- addition of a new ULP module in the IMU.
- replacement of the MicroLogic trip unit or one of the ULP module of the IMU.

Use [EcoStruxure Power Commission software](#) to modify the passwords of the new module to the current IMU passwords.

Password Reset

In case that the Administrator password of the (IMU) is lost or forgotten, the password can be reset to the default password with [EcoStruxure Power Commission software](#), page 13 and the support of the Schneider Electric Customer Care Center.

Command Interface

General Description

The command interface is used to:

- send remote commands.
- send remote control commands.

Remote commands are non-intrusive commands. They are not password-protected and always enabled.

Remote control commands are intrusive commands and can either be dangerous to personnel near the circuit breaker or can cause equipment damage if the protection settings are altered. Therefore, remote control commands are:

- protected by password where a password is required in the command
- protected by configuration with the DCM Modbus SL module, the remote control commands are enabled when the intrusive commands writing mode is unlocked (remote control commands writing is allowed).

Each command has a specific code. For example, command code 904 defines the command to open the circuit breaker.

Executing a Command

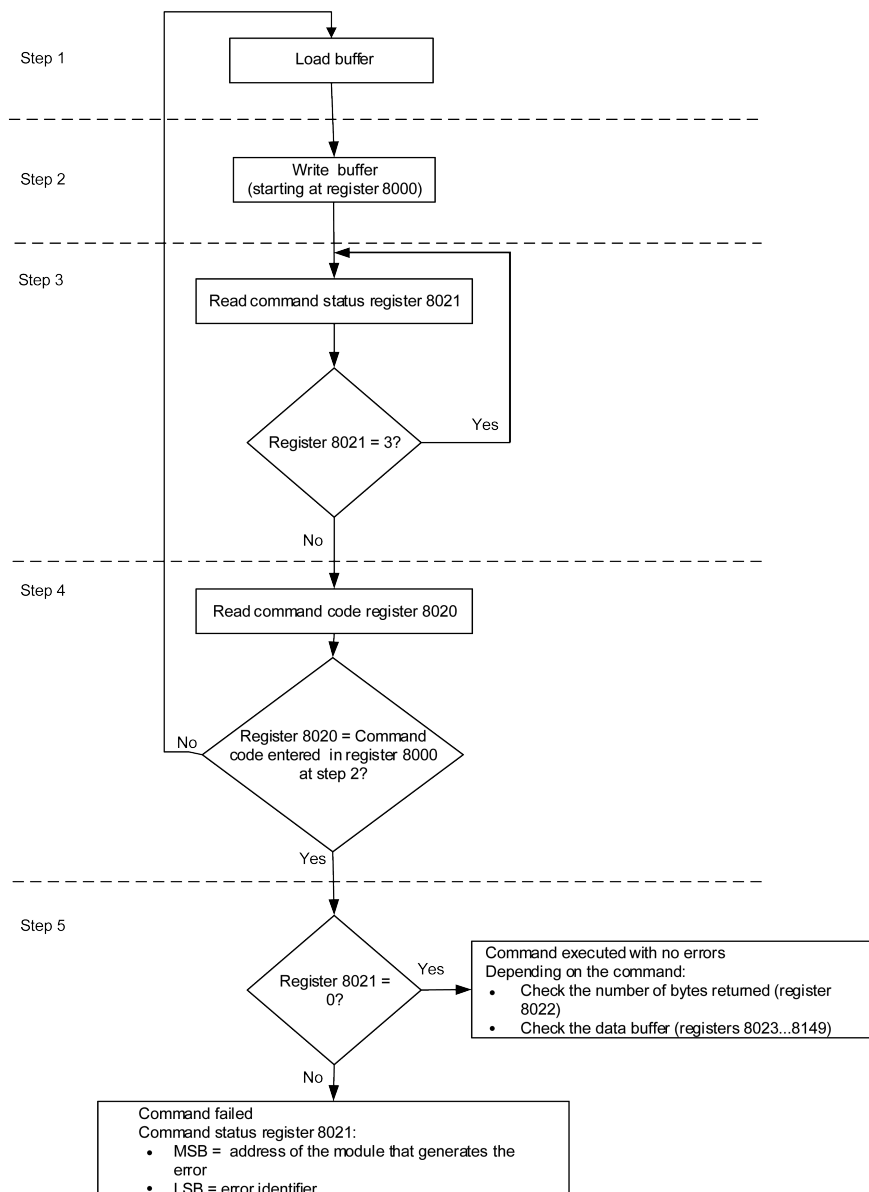
Follow these steps to execute a command:

Step	Action
1	Load a buffer.
2	Write this buffer with a write request (Modbus function 16) starting at register 8000.
3	Read the command status register 8021, and wait while its content shows that the command is still in progress (0x0003).
4	Read the command code register 8020: • If content of register 8020 is the command code entered in register 8000 at step 2, go to next step. • If content of register 8020 is different from the command code entered in register 8000 at step 2, restart at step 1.
5	Read the error code in the LSB of register 8021: • If LSB $\neq$ 0, then the command failed. Check the error code to understand the cause (see next paragraph). For example, if register 8021 returns 4609 (0x1201), then the error code is 1, which means that the password is not correct (insufficient user rights). • If LSB = 0, then the command is executed with no errors.

NOTE: The Modbus application will wait for the complete execution of one command before sending its next command. In case of no response, the Modbus application can resend the command. In this case, the first command will be aborted automatically.

Command Diagram

The following diagram shows the steps to follow in order to execute a command:



Command Data Structure

The command interface uses registers 8000 to 8149:

- The input parameters of a command are written in registers 8000 to 8015. The registers 8016 to 8019 are reserved.
- The data returned after command execution are written in registers 8020 to 8149.

The input parameters of a command are detailed in the following table:

Address	Register	Description	Comments
0x1F3F	8000	Command code	Writing at this register triggers the command using the parameters in the following registers.
0x1F40	8001	Parameter length	Number of bytes used for the parameters including this one (from 10 to 30). This value is provided for each command.

Address	Register	Description	Comments
0x1F41	8002	Destination	A constant value provided for each command. Factory setting: 0x0000
0x1F42	8003	Security type	A constant value provided for each command: 0 for non-intrusive commands not protected by password 1 for intrusive commands protected by a password
0x1F43	8004	Password	The password is composed of 4 ASCII bytes.
0x1F44	8005		The password to use depends on the command. This information is provided for each command.
0x1F45–0x1F4E	8006–8015	Additional parameters	Additional parameters define how the command is performed. Some commands have no additional parameters.
0x1F4F	8016	Reserved	Must be set to 0 (factory setting).
0x1F50	8017	Reserved	Must be set to 8019 (factory setting).
0x1F51	8018	Reserved	Must be set to 8020 (factory setting).
0x1F52	8019	Reserved	Must be set to 8021 (factory setting).

The data returned after command execution are detailed in the following table:

Address	Register	Description	Comments
0x1F53	8020	Last command code	When the command has been executed, it holds the last command code.
0x1F54	8021	Command status	When the command exits the busy state, it holds the completion code.
0x1F55	8022	Data buffer size	Number of bytes returned.
0x1F56–0x1FD4	8023–8149	Data buffer	Returned values. It is empty if the previous register is 0.

Command Status

When the command is successful, the command status is 0.

When the command is in progress, the command status is 3.

When the command generates an error, the command status register contains:

- LSB: the error code
- MSB: the address of the module that generates the error

Module Returning the Command Result

The following table lists the addresses of the modules:

Module Address	Module
1 (0x01)	USB maintenance interface
20 (0x14)	MicroLogic trip unit of ComPacT NSX
40 (0x28)	DCM Modbus-SL for one circuit breaker

Result of the Command

The following table lists the codes corresponding to the result of the command.

Code	Description
0 (0x00)	Successful command
1 (0x01)	Insufficient user rights (incorrect password)
2 (0x02)	Access violation (intrusive command mode is locked or remote control command written is disabled)
3 (0x03)	Unable to perform a read access
4 (0x04)	Unable to perform a write access
5 (0x05)	Unable to execute service (remote control commands or intrusive commands are disabled)
6 (0x06)	Not enough memory
7 (0x07)	Allocated memory is too small
8 (0x08)	Resource is not available
9 (0x09)	Resource does not exist
10 (0x0A)	Resource already exists
11 (0x0B)	Resource is out of order
12 (0x0C)	Access out of available memory
13 (0x0D)	String is too long
14 (0x0E)	Buffer is too small
15 (0x0F)	Buffer is too big
16 (0x10)	Input argument is out of range
17 (0x11)	Requested security level is not supported
18 (0x12)	Requested component is not supported
19 (0x13)	Command is not supported
20 (0x14)	Input argument has an unsupported value
21 (0x15)	Internal error during command
22 (0x16)	Timeout during command
23 (0x17)	Checksum error during command
24 (0x18)	Unsupported destination
151 (0x97)	Circuit breaker tripped, reset before commands
152 (0x98)	Circuit breaker already closed
153 (0x99)	Circuit breaker already open
154 (0x9A)	Circuit breaker already reset
155 (0x9B)	Actuator in manual mode
156 (0x9C)	Actuator not present
157 (0x9D)	Bad ASIC configuration
158 (0x9E)	Previous command in progress
159 (0x9F)	Reset command forbidden
160 (0xA0)	Inhibit mode on
169 (0xA9)	Already in asked state
170 (0xAA)	Unable to preset counters
171 (0xAB)	Output command rejected, already assigned
172 (0xAC)	Emitter not allowed to perform the command
173 (0xAD)	Mode not relevant with requested command
174 (0xAE)	Session key is invalid
175 (0xAF)	Out of session scope

Code	Description
176 (0xB0)	Session is already opened
177 (0xB1)	No session is open
178 (0xB2)	No valid setting was submitted
180 (0xB4)	Wireless component not started
190 (0xBE)	Read and get an invalid value
191 (0xBF)	License is not installed

Command Not Supported

The guide describes the commands available for each IMU module with the latest firmware version. When a command described in the guide is not implemented in an IMU module that has an old firmware version, the command status is returned with the error code 19 (0x13), command is not supported.

You can update the firmware of the IMU modules by using EcoStruxure Power Commission software.

Command Examples

Open Circuit Breaker

The following table details the steps to perform in the client remote device to send a remote command to the DCM Modbus SL module to open the circuit breaker. The command itself has no parameters.

Step	Action
1	Load a buffer of 20 registers, word0 to word19. - Load into word0 the value 904, the code corresponding to the open circuit breaker command. - Load into word1 the value 10, the length of the input parameters. The command itself has no parameters, 10 is the length of the fixed part. - Load into word2 the value 10241 (0x2801), the destination. This value is a constant for the command. It is provided in the command description. - Load into word3 the value 1. - Load into word4 and word5 the 4 ASCII bytes of the Administrator or Operator password. Assuming this password is 'ABcd', load 16706 (0x4142) into word #4 and 25444 (0x6364) into word #5. - Load into word6 to word16 the value 0. - Load into word17 the value 8019, a command setup constant. - Load into word18 the value 8020, a command setup constant. - Load into word19 the value 8021, a command setup constant.
2	Write this buffer with a write request (Modbus function 16) of 20 registers, starting at register 8000.
3	Read the command status register 8021, and wait while its content shows that the command is still in progress (0x0003). If the command status does not change after a timeout (1 s), check the Modbus connection.
4	Read the command status register 8020: - If content of register 8020 is the command code entered in register 8000 at step 2, go to next step. - If content of register 8020 is different from the command code entered in register 8000 at step 2, restart at step 1.
5	Read the error code in the LSB of register 8021: - If LSB ≠ 0, then the command failed. Check the error code to understand the cause (see next paragraph). For example, if register 8021 returns 4609 (0x1201), then the error code is 1, which means that the password is not correct (insufficient user rights). - If LSB = 0, then the command was executed with no errors.

Reset Energy Measurements

The following table details the steps to perform to send a command to the MicroLogic trip unit to reset the minimum/maximum energy measurements. The command itself has one parameter.

Step	Action
1	Load a buffer of 20 registers, word0 to word19. - Load into word0 the value 46728, the code corresponding to the reset minimum/maximum command. - Load into word1 the value 12, the length of the input parameters. The command itself has one parameter, add 2 bytes to 10 which is the length of the fixed part. - Load into word2 the value 5121 (0x1401), the destination. This value is a constant for the command. It is provided in the command description. - Load into word3 the value 1. - Load into word4 and word5 the 4 ASCII bytes of the Administrator or Operator password. Assuming this password is 'Pw57', load 20599 (0x5077) into word #4 and 13623 (0x3537) into word #5. - Load into word6 the value 512 (bit 9 set to one). This value requests that the energy measurement minimum/maximum be reset. - Load into word7 to word16 the value 0. - Load into word17 the value 8019, a command setup constant. - Load into word18 the value 8020, a command setup constant. - Load into word19 the value 8021, a command setup constant.
2	Write this buffer with a write request (Modbus function 16) of 20 registers, starting at register 8000.
3	Read the command status register 8021, and wait while its content shows that the command is still in progress (0x0003). If the command status does not change after a timeout (1 s), check the Modbus connection.
4	Read the command status register 8020: - If content of register 8020 is the command code entered in register 8000 at step 2, go to next step. - If content of register 8020 is different from the command code entered in register 8000 at step 2, restart at step 1.
5	Read the error code in the LSB of register 8021: - If LSB $\neq$ 0, then the command failed. Check the error code to understand the cause (see next paragraph). For example, if register 8021 returns 4609 (0x1201), then the error code is 1, which means that the password is not correct (insufficient user rights). - If LSB = 0, then the command was executed with no errors.

Read Date and Time

The following table details the steps to perform to send a command to the IFM communication interface to read the date and time. The command itself has no parameters. The date and time are returned in a buffer.

Step	Action
1	Load a buffer of 20 registers, word0 to word19. - Load into word0 the value 768, the code corresponding to the read date/time command. - Load into word1 the value 10, the length of the input parameters. The command itself has no parameters, the length is the length of the fixed part which is 10. - Load into word2 the value 10240 (0x2800), the destination. This value is a constant for the command. It is provided in the command description. NOTE: Load into word2 the value 8704 (0x2200) for IFE communication interface as destination. - Load into word3 the value 0. - Load into word4 and word5 the value 0x0000 (no password required). - Load into word6 to word16 the value 0. - Load into word17 the value 8019, a command setup constant. - Load into word18 the value 8020, a command setup constant. - Load into word19 the value 8021, a command setup constant.
2	Write this buffer with a write request (Modbus function 16) of 20 registers, starting at register 8000.
3	Read the command status register 8021, and wait while its content shows that the command is still in progress (0x0003). If the command status does not change after a timeout (1 s), check the Modbus connection.
4	Read the command status register 8020: - If content of register 8020 is the command code entered in register 8000 at step 2, go to next step. - If content of register 8020 is different from the command code entered in register 8000 at step 2, restart at step 1.
5	Read the error code in the LSB of register 8021:

Step	Action
	- If LSB ≠ 0, then the command failed. Check the error code to understand the cause (see next paragraph). For example, if register 8021 returns 783 (0x030F), then the error code is 15 (0x0F), which means that the input argument is out of range (too many parameters). - If LSB = 0, then the command was executed with no errors.
6	If there were no errors, read the data buffer length in register 8022. Its value must be 8 for this command.
7	In the data buffer: - register 8023 holds the month in the MSB, the day in the LSB. - register 8024 holds the year offset in the MSB (add 2000 to get the year) and the hour in the LSB. - register 8025 holds the minutes in the MSB, the seconds in the LSB. - register 8026 holds the milliseconds.

Configure Modbus Address Using DCM Serial Number

The following table details the steps to perform to send a command to the DCM interface to configure DCM Modbus address. The command itself has no parameters.

Steps	Action
1	Load a buffer of 20 registers, word0 to word19. - Load into word0 the value 38667 (0x970B), the code corresponding to the configure Modbus address command. - Load into word1 the value 30 (0x001E), the length of the input parameters. - Load into word2 the value 0 (0x0000), the destination. This value is a constant for the command. - Load into word3 the value 1 (0x0001). - Load into word4 and word5 the 4 ASCII bytes of the Administrator or Operator password. Assuming this password is '0000', load 12336 (0x3030) into word #4 and 12336 (0x3030) into word #5. Assuming this password is 'ABcd', load 16706 (0x4142) into word #4 and 25444 (0x6364) into word #5. - Load into word6 the desired Modbus address. Assuming the desired Modbus address is 1, load 1 (0x0001) into word #6. Assuming it is compatible with PAS700, Modbus address should be set to 1-99. - Load into word7 the value 1 (0x0001). - Load into word8 to word15 the DCM serial number and end byte 0 (0x00), totally 8 characters. DCM serial number should transformed to ASCII bytes. Assuming SN is WX2342200100031, load 0x5758 (WX) into word #8, 0x3233 (23) into word #9, 0x3432 (42) into word #10, 0x3230 (20) into word #11, 0x3031 (01) into word #12, 0x3030 (00) into word #13, 0x3033 (03) into word #14, 0x3100 (1 and end byte 0x00) into word #15. NOTE: Get the DCM serial number from SN tag or scan the QR code on DCM module to get the serial number. ASCII bytes of DCM serial number are as following: w = 0x57, X = 0x58, 0 = 0x30, 1 = 0x31, 2 = 0x32, 3 = 0x33, 4 = 0x34, 5 = 0x35, 6 = 0x36, 7 = 0x37, 8 = 0x38, 9 = 0x39
2	Write this buffer with a write request (broadcast address 00, Modbus function 16, address to wrote 7999 (0x1F3F), length of data 16) of 16 registers, starting at register 8000. Example, when configuring the Modbus address of the DCM with SN "WX2342200100031" to 1, the corresponding bytes are as following: <pre>00 10 1F 3F 00 10 20 97 0B 00 1E 00 00 00 01 30 30 30 30 00 01 00 01 57 58 32 33 34 32 32 30 30 31 30 30 30 33 31 00 0C FD</pre> 00: broadcast address 10: Modbus function 16 1F 3F: address to wrote 7999 00: the length of data 20: number of bytes 97 0B: word0 of step 1; 00 1E: word1 of step 1; 00 00: word2 of step 1; 00 01: word3 of step 1 30 30 30 30: word4 and word5 of step 1 00 01: word6 of step 1, that is the desired address of Modbus slave 00 01: word7 of step 1 57 58 32 33 34 32 32 30 30 31 30 30 30 33 31 00: word8 to word15 of step 1, that is DCM serial number and end byte 0C FD: Modbus CRC verification code

3	Read the command status register 8021, and wait while its content shows that the command is still in progress (0x0003). If the command status does not change after a timeout (1 s), check the Modbus connection.
4	Read the command status register 8020: • If content of register 8020 is the command code entered in register 8000 at step 2, go to next step. • If content of register 8020 is different from the command code entered in register 8000 at step 2, restart at step 1.
5	Read the error code in the LSB of register 8021: • If LSB $\neq$ 0, then the command failed. Check the error code to understand the cause (see next paragraph). For example, if register 8021 returns 783 (0x030F), then the error code is 15 (0x0F), which means that the input argument is out of range (too many parameters). • If LSB = 0, then the command was executed with no errors.

Date Management

Introduction

Each module of the IMU uses its date to time-stamp events and history registers.

The date of the IMU modules is updated in two steps:

1. External synchronization: The Modbus client synchronizes the DCM Modbus SL module.
2. Internal synchronization:
 - The DCM module synchronizes all ULP modules connected in the IMU, eg, MicroLogic trip units.

External Synchronization

There are two ways to externally synchronize the the DCM Modbus SL module:

- Manually with *EcoStruxure Power Commission* software, page 13.
- By programming of the Modbus client using:
 - either the Modbus function set date and time: function code 43-16 *Write Functions*, page 30
 - or the set absolute time command through the DCM Modbus SL module.

The communication interface is considered as externally synchronized if the last synchronization has occurred within the last 2 hours.

Internal Synchronization

When the DCM module receives the date and time, it broadcasts the date and time to all the ULP modules connected in the IMU.

History Mechanism

General Description

The Modbus history registers enable the user to track the occurrence of specific events and their corresponding dates.

4 event histories are available:

- Alarm history: the alarm history format corresponds to a series of 10 records. Each record is composed of 5 registers describing one alarm. See [Alarm History](#), page 118.
- Trip history: the trip history format corresponds to a series of 17 records. Each record is composed of 7 registers describing one trip. See [Trip History](#), page 120.
- Maintenance operation history: the maintenance operation history format corresponds to a series of 10 records. Each record is composed of 5 registers describing one maintenance operation. See [Maintenance Operation History](#), page 125.
- DCM Modbus SL module event history: the DCM Modbus SL module event history format corresponds to a series of 10 records. Each record is composed of 5 registers describing one DCM Modbus SL module event. See [Event History](#), page 176.

History Mechanism

Each event is time-stamped using the ULP DATE format, page 54.

When the history is full, the oldest event record is discarded to make room for the most recent event record, which is pushed to the top of the history.

The records are ordered in decreasing occurrence time, the most recent occurrence is in the first record.

The following tables describe the history mechanism for a 10-record history format:

Before event E

Record	1	2	3	4	5	6	7	8	9	10
Event	E-1 (most recent event)	E-2	E-3	E-4	E-5	E-6	E-7	E-8	E-9	E-10 (oldest event)

After event E

Record	1	2	3	4	5	6	7	8	9	10
Event	E (most recent event)	E-1	E-2	E-3	E-4	E-5	E-6	E-7	E-8	E-9 (oldest event)

After event E, event E-10 is lost.

History Reading

A read request is necessary to read a history record (see Modbus function read n input words, function code = 4, in [Read Functions](#), page 28). For example, a read request of 5 registers is necessary to read the most recent alarm record of the alarm history format (see [Alarm History](#), page 118).

Furthermore, to read the last n records of a history format, a read request of (m) x (n) registers is necessary, where m is the number of registers that compose the record.

For example, a read request of $7 \times 3 = 21$ registers is necessary to read the most recent 3 trip records of the trip history format (see [Trip History](#), page 120):

- The first 7 registers describe the first record of the trip history format (most recent trip).
- The next 7 registers describe the second record of the trip history format.
- The last 7 registers describe the third record of the trip history format.

When not used, history registers return 32768 (0x8000).

Modbus Registers Tables

General Description

The following chapters describe the Modbus registers of the MicroLogic trip unit and the Modbus registers of the modules connected to it. These registers provide information that can be read, like electrical measures, protection configuration, and monitoring information. The command interface enables the user to modify these registers in a controlled way.

The presentation rules of the Modbus registers are as follows:

- The registers are grouped according to the module they relate to:
 - MicroLogic 5A/E, 6A/E, and 7E Trip Unit Registers, page 99
 - DCM Modbus SL Module Registers, page 170
- For each module, the registers are grouped in tables of logically related information. The tables are presented in increasing address.
- For each module, the commands are described separately:
 - MicroLogic 5A/E, 6A/E, and 7E Trip Unit Commands, page 156
 - DCM Modbus SL Module Commands, page 180

To find a register, refer to the ordered list of the registers with a cross reference to the page where these registers are described, page 197.

Table Format

Register tables have the following columns:

Address	Register	RW	X	Unit	Type	Range	A/E	Description

- **Address:** a 16-bit register address in hexadecimal. The address is data used in the Modbus frame.
- **Register:** a 16-bit register number in decimal (register = address + 1).
- **RW:** register read-write status
 - R: the register can be read by using Modbus functions
 - W: the register can be written by using Modbus functions
 - RW: the register can be read and written by using Modbus functions
 - RC: the register can be read by using the command interface
 - WC: the register can be written by using the command interface
- **X:** the scale factor. A scale of 10 means that the register contains the value multiplied by 10. So, the real value is the value in the register divided by 10.

Example:

Register 1054 contains the system frequency, page 103. The unit is Hz and the scale factor is 10.

If the register returns 503, this means that the system frequency is $503/10 = 50.3$ Hz.

- **Unit:** the unit the information is expressed in.
- **Type:** the encoding data type (see data type description below).
- **Range:** the permitted values for this variable, usually a subset of what the format allows.

- **A/E** types of MicroLogic trip unit:
 - Type A (Ammeter): current measurements
 - Type E (Energy): current, voltage, power and energy measurements
- **Description:** provides information about the register and restrictions that apply.

Data Types

Data Types	Description	Range
INT16U	16-bit unsigned integer	0 to 65535
INT16	16-bit signed integer	-32768 to +32767
INT32U	32-bit unsigned integer	0 to 4 294 967 295
INT32	32-bit signed integer	-2 147 483 648 to +2 147 483 647
INT64	64-bit signed integer	-9 223 372 036 854 775 808 to +9 223 372 036 854 775 807
FLOAT32	32-bit signed integer with a floating point	2^{-126} (1.0) to 2^{127} ($2 - 2^{-23}$)
OCTET STRING	Text string	1 byte per character
DATETIME	Date and time in the IEC 60870-5 format , page 53	–
ULP DATE	Date and time in ULP DATE format, page 54	–

Big-Endian Format

INT32, INT32U, INT64, and INT64U variables are stored in big-endian format: the most significant register is transmitted first, the least significant register is transmitted at last place.

INT32, INT32U, INT64, and INT64U variables are made of INT16U variables.

The formulas to calculate the decimal value of these variables are:

- INT32: $(0\text{-bit}31) \times 2^{31} + \text{bit}30 \times 2^{30} + \text{bit}29 \times 2^{29} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$
- INT32U: $\text{bit}31 \times 2^{31} + \text{bit}30 \times 2^{30} + \text{bit}29 \times 2^{29} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$
- INT64: $(0\text{-bit}63) \times 2^{63} + \text{bit}62 \times 2^{62} + \text{bit}61 \times 2^{61} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$
- INT64U: $\text{bit}63 \times 2^{63} + \text{bit}62 \times 2^{62} + \text{bit}61 \times 2^{61} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$

Example 1:

The total active energy in the standard dataset is an INT64 variable coded in registers 32096 to 32099.

If the values in the registers are:

- register 32096 = 0
- register 32097 = 0
- register 32098 = 0x0017 or 23
- register 32099 = 0x9692 or 38546 as INT16U variable and -26990 as INT16 variable (use the INT16U value to calculate the value of the total active energy).

Then the total active energy is equal to $0 \times 2^{48} + 0 \times 2^{32} + 23 \times 2^{16} + 38546 \times 2^0 = 1545874$ Wh.

Example 2:

The reactive energy in the legacy dataset is an INT32 variable coded in registers 12052 to 12053.

If the values in the registers are:

- register 12052 = 0xFFF2 = 0x8000 + 0x7FF2 or 32754
- register 12053 = 0xA96E or 43374 as INT16U variable and -10606 as INT16 variable (use the INT16U value to calculate the value of the reactive energy).

Then the reactive energy is equal to $(0-1) \times 2^{31} + 32754 \times 2^{16} + 43374 \times 2^0 = -874130$ kVARh.

Data Type: FLOAT32

Data type FLOAT32 is represented in the single precision IEEE 754 (IEEE standard for floating-point arithmetic). A value N is calculated as indicated below:

$$N = (-1)^S \times 2^{E-127} \times (1+M)$$

Coefficient	Stands for	Description	Number of Bits
S	Sign	Defines the sign of the value: 0 = positive 1 = negative	1 bit
E	Exponent	Excess 127 binary integer added. When $0 < E < 255$, the actual exponent is: $e = E - 127$.	8 bits
M	Mantissa	Magnitude, normalized binary significant	23 bits

Example:

0 = 0 **00000000** 000000000000000000000000

-1.5 = 1 **01111111** 100000000000000000000000

with:

- $S = 1$
- $E = 01111111 = 127$
- $M = 100000000000000000000000 = 1 \times 2^{-1} + 0 \times 2^{-2} + \dots + 0 \times 2^{-23} = 0.5$
- $N = (-1) \times 2^0 \times (1+0.5) = -1.5$

Data Type: DATETIME

DATETIME is a data type used to code date and time defined by the IEC 60870-5 standard.

Register	Type	Bit	Range	Description
1	INT16U	0–6	0x00–0x7F	Year: 0x00 (00) to 0x7F (127) correspond to years 2000 to 2127 For example, 0x0D (13) corresponds to year 2013.
		7–15	–	Reserved
2	INT16U	0–4	0x01–0x1F	Day
		5–7	–	Reserved
		8–11	0x00–0x0C	Month
		12–15	–	Reserved
3	INT16U	0–5	0x00–0x3B	Minutes
		6–7	–	Reserved
		8–12	0x00–0x17	Hours
		13–15	–	Reserved
4	INT16U	0–15	0x0000–0xEA5F	Milliseconds

Quality of DATETIME Timestamps

The quality of timestamps coded with the DATETIME data type can be indicated in the register following the 4 registers of the timestamp. In this case, the timestamp quality is coded as follows:

Bit	Description
0–11	Reserved
12	Externally synchronized: 0 = Invalid 1 = Valid
13	Synchronized: 0 = Invalid 1 = Valid
14	Date and time is set: 0 = Invalid 1 = Valid
15	Reserved

Quality of Bits in Registers

The quality of each bit of a register coded as INT16U data type as an enumeration of bits can be indicated in the register preceding the register.

Example:

The quality of each bit of the register 32001, circuit breaker status, is given in the preceding register, 32000.

The quality of the data corresponding to the bit 0 of register 32001, OF status indication contact, is given in the bit 0 of register 32000:

- bit 0 of register 32000 = quality of OF status indication
- bit 0 of register 32001 = OF status indication contact

If	Then
If bit 0 of register 32000 = 1 AND bit 0 of register 32001 = 0	The OF contact indicates that the device is open
If bit 0 of register 32000 = 1 AND bit 0 of register 32001 = 1	The OF contact indicates that the device is closed
If bit 0 of register 32000 = 0	The OF contact indication is invalid

Data Type: ULP DATE

ULP DATE is a data type used to code date and time. This table presents the ULP DATE data type.

Register	Type	Bit	Range	Description
1 2	INT32U	–	0x00000000– 0xFFFFFFFF	Number of seconds since January 1 2000
3	INT16U	–	–	Complement in milliseconds
		0–9	–	Encodes the milliseconds
		10–11	–	Not used
		12	0–1	DCM, IFM or IFE communication interface external synchronization status 0 = The communication interface has not been externally synchronized within the last 2 hours. 1 = The communication interface has been externally synchronized within the last 2 hours.
		13	0–1	ULP module internal synchronization status 0 = The ULP module has not been internally synchronized. 1 = The ULP module has been internally synchronized.
		14	0–1	Absolute date is set since last power on 0 = No 1 = Yes
		15	–	Reserved

ULP Date Counter

The date in ULP DATE format is counted in number of seconds since January 1 2000.

In case of a power loss for an IMU module, the time counter is reset to January 1 2000.

If an external synchronization occurs after a power loss, the time counter is updated and converts the synchronization date to the corresponding number of seconds since January 1 2000.

ULP Date Conversion Principle

To convert the date from number of seconds since January 1 2000 to current date, the following rules apply:

- 1 non-leap year = 365 days
- 1 leap year = 366 days

Years 2000, 2004, 2008, 2012,...(multiple of 4) are leap years (except year 2100).

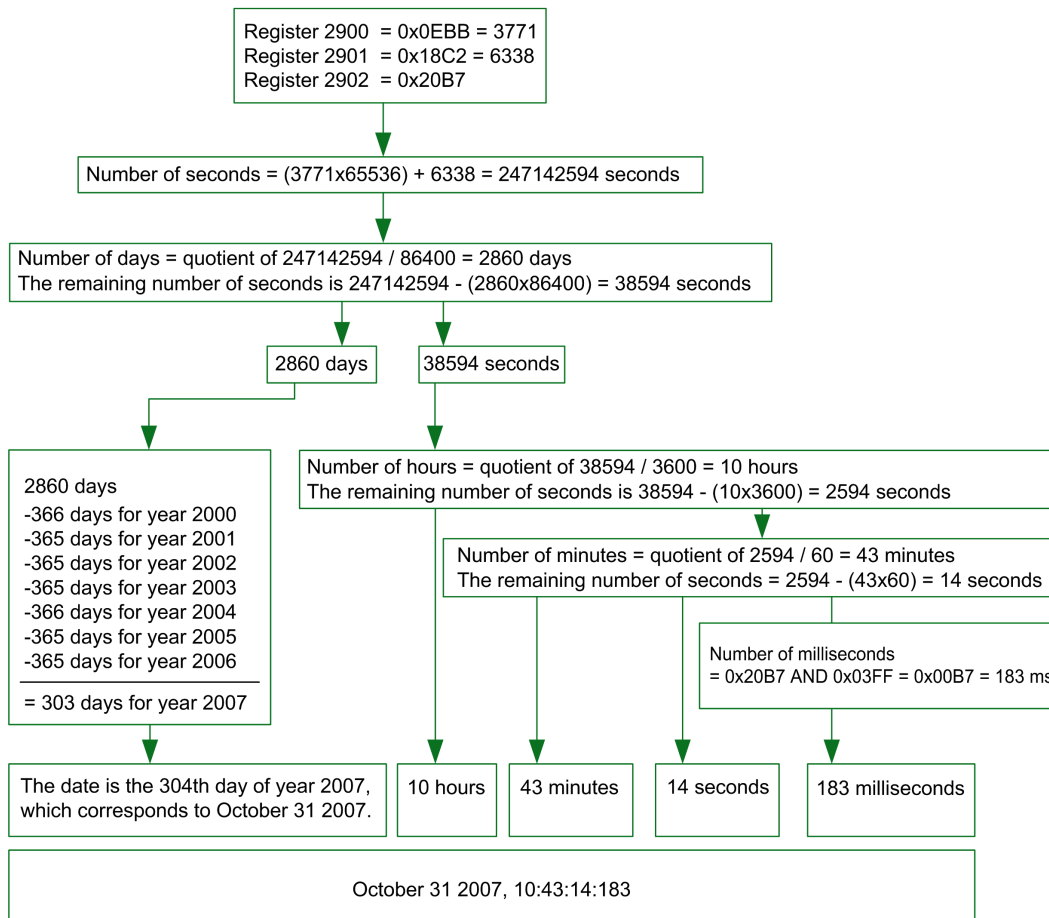
- 1 day = 86,400 seconds
- 1 hour = 3,600 seconds
- 1 minute = 60 seconds

The following table describes the steps to follow to convert the date from number of seconds since January 1 2000 to current date:

Step	Action
1	Calculate the number of seconds since January 1 2000: $S = (\text{content of register 1} \times 65536) + (\text{content of register 2})$
2	Calculate the number of days since January 1 2000: $D = \text{integer value of the quotient of } S / 86,400$ Calculate the remaining number of seconds: $s = S - (D \times 86,400)$
3	Calculate the number of days elapsed for the current year: $d = D - (NL \times 365) - (L \times 366)$ with NL = number of non-leap years since year 2000 and L = number of leap years since year 2000
4	Calculate the number of hours: $h = \text{integer value of the quotient of } s / 3600$ Calculate the remaining number of seconds: $s' = s - (h \times 3600)$
5	Calculate the number of minutes: $m = \text{integer value of the quotient of } s' / 60$ Calculate the remaining number of seconds: $s'' = s' - (m \times 60)$
6	Calculate the number of milliseconds: $ms = (\text{content of register 3}) \text{ AND } 0x03FF$
7	Result: • The current date is $\text{date} = d + 1$. For example, if $d = 303$, the current date corresponds to the 304th day of the year, which corresponds to October 31 2007. • The current time is $h:m:s''$:ms

ULP Date Conversion Example

Registers 2900 and 2901 return the date in number of seconds since January 1 2000. Register 2902 returns the complement in ms with the quality of the date.



Notes

- The Type column tells how many registers to read to get the variable. For instance INT16U requires reading one register, whereas INT32 requires reading 2 registers.
- Some variables must be read as a block of multiple registers, like the energy measurements. Reading the block partially results in an error.
- Reading from an undocumented register results in a Modbus exception. Refer to [Modbus Exception Codes](#), page 33.
- Numerical values are given in decimal. When it is useful to have the corresponding value in hexadecimal, it is shown as a C language type constant: 0xdddd. For example, the decimal value 123 is represented in hexadecimal as: 0x007B.
- For measures that depend on the presence of neutral as identified by [register 3314](#), page 140, reading the value returns 32768 (0x8000) if not applicable. For each table where it occurs, it is explained in a footnote.
- Out of order and not applicable values depend on the data type.

NOTE: As per legacy register implementation, some registers may display different out of order and not applicable values. For example, INT16U registers may return 32768 (0x8000) and INT32U may display 0x80000000.

Data Type	Out of Order and Not Applicable Values
INT16U	65535 (0xFFFF)
INT16	-32768 (0x8000)
INT32U	4294967295 (0xFFFFFFFF)
INT32	0x80000000
INT64U	0xFFFFFFFFFFFFFFFF
INT64	0x8000000000000000
FLOAT32	0xFFC00000

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Standard Dataset

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Standard Dataset

Description

The standard dataset contains the most useful information of each IMU module in one convenient table. The standard dataset is available in the registers 32000 to 32341. It can be read with three read requests.

Each IMU module updates the values in the dataset registers on a regular basis.

The response time of requests to standard dataset registers is shorter than the response time of requests to device registers. Therefore, it is recommended to read the standard dataset registers instead of device registers, to improve the overall performance of the system [Modbus Programming Recommendations](#), page 26.

The standard dataset can be used with:

- the IFE Ethernet communication interface for one circuit breaker
- the IFE Ethernet switchboard server
- the IFM Modbus SL communication interface for one circuit breaker
- the DCM Modbus SL module

Modbus Registers

Table of Standard Dataset Common Registers

The main information needed for remote supervision of a ComPacT NSX, ComPacT NS, MasterPacT NT/NW or MasterPacT MTZ circuit breaker is contained in the table of common registers starting at register 32000.

One Modbus read request is limited to 125 registers maximum. Three Modbus read requests are necessary to read the entire table.

It contains the following information:

- Circuit breaker status
- Tripping causes
- Real-time values of main measurements: current, voltage, power, and energy

The content of this table of registers is detailed in [Standard Dataset Common Registers](#), page 66.

Use of these common registers is highly recommended to optimize response times and simplify the use of data.

Table Format

Register tables have the following columns:

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description

- **Address:** a 16-bit register address in hexadecimal. The address is the data used in the Modbus frame.
- **Register:** a 16-bit register number in decimal (register = address + 1).
- **RW:** register read-write status
 - R: the register can be read by using Modbus functions
 - W: the register can be written by using Modbus functions
 - RW: the register can be read and written by using Modbus functions
 - RC: the register can be read by using the command interface
 - WC: the register can be written by using the command interface
- **Unit:** the unit the information is expressed in.
- **Type:** the encoding data type (see data type description below).
- **Range:** the permitted values for this variable, usually a subset of what the format allows.
- **A/E:** types of ComPact or ComPacT NSX MicroLogic trip unit for which the register is available.
 - Type A (Ammeter): current measurements for ComPact NSX circuit breakers only.
 - Type E (Energy): current, voltage, power, and energy measurements for ComPact or ComPacT NSX circuit breakers.

- **A/E/P/H**: types of MasterPacT NT/NW and ComPacT NS MicroLogic trip unit for which the register is available.
 - Type A (Ammeter): current measurements
 - Type E (Energy): current, voltage, power, and energy measurements
 - Type P (Power): current, voltage, power, energy measurements, and advanced protection
 - Type H (Harmonics): current, voltage, power, energy, energy quality measurements, and advanced protection
- **X**: register available in the MicroLogic X control unit for MasterPacT MTZ circuit breakers.
- **Description**: provides information about the register and restrictions that apply.

Data Types

Data Types	Description	Range
INT16U	16-bit unsigned integer	0 to 65535
INT64	64-bit signed integer	- 9 223 372 036 854 775 808 to + 9 223 372 036 854 775 807
INT64U	64-bit unsigned integer	0 to 18 446 744 073 709 600 000
FLOAT32	32-bit signed integer with a floating point	$2^{-126} (1.0)$ to $2^{127} (2 - 2^{-23})$

Big-Endian Format

INT64 and INT64U variables are stored in big-endian format: the most significant register is transmitted first and the least significant register is transmitted at last place.

INT64 and INT64U variables are made of INT16U variables.

The formulas to calculate the decimal value of these variables are:

- INT64: $(0\text{-bit}63) \times 2^{63} + \text{bit}62 \times 2^{62} + \text{bit}61 \times 2^{61} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$
- INT64U: $\text{bit}63 \times 2^{63} + \text{bit}62 \times 2^{62} + \text{bit}61 \times 2^{61} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$

Example:

The total active energy in the standard dataset is an INT64 variable coded in registers 32096 to 32099.

If the values in the registers are:

- register 32096 = 0
- register 32097 = 0
- register 32098 = 70 (0x0046) 0x0017 or 23
- register 32099 = 2105 (0x0839) 0x9692 or 38546 as INT16U variable and -26990 as INT16 variable (use the INT16U value to calculate the value of the total active energy).

Then the total active energy is equal to $0 \times 2^{48} + 0 \times 2^{32} + 23 \times 2^{16} + 38546 \times 2^0 = 1545874$ Wh.

Data Type: FLOAT32

Data type FLOAT32 is represented in the single precision IEEE 754 (IEEE standard for floating-point arithmetic). A value N is calculated as indicated below:

$$N = (-1)^S \times 2^{E-127} \times (1+M)$$

Coefficient	Stands for	Description	Number of Bits
S	Sign	Defines the sign of the value: 0 = positive 1 = negative	1 bit
E	Exponent	Excess 127 binary integer added. When $0 < E < 255$, the actual exponent is: $e = E - 127$.	8 bits
M	Mantissa	Magnitude, normalized binary significant	23 bits

Example:

0 = 0 **00000000** 000000000000000000000000

-1.5 = 1 **01111111** 100000000000000000000000

with:

- S = 1
- E = **01111111** = 127
- M = 100000000000000000000000 = $1 \times 2^{-1} + 0 \times 2^{-2} + \dots + 0 \times 2^{-23} = 0.5$
- $N = (-1) \times 2^0 \times (1+0.5) = -1.5$

Quality of Bits in Registers

The quality of each bit of a register coded as INT16U data type as an enumeration of bits can be indicated in the register preceding the register.

Example:

The quality of each bit of the register 32001, circuit breaker status, is given in the preceding register, 32000.

The quality of the data corresponding to the bit 0 of register 32001, OF status indication contact, is given in the bit 0 of register 32000:

- bit 0 of register 32000 = quality of OF status indication
- bit 0 of register 32001 = OF status indication contact

If	Then
If bit 0 of register 32000 = 1 AND bit 0 of register 32001 = 0	The OF contact indicates that the device is open
If bit 0 of register 32000 = 1 AND bit 0 of register 32001 = 1	The OF contact indicates that the device is closed
If bit 0 of register 32000 = 0	The OF contact indication is invalid

Readout Examples

Readout Example of a Modbus Register

The table below shows how to read the RMS current on phase 1 (I1) in registers 32028 and 32029 (coded in FLOAT32).

- The address of register 32028 equals $32028 - 1 = 32027 = 0x7D1B$.
- The Modbus address of the Modbus server is $255 = 0xFF$.

Request from the Client		Response from the Server	
Field Name	Example	Field Name	Example
Modbus server address	0xFF	Modbus server address	0xFF
Function code	0x03	Function code	0x03
Address of first register to be read (MSB)	0x7D	Data length in bytes	0x04
Address of first register to be read (LSB)	0x1B	Value read at address 0x7D1B (register 32028) (MSB)	0x44
Number of registers (MSB)	0x00	Value read at address 0x7D1B (register 32028) (LSB)	0x0A
Number of registers (LSB)	0x02	Value read at address 0x7D1C (register 32029) (MSB)	0xC0
CRC (MSB)	0XX	Value read at address 0x7D1C (register 32029) (LSB)	0x00
CRC (LSB)	0XX	CRC (MSB)	0XX
-	-	CRC (LSB)	0XX

The converted value of the FLOAT32 registers 32028 and 32029 is 555.

The RMS current on phase 1 (I1) is thus 555 A.

Readout Example of the Table of Standard Dataset Common Registers

Since there are more than 125 registers in the standard dataset, at least three Modbus read requests are needed to read the entire table.

Request to read registers 32000 to 32123:

- The address of register 32000 is $0x7CFF$.
- The length is 124 registers = $0x7C$.
- The number of bytes is $124 \times 2 = 248$ bytes = $0xF8$.
- The Modbus address of the server is $255 = 0xFF$.

Request to read registers 32124 to 32241:

- The address of register 32124 is $0x7D7B$.
- The length is 118 registers = $0x76$.
- The number of bytes is $118 \times 2 = 236$ bytes = $0xEC$.
- The Modbus address of the server is $255 = 0xFF$.

Request to read registers 32340 to 32435:

- The address of register 32340 is $0x7E53$.
- The length is 96 registers = $0x60$.
- The number of bytes is $2 \times 96 = 192$ bytes = $0xC0$.
- The Modbus address of the server is $255 = 0xFF$.

Request from the Client		Response from the Server	
Field Name	Example	Field Name	Example
Modbus server address	0xFF	Modbus server address	0xFF
Function code	0x03	Function code	0x03
Address of the first register to be read (MSB)	0x7C	Data length in bytes	0x8F
Address of the first register to be read (LSB)	0xFF	Value of register 32000 (MSB)	0XX
Number of registers (MSB)	0x00	Value of register 32000 (LSB)	0XX
Number of registers (LSB)	0x7C	Value of register 32001 (MSB)	0XX
CRC (MSB)	0XX	Value of register 32001 (LSB)	0XX
CRC (LSB)	0XX	–	0XX
–	–	–	0XX
–	–	Value of register 32123 (MSB)	0XX
–	–	Value of register 32123 (LSB)	0XX
–	–	CRC (MSB)	0XX
–	–	CRC (LSB)	0XX

Standard Dataset Common Registers

Circuit Breaker Status Register

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x7CFF	32000	R	–	INT16U	–	A/E	A/E/P/H	X	–	Quality of each bit of register 32001: 0 = Invalid 1 = Valid
0x7D00	32001	R	–	INT16U	–	A/E	A/E/P/H	X	–	Circuit breaker status register
						A/E	A/E/P/H	X	0	OF status indication contact 0 = The circuit breaker is open. 1 = The circuit breaker is closed.
						A/E	A/E/P/H	X	1	SD trip indication contact 0 = Circuit breaker is not tripped. 1 = Circuit breaker is tripped due to electrical fault, shunt trip, or push-to-trip. Bit always equal to 0 for MasterPacT and ComPacT NS circuit breakers with motor mechanism.
						A/E	A/E/P/H	X	2	SDE fault trip indication contact 0 = Circuit breaker is not tripped on electrical fault. 1 = Circuit breaker is tripped due to electrical fault (including ground-fault test and earth-leakage test).
						–	A/E/P/H	X	3	CH spring charged contact (only with MasterPacT) 0 = Spring discharged 1 = Spring charged Bit always equal to 0 for MasterPacT and ComPacT NS circuit breakers with motor mechanism.
						–	–	–	4	Reserved
						–	A/E/P/H	X	5	PF ready to close contact (only with MasterPacT) 0 = Not ready to close 1 = Ready to close Bit always equal to 0 for MasterPacT and ComPacT NS circuit breakers with motor mechanism.
						–	–	–	6–14	Reserved
						A/E	A/E/P/H	–	15	Data availability If this bit is set at 1, all other bits of the register are not significant.

IO Status Registers

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x7D01	32002	R	–	INT16U	–	A/E	A/E/P/H	X	–	Quality of each bit of register 32003: 0 = Invalid 1 = Valid
0x7D02	32003	R	–	INT16U	–	A/E	A/E/P/H	X	–	IO1 module and M2C status
						A/E	A/E/P/H	X	0	Digital input 1 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	1	Digital input 2 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	2	Digital input 3 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	3	Digital input 4 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	4	Digital input 5 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	5	Digital input 6 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	6	Digital output 1 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	7	Digital output 2 status: 0 = Off 1 = On
						A/E	A/E/P/H	X	8	Digital output 3 status: 0 = Off 1 = On
						–	–	X	9	Digital M2C output 1 status: 0 = Off 1 = On
						–	–	X	10	Digital M2C output 2 status: 0 = Off 1 = On
						–	–	–	11–14	Reserved
						A/E	A/E/P/H	–	15	Data availability If this bit is set at 1, all other bits of the register are not significant.
0x7D03	32004	R	–	INT16U	–	A/E	A/E/P/H	X	–	Quality of each bit of register 32005: 0 = Invalid 1 = Valid
0x7D04	32005	R	–	INT16U	–	A/E	A/E/P/H	X	–	IO2 module status
									0	Digital input 1 status:

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
										0 = Off 1 = On
									1	Digital input 2 status: 0 = Off 1 = On
									2	Digital input 3 status: 0 = Off 1 = On
									3	Digital input 4 status: 0 = Off 1 = On
									4	Digital input 5 status: 0 = Off 1 = On
									5	Digital input 6 status: 0 = Off 1 = On
									6	Digital output 1 status: 0 = Off 1 = On
									7	Digital output 2 status: 0 = Off 1 = On
									8	Digital output 3 status: 0 = Off 1 = On
								–	9–14	Reserved
								–	15	Data availability If this bit is set at 1, all other bits of the register are not significant.

Tripping Cause

The tripping cause register provides information about the cause of the trip for the standard protection functions. When a tripping cause bit is at 1 in the tripping cause register, it indicates that a trip has occurred and has not been reset.

- For MicroLogic trip units for ComPacT NSX circuit breakers, the tripping cause bit is reset by pressing the key OK (keypad of the MicroLogic trip unit) twice (validation and confirmation).
- For MicroLogic A/E/P/H trip units for MasterPacT NT/NW and ComPacT NS circuit breakers, the tripping cause bit is reset as soon as the circuit breaker is closed again.
- For MicroLogic X control units for MasterPacT MTZ circuit breakers, the tripping cause bit is reset by pressing the test/reset button (located beside the trip cause LEDs on the MicroLogic X control unit). Press and hold the button for 3 to 15 seconds to reset all the trip causes.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x7D05	32006	R	–	INT16U	–	A/E	A/E/P/H	–	–	Quality of each bit of register 32007: • 0 = Invalid • 1 = Valid
0x7D06	32007	R	–	INT16U	–	A/E	A/E/P/H	X		Tripping cause for the standard protection functions
						A/E	A/E/P/H	X	0	Long-time protection Ir
						A/E	A/E/P/H	X	1	Short-time protection Isd
						A/E	A/E/P/H	X	2	Instantaneous protection Ii
						A/E	A/E/P/H	X	3	Ground-fault protection Ig
						E	A/P/H	X	4	Earth-leakage protection IΔn
						A/E	A/E/P/H	X	5	Integrated instantaneous protection (SELLIM and DIN/ DINF)
						A/E	–	X	6	Internal failure (STOP)
						–	A/E	–		Other protections
						–	P/H	–		Internal failure (temperature)
						–	A/E/P/H	–	7	Internal failure (overvoltage)
						–	P/H	X	8	Other protection (see register 32009)
						–	–	–	9	Reserved
						E	–	–	10	Unbalance motor protection
						E	–	–	11	Jam motor protection
						E	–	–	12	Underload motor protection
						E	–	–	13	Long-start motor protection
						A/E	–	–	14	Reflex tripping protection
						A/E	A/E/P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x7D07	32008	R	–	INT16U	–	–	P/H	–		Quality of each bit of register 32009: • 0 = Invalid • 1 = Valid
0x7D08	32009	R	–	INT16U	–	–	P/H	–	–	Tripping causes for the advanced protection functions
						–	P/H	–	0	Current unbalance
						–	P/H	–	1	Overcurrent on phase 1
						–	P/H	–	2	Overcurrent on phase 2
						–	P/H	–	3	Overcurrent on phase 3
						–	P/H	–	4	Overcurrent on Neutral
						–	P/H	X	5	Undervoltage
						–	P/H	X	6	Overvoltage
						–	P/H	–	7	Voltage unbalance
						–	P/H	–	8	Overpower
						–	P/H	X	9	Reverse power
						–	P/H	X	10	Underfrequency
						–	P/H	X	11	Overfrequency
						–	P/H	–	12	Phase rotation

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
						–	P/H	–	13	Load shedding based on current
						–	P/H	–	14	Load shedding based on power
						–	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x7D09– 0x7D0C	32010– 32013	–	–	–	–	–	–	–	–	Reserved

Overrun of the Protection Setpoints

The alarm setpoint registers provide information about overrun of the standard and advanced protection setpoints. A bit is at 1 once a setpoint overrun has occurred, even if the time delay has not expired.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x7D0D	32014	R	–	INT16U	–	A/E	P/H	–	–	Quality of each bit of register 32015: 0 = Invalid 1 = Valid
0x7D0E	32015	R	–	INT16U	–	A/E	P/H	–	–	Overrun of the standard protection setpoints
						A/E	P/H	–	0	Long-time protection pick-up
						–	–	–	1–14	Reserved
						A/E	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x7D0F	32016	R	–	INT16U	–	A/E	P/H	–	–	Quality of each bit of register 32017: 0 = Invalid 1 = Valid
0x7D10	32017	R	–	INT16U	–	A/E	P/H	–	–	Overrun of the advanced protection setpoints
						–	P/H	–	0	Current unbalance
						–	P/H	–	1	Maximum current on phase 1
						–	P/H	–	2	Maximum current on phase 2
						–	P/H	–	3	Maximum current on phase 3
						–	P/H	–	4	Maximum current on the neutral
						–	P/H	–	5	Minimum voltage
						–	P/H	–	6	Maximum voltage
						–	P/H	–	7	Voltage unbalance
						–	P/H	–	8	Maximum power
						–	P/H	–	9	Reverse power
						–	P/H	–	10	Minimum frequency
						–	P/H	–	11	Maximum frequency
						–	P/H	–	12	Phase rotation
						–	P/H	–	13	Load shedding based on the current

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
						–	P/H	–	14	Load shedding based on the power
						–	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x7D11	32018	R	–	INT16U	–	–	P/H	–	–	Quality of each bit of register 32019: 0 = Invalid 1 = Valid
0x7D12	32019	R	–	INT16U	–	–	P/H	–	–	Advanced protection settings extended
						–	P/H	–	0	Ground-fault alarm
						E	P/H	–	1	Earth-leakage alarm
						–	–	–	2–14	Reserved
						–	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.

Alarms

The alarm register provides information about the pre-alarms and the user-defined alarms. A bit is set to 1 once an alarm is active.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x7D13	32020	R	–	INT16U	–	A/E	–	–	–	Quality of each bit of register 32021: 0 = Invalid 1 = Valid
0x7D14	32021	R	–	INT16U	–	A/E	–	–	–	Pre-alarm extended register
						A/E	–	X	0	Long-time protection time pre-alarm (PAL Ir)
						E	–	–	1	Earth-leakage protection pre-alarm (PAL IΔn)
						–	–	X		Earth-leakage alarm ⁽¹⁾
						A/E	–	–	2	Ground-fault protection pre-alarm (PAL Ig)
						–	–	X		Ground-fault alarm ⁽²⁾
						–	–	–	3–14	Reserved
						A/E	–	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x7D15	32022	R	–	INT16U	–	A/E	–	–	–	Quality of each bit of register 32023: 0 = Invalid 1 = Valid
0x7D16	32023	R	–	INT16U	–	A/E	–	–	–	Register of user-defined alarms
						A/E	–	–	0	User-defined alarm 201
						A/E	–	–	1	User-defined alarm 202
						A/E	–	–	2	User-defined alarm 203
						A/E	–	–	3	User-defined alarm 204
						A/E	–	–	4	User-defined alarm 205
						A/E	–	–	5	User-defined alarm 206
						A/E	–	–	6	User-defined alarm 207
						A/E	–	–	7	User-defined alarm 208

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
						A/E	–	–	8	User-defined alarm 209
						A/E	–	–	9	User-defined alarm 210
						–	–	–	10–14	Reserved
						A/E	–	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x7D17– 0x7D1A	32024– 32027	–	–	–	–	–	–	–	–	Reserved

(1) Value available on MicroLogic 7.0 X control unit only when the Digital Module ANSI 51N/51G Ground-fault alarm is installed.

(2) Value available on MicroLogic 2.0 X, 3.0 X, 5.0 X, and 6.0 X control units only when the Digital Module ANSI 51N/51G Ground-fault alarm is installed.

Current

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D1B– 0x7D1C	32028– 32029	R	A	FLOAT32	–	A/E	A/E/P/H	X	RMS current on phase 1
0x7D1D– 0x7D1E	32030– 32031	R	A	FLOAT32	–	A/E	A/E/P/H	X	RMS current on phase 2
0x7D1F– 0x7D20	32032– 32033	R	A	FLOAT32	–	A/E	A/E/P/H	X	RMS current on phase 3
0x7D21– 0x7D22	32034– 32035	R	A	FLOAT32	–	A/E	A/E/P/H	X	RMS current on the neutral ⁽¹⁾
0x7D23– 0x7D24	32036– 32037	R	A	FLOAT32	–	A/E	A/E/P/H	X	Maximum of RMS current of phases 1, 2, 3 and N (most loaded phase) ⁽³⁾
0x7D25– 0x7D26	32038– 32039	R	–	FLOAT32	–	A/E	A/E/P/H	X	Current ratio on ground (I _g setting ratio)
0x7D27– 0x7D28	32040– 32041	R	–	FLOAT32	–	E	A/P/H	X	Current ratio on earth-leakage (I _{Δn} setting ratio) ⁽²⁾

(1) Value available when system type register returns 30 or 41.

(2) Value available with MicroLogic 7.0 X.

(3) Value reset with the reset minimum/maximum command.

Maximum Current Values

Maximum current values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D29– 0x7D2A	32042– 32043	R	A	FLOAT32	–	A/E	A/E/P/H	X	Maximum RMS current on phase 1
0x7D2B– 0x7D2C	32044– 32045	R	A	FLOAT32	–	A/E	A/E/P/H	X	Maximum RMS current on phase 2
0x7D2D– 0x7D2E	32046– 32047	R	A	FLOAT32	–	A/E	A/E/P/H	X	Maximum RMS current on phase 3
0x7D2F– 0x7D30	32048– 32049	R	A	FLOAT32	–	A/E	A/E/P/H	X	Maximum RMS current on the neutral ⁽¹⁾
0x7D31– 0x7D32	32050– 32051	R	A	FLOAT32	–	A/E	A/E/P/H	X	This is the highest (i.e. maximum) maximum current value since this measurement was last reset. The measurement looks at all 3 currents, MaxI1, MaxI2, MaxI3 and MaxIN and keeps track of the

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
									highest value of any of them over time.
0x7D33–0x7D36	32052–32055	–	–	–	–	–	–	–	Reserved

(1) Value available when system type register returns 30 or 41.

Voltage

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D37–0x7D38	32056–32057	R	V	FLOAT32	41.6–2250	E	E/P/H	X	RMS phase-to-phase voltage V12
0x7D39–0x7D3A	32058–32059	R	V	FLOAT32	41.6–2250	E	E/P/H	X	RMS phase-to-phase voltage V23
0x7D3B–0x7D3C	32060–32061	R	V	FLOAT32	41.6–2250	E	E/P/H	X	RMS phase-to-phase voltage V31
0x7D3D–0x7D3E	32062–32063	R	V	FLOAT32	24-1500	E	E/P/H	X	RMS phase-to-neutral voltage V1N ⁽¹⁾
0x7D3F–0x7D40	32064–32065	R	V	FLOAT32	24-1500	E	E/P/H	X	RMS phase-to-neutral voltage V2N ⁽¹⁾
0x7D41–0x7D42	32066–32067	R	V	FLOAT32	24-1500	E	E/P/H	X	RMS phase-to-neutral voltage V3N ⁽¹⁾

(1) Value available when system type register returns 40 or 41.

Frequency

When the MicroLogic trip unit cannot calculate the frequency, it returns Not applicable = 0xFFC00000.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D43–0x7D44	32068–32069	R	Hz	FLOAT32	40.0–70.0	E	P/H	X	Frequency
0x7D45–0x7D46	32070–32071	R	Hz	FLOAT32	40.0–70.0	E	P/H	X	Maximum frequency ⁽¹⁾

(1) This value can be reset with the reset minimum/maximum command.

Power

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D47–0x7D48	32072–32073	R	W	FLOAT32	-16000000–16000000	E	E/P/H	X	Active power on phase 1 ⁽¹⁾ ⁽²⁾
0x7D49–0x7D4A	32074–32075	R	W	FLOAT32	-16000000–16000000	E	E/P/H	X	Active power on phase 2 ⁽¹⁾ ⁽²⁾
0x7D4B–0x7D4C	32076–32077	R	W	FLOAT32	-16000000–16000000	E	E/P/H	X	Active power on phase 3 ⁽¹⁾ ⁽²⁾
0x7D4D–0x7D4E	32078–32079	R	W	FLOAT32	-16000000–16000000	E	E/P/H	X	Total active power ⁽²⁾
0x7D4F–0x7D50	32080–32081	R	VA _r	FLOAT32	-16000000–16000000	E	E/P/H	X	Reactive power on phase 1 ⁽¹⁾ ⁽²⁾
0x7D51–0x7D52	32082–32083	R	VA _r	FLOAT32	-16000000–16000000	E	E/P/H	X	Reactive power on phase 2 ⁽¹⁾ ⁽²⁾
0x7D53–0x7D54	32084–32085	R	VA _r	FLOAT32	-16000000–16000000	E	E/P/H	X	Reactive power on phase 3 ⁽¹⁾ ⁽²⁾

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D55– 0x7D56	32086– 32087	R	VA _r	FLOAT32	-16000000– 16000000	E	E/P/H	X	Total reactive power ⁽²⁾
0x7D57– 0x7D58	32088– 32089	R	VA	FLOAT32	0–16000000	E	E/P/H	X	Apparent power on phase 1 ⁽¹⁾
0x7D59– 0x7D5A	32090– 32091	R	VA	FLOAT32	0–16000000	E	E/P/H	X	Apparent power on phase 2 ⁽¹⁾
0x7D5B– 0x7D5C	32092– 32093	R	VA	FLOAT32	0–16000000	E	E/P/H	X	Apparent power on phase 3 ⁽¹⁾
0x7D5D– 0x7D5E	32094– 32095	R	VA	FLOAT32	0–16000000	E	E/P/H	X	Total apparent power

(1) Value available when system type register returns 40 or 41.

(2) The sign for the active and reactive power depends on the configuration of:

- register 3316 for ComPacT NSX, ComPacT NS and MasterPacT NT/NW circuit breakers.
- register 8405 for MasterPacT MTZ circuit breakers.

Energy

Energy is stored in big-endian format: the most significant register is transmitted first.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D5F– 0x7D62	32096– 32099	R	Wh	INT64	–	E	E/P/H	X	Total active energy ⁽²⁾
0x7D63– 0x7D66	32100– 32103	R	VARh	INT64	–	E	E/P/H ⁽¹⁾	X	Total reactive energy ⁽²⁾
0x7D67– 0x7D6A	32104– 32107	R	Wh	INT64U	–	E	P/H	X	Total active energy delivered (into the load, counted positively) ⁽²⁾
0x7D6B– 0x7D6E	32108– 32111	R	Wh	INT64U	–	E	P/H	X	Total active energy received (out of the load, counted negatively) ⁽²⁾
0x7D6F– 0x7D72	32112– 32115	R	VARh	INT64U	–	E	P/H	X	Total reactive energy delivered (into the load, counted positively) ⁽²⁾
0x7D73– 0x7D76	32116– 32119	R	VARh	INT64U	–	E	P/H	X	Total reactive energy received (out of the load, counted negatively) ⁽²⁾
0x7D77– 0x7D7A	32120– 32123	R	VAh	INT64U	–	E	–	X	Total apparent energy ⁽²⁾
0x7D7B– 0x7D7E	32124– 32127	R	Wh	INT64U	–	E	–	X	Total cumulative active energy delivered (into the load, counted positively, not resettable)
0x7D7F– 0x7D82	32128– 32131	R	Wh	INT64U	–	E	–	X	Total cumulative active energy received (out of the load, counted negatively, not resettable)

(1) This value is always positive with MasterPacT MicroLogic E trip unit.

(2) Value reset with the reset energies command.

Average Values

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D83– 0x7D84	32132– 32133	R	A	FLOAT32	–	–	–	X	Average of 3-phase RMS currents
0x7D85– 0x7D86	32134– 32135	R	V	FLOAT32	–	–	–	X	Average of 3 RMS phase-to-phase voltages: $(V12+V23+V31)/3$
0x7D87– 0x7D88	32136– 32137	R	V	FLOAT32	–	–	–	X	Average of 3 RMS phase-to-neutral voltages: $(V1N+V2N+V3N)/3^{(1)}$

(1) Value available when system type register returns 40 or 41.

Maximum Power Values

Maximum power values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D89– 0x7D8A	32138– 32139	R	W	FLOAT32	–	–	–	X	Maximum total active power
0x7D8B– 0x7D8C	32140– 32141	R	VAr	FLOAT32	–	–	–	X	Maximum total reactive power
0x7D8D– 0x7D8E	32142– 32143	R	VA	FLOAT32	–	–	–	X	Maximum total apparent power

Maximum Average Values

Maximum average values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D8F– 0x7D90	32144– 32145	R	A	FLOAT32	–	–	–	X	Maximum of average of 3-phase RMS currents
0x7D91– 0x7D92	32146– 32147	R	V	FLOAT32	–	–	–	X	Maximum of average of 3 RMS phase-to-phase voltages
0x7D93– 0x7D94	32148– 32149	R	V	FLOAT32	–	–	–	X	Maximum of average of 3 RMS phase-to-neutral voltages

Ground and Earth-Leakage Current

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D95– 0x7D96	32150– 32151	R	A	FLOAT32	–	–	–	X	Ground-fault current
0x7D97– 0x7D98	32152– 32153	R	A	FLOAT32	–	–	–	X	Earth-leakage current ⁽¹⁾
0x7D99– 0x7D9A	32154– 32155	–	–	–	–	–	–	–	Reserved

(1) Value available with MicroLogic 7E

Current Demand Values

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7D9B– 0x7D9C	32156– 32157	R	A	FLOAT32	–	E	E/P/H	X	Current demand value on phase 1: I1 Dmd
0x7D9D– 0x7D9E	32158– 32159	R	A	FLOAT32	–	E	E/P/H	X	Current demand value on phase 2: I2 Dmd
0x7D9F– 0x7DA0	32160– 32161	R	A	FLOAT32	–	E	E/P/H	X	Current demand value on phase 3: I3 Dmd
0x7DA1– 0x7DA2	32162– 32163	R	A	FLOAT32	–	E	E/P/H	X	Current demand value on the neutral: IN Dmd ⁽¹⁾

(1) Value available when system type register returns 30 or 41.

Power Demand Values

- For the block window, the demand value is updated at the end of the window.
- For the sliding window,
 - If window duration is configured for less or equal to 15 minutes, the demand value is updated every 15 seconds
 - If window duration is configured for more than 15 minutes, the demand value is updated every 1 minute.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DA3– 0x7DA4	32164– 32165	R	W	FLOAT32	–	E	E/P/H	X	Total active power demand: P Dmd
0x7DA5– 0x7DA6	32166– 32167	R	VAR	FLOAT32	–	E	P/H	X	Total reactive power demand: Q Dmd
0x7DA7– 0x7DA8	32168– 32169	R	VA	FLOAT32	–	E	P/H	X	Total apparent power demand: S Dmd

Current Peak Demand Values

Current peak demand values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DA9– 0x7DAA	32170– 32171	R	A	FLOAT32	–	–	–	X	Current peak demand value on phase 1: I1 dmd max
0x7DAB– 0x7DAC	32172– 32173	R	A	FLOAT32	–	–	–	X	Current peak demand value on phase 2: I2 dmd max
0x7DAD– 0x7DAE	32174– 32175	R	A	FLOAT32	–	–	–	X	Current peak demand value on phase 3: I3 dmd max
0x7DAF– 0x7DB0	32176– 32177	R	A	FLOAT32	–	–	–	X	Current peak demand value on the neutral: IN dmd max ⁽¹⁾

(1) Value available when system type register returns 30 or 41.

Power Peak Demand Values

Power peak demand values are updated every 15 seconds. Power peak demand values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DB1– 0x7DB2	32178– 32179	R	W	FLOAT32	–	–	–	X	Total active power peak demand: P dmd max
0x7DB3– 0x7DB4	32180– 32181	R	VAR	FLOAT32	–	–	–	X	Total reactive power peak demand: Q dmd max
0x7DB5– 0x7DB6	32182– 32183	R	VA	FLOAT32	–	–	–	X	Total apparent power peak demand: S dmd max

Maximum Ground and Earth-Leakage Current Values

Maximum ground and earth-leakage current values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DB7– 0x7DB8	32184– 32185	R	A	FLOAT32	–	–	–	X	Maximum ground-fault current
0x7DB9– 0x7DBA	32186– 32187	R	V	FLOAT32	–	E	–	X	Maximum earth-leakage current ⁽¹⁾
0x7DBB– 0x7DC0	32188– 32193	–	–	–	–	–	–	–	Reserved

(1) Value available with MicroLogic 7E.

Maximum Voltage Values

Maximum voltage values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DC1– 0x7DC2	32194– 32195	R	V	FLOAT32	41.6– 2250	E	E/P/H	X	Maximum RMS phase-to-phase voltage V12
0x7DC3– 0x7DC4	32196– 32197	R	V	FLOAT32	41.6– 2250	E	E/P/H	X	Maximum RMS phase-to-phase voltage V23
0x7DC5– 0x7DC6	32198– 32199	R	V	FLOAT32	41.6– 2250	E	E/P/H	X	Maximum RMS phase-to-phase voltage V31
0x7DC7– 0x7DC8	32200– 32201	R	V	FLOAT32	24–1500	E	E/P/H	X	Maximum RMS phase-to-neutral voltage V1N ⁽¹⁾
0x7DC9– 0x7DCA	32202– 32203	R	V	FLOAT32	24–1500	E	E/P/H	X	Maximum RMS phase-to-neutral voltage V2N ⁽¹⁾
0x7DCB– 0x7DCC	32204– 32205	R	V	FLOAT32	24–1500	E	E/P/H	X	Maximum RMS phase-to-neutral voltage V3N ⁽¹⁾

(1) Value available when system type register returns 40 or 41.

Power Factor

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DCD– 0x7DCE	32206– 32207	R	–	FLOAT32	–	E	E/P/H	X	Power factor on phase 1 ⁽¹⁾
0x7DCF– 0x7DD0	32208– 32209	R	–	FLOAT32	–	E	E/P/H	X	Power factor on phase 2 ⁽¹⁾
0x7DD1– 0x7DD2	32210– 32211	R	–	FLOAT32	–	E	E/P/H	X	Power factor on phase 3 ⁽¹⁾
0x7DD3– 0x7DD4	32212– 32213	R	–	FLOAT32	–	E	E/P/H	X	Total power factor

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DD5– 0x7DD6	32214– 32215	R	–	FLOAT32	–	E	H	X	Fundamental power factor on phase 1 ($\cos\phi 1$) ⁽¹⁾⁽²⁾
0x7DD7– 0x7DD8	32216– 32217	R	–	FLOAT32	–	E	H	X	Fundamental power factor on phase 2 ($\cos\phi 2$) ⁽¹⁾⁽²⁾
0x7DD9– 0x7DDA	32218– 32219	R	–	FLOAT32	–	E	H	X	Fundamental power factor on phase 3 ($\cos\phi 3$) ⁽¹⁾⁽²⁾
0x7DDB– 0x7DDC	32220– 32221	R	–	FLOAT32	–	E	H	X	Total fundamental power factor ⁽²⁾

(1) Value available when system type register returns 40 or 41.

(2) The sign for the fundamental power factor ($\cos\phi$) depends on the configuration:

- register 3318 for ComPacT NSX, ComPacT NS and MasterPacT NT/NW circuit breakers.
- register 8404 for MasterPacT MTZ circuit breakers.

Total Harmonic Distortion (THD)

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DDD– 0x7DDE	32222– 32223	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of phase-to phase voltage V12 compared to the fundamental
0x7DDF– 0x7DE0	32224– 32225	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of phase-to phase voltage V23 compared to the fundamental
0x7DE1– 0x7DE2	32226– 32227	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of phase-to phase voltage V31 compared to the fundamental
0x7DE3– 0x7DE4	32228– 32229	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of phase-to-neutral voltage V1N compared to the fundamental ⁽¹⁾
0x7DE5– 0x7DE6	32230– 32231	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of phase-to-neutral voltage V2N compared to the fundamental ⁽¹⁾
0x7DE7– 0x7DE8	32232– 32233	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of phase-to-neutral voltage V3N compared to the fundamental ⁽¹⁾
0x7DE9– 0x7DEA	32234– 32235	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of current on phase 1 compared to the fundamental
0x7DEB– 0x7DEC	32236– 32237	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of current on phase 2 compared to the fundamental
0x7DED– 0x7DEE	32238– 32239	R	–	FLOAT32	0–2	E	H	X	Total harmonic distortion (THD) of current on phase 3 compared to the fundamental
0x7DEF– 0x7DF0	32240– 32241	R	–	FLOAT32	0–2	E	H	X	Average of 3-phase current Total harmonic distortions (THD) compared to the fundamental

(1) Value available when system type register returns 40 or 41.

Maximum Power Factor

Maximum power factor can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x7DF1– 0x7DF2	32242– 32243	R	–	FLOAT32	–	–	–	X	Maximum total power factor
0x7DF3– 0x7E52	32244– 32339	–	–	–	–	–	–	–	Reserved

Inhibit Close Order

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x7E53	32340	R	–	INT16U	–	A/E	A/E/P/H	X	–	Quality of each bit of register 32341: • 0 = Invalid • 1 = Valid
0x7E54	32341	R	–	INT16U	–	A/E	A/E/P/H	X	–	Inhibit close order status
									0	Close breaker inhibited by IO module • 0 = Disable • 1 = Enable
									1	Close breaker inhibited by communication • 0 = Disable • 1 = Enable
						–	–	–	2–15	Reserved

Legacy Dataset

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Legacy Dataset

Description

The legacy dataset contains the most useful information of each IMU module in one convenient table. The legacy dataset is available in the registers 12000 to 12200. It can be read with two read requests.

Each IMU module updates the values in the dataset registers on a regular basis.

The response time of requests to legacy dataset registers is shorter than the response time of requests to device registers. Therefore, it is recommended to read the legacy dataset registers instead of device registers, to improve the overall performance of the system [Modbus Programming Recommendations](#), page 26.

NOTE:

- The legacy dataset is compatible with legacy versions of the MicroLogic trip unit for ComPacT NSX, PowerPacT H-, J-, and L-Frame, ComPacT NS, PowerPacT P- and R-frame or MasterPacT NT/NW circuit breaker. For this reason, data read directly in the Modbus registers is organized in a different way than in the standard dataset.
- For new applications, it is recommended to use the standard dataset instead of the legacy dataset.

Modbus Registers

Table of Legacy Dataset Common Registers

The main information needed for remote supervision of a ComPacT NSX, ComPacT NS, MasterPacT NT/NW, or MasterPacT MTZ circuit breaker is contained in the table of common registers starting at register 12000.

This compact table of 114 registers can be read with a single Modbus request.

It contains the following information:

- Circuit breaker status
- Tripping causes
- Real-time values of main measurements: current, voltage, power, energy, total harmonic distortion

The content of this table of registers is detailed in [Legacy Dataset Common Registers](#), page 86.

Use of these common registers is highly recommended to optimize response times and simplify the use of data.

Table Format

Register tables have the following columns:

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description

- **Address:** a 16-bit register address in hexadecimal. The address is the data used in the Modbus frame.
- **Register:** a 16-bit register number in decimal (register = address + 1).
- **RW:** register read-write status
 - R: the register can be read by using Modbus functions
 - W: the register can be written by using Modbus functions
 - RW: the register can be read and written by using Modbus functions
 - RC: the register can be read by using the command interface
 - WC: the register can be written by using the command interface
- **Unit:** the unit the information is expressed in.
- **Type:** the encoding data type (see data type description below).
- **Range:** the permitted values for this variable, usually a subset of what the format allows.
- **A/E:** types of ComPacT or ComPacT NSX MicroLogic trip unit for which the register is available.
 - Type A (Ammeter): current measurements for ComPacT NSX circuit breakers only.
 - Type E (Energy): current, voltage, power, and energy measurements for ComPacT or ComPacT NSX circuit breakers.

- **A/E/P/H**: types of MasterPacT NT/NW and ComPacT NS MicroLogic trip unit for which the register is available.
 - Type A (Ammeter): current measurements
 - Type E (Energy): current, voltage, power, and energy measurements
 - Type P (Power): current, voltage, power, energy measurements, and advanced protection
 - Type H (Harmonics): current, voltage, power, energy, energy quality measurements, and advanced protection
- **X**: register available in the MicroLogic X control unit for MasterPacT MTZ circuit breakers when the Modbus legacy dataset Digital Module is purchased and installed on the MicroLogic X control unit.
- **Description**: provides information about the register and restrictions that apply.

Data Types

Data Types	Description	Range
INT16U	16-bit unsigned integer	0 to 65535
INT16	16-bit signed integer	-32768 to +32767
INT32U	32-bit unsigned integer	0 to 4 294 967 295
INT32	32-bit signed integer	-2 147 483 648 to +2 147 483 647

Big-Endian Format

INT32 and INT32U variables are stored in big-endian format: the most significant register is transmitted first and the least significant register is transmitted at last place.

INT32 and INT32U variables are made of INT16U variables.

The formulas to calculate the decimal value of these variables are:

- INT32: $(0\text{-bit}31) \times 2^{31} + \text{bit}30 \times 2^{30} + \text{bit}29 \times 2^{29} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$
- INT32U: $\text{bit}31 \times 2^{31} + \text{bit}30 \times 2^{30} + \text{bit}29 \times 2^{29} + \dots \text{bit}1 \times 2^1 + \text{bit}0 \times 2^0$

Example:

The reactive energy in the legacy dataset is an INT32 variable coded in registers 12052 to 12053.

If the values in the registers are:

- register 12052 = 0xFFF2 = 0x8000 + 0x7FF2 or 32754
- register 12053 = 0xA96E or 43374 as INT16U variable and -10606 as INT16 variable (use the INT16U value to calculate the value of the reactive energy).

Then the reactive energy is equal to $(0-1) \times 2^{31} + 32754 \times 2^{16} + 43374 \times 2^0 = -874130$ kVARh.

Readout Examples

Readout Example of a Modbus Register

The table below shows how to read the rms current on phase 1 (I1) in register 12016.

- The address of register 12016 equals $12016 - 1 = 12015 = 0x2EEF$.
- The Modbus address of the Modbus server is $47 = 0x2F$.

Request from the Client		Response from the Server	
Field Name	Example	Field Name	Example
Modbus server address	0x2F	Modbus server address	0x2F
Function code	0x03	Function code	0x03
Address of register to be read (MSB)	0x2E	Data length in bytes	0x02
Address of register to be read (LSB)	0xEF	Register value (MSB)	0x02
Number of registers (MSB)	0x00	Register value (LSB)	0x2B
Number of registers (LSB)	0x01	CRC (MSB)	0xFF
CRC (MSB)	0xFF	CRC (LSB)	0xFF
CRC (LSB)	0xFF	–	–

The content of register 12016 (address 0x2EEF) is $0x022B = 555$.

The rms current on phase 1 (I1) is thus 555 A.

Readout Example of the Table of Legacy Dataset Common Registers

The table below shows how to read the table of legacy dataset common registers. This table starts at register 12000 and consists of 113 registers.

- The address of register 12000 = $0x2EDF$.
- The table length is 113 registers = $0x71$.
- The number of bytes is $113 \times 2 = 226$ bytes = $0xE2$.
- The Modbus address of the server is $47 = 0x2F$.

Request from the Client		Response from the Server	
Field Name	Example	Field Name	Example
Modbus server address	0x2F	Modbus server address	0x2F
Function code	0x03	Function code	0x03
Address of the first register to be read (MSB)	0x2E	Data length in bytes	0xE2
Address of the first register to be read (LSB)	0xDF	Value of register 12000 (MSB)	0xFF
Number of registers (MSB)	0x00	Value of register 12000 (LSB)	0xFF
Number of registers (LSB)	0x71	Value of register 12001 (MSB)	0xFF
CRC (MSB)	0xFF	Value of register 12001 (LSB)	0xFF
CRC (LSB)	0xFF	–	0xFF
–	–	–	0xFF
–	–	Value of register 12112 (MSB)	0xFF
–	–	Value of register 12112 (LSB)	0xFF

Request from the Client		Response from the Server	
Field Name	Example	Field Name	Example
–	–	CRC (MSB)	0xXX
–	–	CRC (LSB)	0xXX

Legacy Dataset Common Registers

Circuit Breaker Status Register

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x2EDF	12000	R	–	INT16U	–	A/E	A/E/P/H	X	–	Validity of each bit in the circuit breaker status register.
0x2EE0	12001	R	–	INT16U	–	A/E	A/E/P/H	X	–	Circuit breaker status register
						A/E	A/E/P/H	X	0	OF status indication contact 0 = The circuit breaker is open. 1 = The circuit breaker is closed.
						A/E	A/E/P/H	X	1	SD trip indication contact 0 = Circuit breaker is not tripped. 1 = Circuit breaker is tripped due to electrical fault, shunt trip or push-to-trip. Bit always equal to 0 for MasterPacT NT/NW and ComPacT NS circuit breakers with motor mechanism.
						A/E	A/E/P/H	X	2	SDE fault trip indication contact 0 = Circuit breaker is not tripped on electrical fault. 1 = Circuit breaker is tripped due to electrical fault (including ground-fault test and earth-leakage test).
						–	A/E/P/H	X	3	CH spring charged contact (only with MasterPacT) 0 = Spring discharged 1 = Spring charged Bit always equal to 0 for ComPacT NS circuit breaker.
						–	–	–	4	Reserved
						–	A/E/P/H	X	5	PF ready to close contact (only with MasterPacT) 0 = Not ready to close 1 = Ready to close Bit always equal to 0 for ComPacT NS circuit breaker.
						–	A/E/P/H	X	6	Distinction between ComPacT NS and MasterPacT NT/NW 0 = ComPacT NS 1 = MasterPacT NT/NW
						–	–	–	7–14	Reserved

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
						A/E	–	X	15	Data availability If this bit is set at 1, all other bits of the register are not significant.

IO Status Registers

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x2EE1	12002	R	–	INT16U	–	A/E	A/E/P/H	X	–	Status IO 1
									0	Input 1 status • 0 = Off • 1 = On
									1	Input 2 status • 0 = Off • 1 = On
									2	Input 3 status • 0 = Off • 1 = On
									3	Input 4 status • 0 = Off • 1 = On
									4	Input 5 status • 0 = Off • 1 = On
									5	Input 6 status • 0 = Off • 1 = On
									6	Output 1 status • 0 = Off • 1 = On
									7	Output 2 status • 0 = Off • 1 = On
									8	Output 3 status • 0 = Off • 1 = On
									9–14	Reserved
0x2EE2	12003	R	–	INT16U	–	A/E	A/E/P/H	X	–	Status IO 2
									0	Input 1 status • 0 = Off • 1 = On
									1	Input 2 status • 0 = Off • 1 = On
									2	Input 3 status • 0 = Off • 1 = On

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
									3	Input 4 status • 0 = Off • 1 = On
									4	Input 5 status • 0 = Off • 1 = On
									5	Input 6 status • 0 = Off • 1 = On
									6	Output 1 status • 0 = Off • 1 = On
									7	Output 2 status • 0 = Off • 1 = On
									8	Output 3 status • 0 = Off • 1 = On
									9–14	Reserved
									15	Data availability If this bit is set at 1, all other bits of the register are not significant.

Tripping Cause

The tripping cause register provides information about the cause of the trip for the standard protection functions. When a tripping cause bit is at 1 in the tripping cause register, it indicates that a trip has occurred and has not been reset.

- For MicroLogic trip units for ComPacT NSX circuit breakers, the tripping cause bit is reset by pressing the OK key (keypad of the MicroLogic trip unit) twice (validation and confirmation).
- For MicroLogic A/E/P/H trip units for MasterPacT NT/NW and ComPacT NS circuit breakers, the tripping cause bit is reset as soon as the circuit breaker is closed again.
- For MicroLogic X control units for MasterPacT MTZ circuit breakers, the tripping cause bit is reset by pressing the test/reset button (located beside the trip cause LEDs on the MicroLogic X control unit). Press and hold the button for 3 to 15 seconds to reset all the trip causes.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x2EE3	12004	R	–	INT16U	–	A/E	A/E/P/H	X	–	Tripping cause for the standard protection functions
						A/E	A/E/P/H	X	0	Long-time protection I _r
						A/E	P/H	X	1	Short-time protection I _{sd}
						–	A/E	X	1	Short-time protection I _{sd} or Instantaneous protection I _i
						A/E	P/H	X	2	Instantaneous protection I _i
						A/E	A/E/P/H	X	3	Ground-fault protection I _g
						E	A/P/H	X	4	Earth-leakage protection I _{Δn}
						A/E	A/E/P/H	X	5	Integrated instantaneous protection for:

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
										- MasterPacT NT06L1, NT08L1, NT10L1, and equivalent ComPacT NS - ComPacT NSX
						A/E	–	X	6	Internal failure (STOP)
						–	A/E	–		Other protections or integrated instantaneous protection
						–	P/H	–		Internal failure (temperature)
						–	A/E/P/H	–	7	Internal failure (overvoltage)
						–	P/H	X	8	Other protection (see register 12005)
						E	–	–	9	Instantaneous with earth-leakage protection on the trip unit.
						E	–	–	10	Unbalance motor protection
						E	–	–	11	Jam motor protection
						E	–	–	12	Underload motor protection
						E	–	–	13	Long-start motor protection
						A/E	–	–	14	Reflex tripping protection
						A/E	A/E/P/H	X	15	If this bit is at 1, bits 0 to 14 are not valid.
0x2EE4	12005	R	–	INT16U	–	–	P/H	X	–	Tripping causes for the advanced protection functions
						–	P/H	–	0	Current unbalance
						–	P/H	–	1	Overcurrent on phase 1
						–	P/H	–	2	Overcurrent on phase 2
						–	P/H	–	3	Overcurrent on phase 3
						–	P/H	–	4	Overcurrent on Neutral
						–	P/H	X	5	Undervoltage
						–	P/H	X	6	Overvoltage
						–	P/H	–	7	Voltage unbalance
						–	P/H	–	8	Overpower
						–	P/H	X	9	Reverse power
						–	P/H	X	10	Underfrequency
						–	P/H	X	11	Overfrequency
						–	P/H	–	12	Phase rotation
						–	P/H	–	13	Load shedding based on current
						–	P/H	–	14	Load shedding based on power
						–	P/H	X	15	If this bit is at 1, bits 0 to 14 are not valid.
0x2EE5–0x2EE6	12006–12007	–	–	–	–	–	–	–	–	Reserved

Overrun of the Protection Setpoints

The alarm setpoint registers provide information about overrun of the standard and advanced protection setpoints. A bit is at 1 once a setpoint overrun has occurred, even if the time delay has not expired.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x2EE7	12008	R	–	INT16U	–	A/E	P/H	–	–	Overrun of the standard protection setpoints
						A/E	P/H	–	0	Long-time protection pick-up
						–	–	–	1–14	Reserved
						A/E	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x2EE8	12009	R	–	INT16U	–	–	P/H	–	–	Overrun of the advanced protection setpoints
						–	P/H	–	0	Current unbalance
						–	P/H	–	1	Maximum current on phase 1
						–	P/H	–	2	Maximum current on phase 2
						–	P/H	–	3	Maximum current on phase 3
						–	P/H	–	4	Maximum current on the neutral
						–	P/H	–	5	Minimum voltage
						–	P/H	–	6	Maximum voltage
						–	P/H	–	7	Voltage unbalance
						–	P/H	–	8	Maximum power
						–	P/H	–	9	Reverse power
						–	P/H	–	10	Minimum frequency
						–	P/H	–	11	Maximum frequency
						–	P/H	–	12	Phase rotation
						–	P/H	–	13	Load shedding based on the current
						–	P/H	–	14	Load shedding based on the power
						–	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x2EE9	12010	R	–	INT16U	–	–	P/H	–	–	Continuation of the previous register
						–	P/H	–	0	Ground-fault alarm
						E	P/H	–	1	Earth-leakage alarm
						–	–	–	2–14	Reserved
						–	P/H	–	15	If this bit is at 1, bits 0 to 14 are not valid.

Alarms

The alarm register provides information about the pre-alarms and the user-defined alarms. A bit is set to 1 once an alarm is active.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x2EEA	12011	R	–	INT16U	–	A/E	–	X	–	Pre-alarm register
						A/E	–	X	0	Long-time protection time pre-alarm (PAL Ir)
						E	–	–	1	Earth-leakage protection pre-alarm (PAL IΔn)
						–	–	X	–	Earth-leakage alarm ⁽¹⁾
						A/E	–	–	2	Ground-fault protection pre-alarm (PAL Ig)

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
						–	–	X		Ground-fault alarm ⁽²⁾
						–	–	–	3–14	Reserved
						A/E	–	X	15	If this bit is at 1, bits 0 to 14 are not valid.
0x2EEB	12012	R	–	INT16U	–	A/E	–	–	–	Register of user-defined alarms
						A/E	–	–	0	User-defined alarm 201
						A/E	–	–	1	User-defined alarm 202
						A/E	–	–	2	User-defined alarm 203
						A/E	–	–	3	User-defined alarm 204
						A/E	–	–	4	User-defined alarm 205
						A/E	–	–	5	User-defined alarm 206
						A/E	–	–	6	User-defined alarm 207
						A/E	–	–	7	User-defined alarm 208
						A/E	–	–	8	User-defined alarm 209
						A/E	–	–	9	User-defined alarm 210
						–	–	–	10–14	Reserved
						A/E	–	–	15	If this bit is at 1, bits 0 to 14 are not valid.
0x2EEC–0x2EEE	12013–12015	–	–	–	–	–	–	–	–	Reserved

(1) Value available on MicroLogic 7.0 X control unit only when the Digital Module ANSI 51N/51G Ground-fault alarm is installed.

(2) Value available on MicroLogic 2.0 X, 3.0 X, 5.0 X, and 6.0 X control units only when the Digital Module ANSI 51N/51G Ground-fault alarm is installed.

Current

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2EEF	12016	R	A	INT16U	0–32767	A/E	A/E/P/H	X	RMS current on phase 1: I1
0x2EF0	12017	R	A	INT16U	0–32767	A/E	A/E/P/H	X	RMS current on phase 2: I2
0x2EF1	12018	R	A	INT16U	0–32767	A/E	A/E/P/H	X	RMS current on phase 3: I3
0x2EF2	12019	R	A	INT16U	0–32767	A/E	A/E/P/H	X	RMS current on the neutral: IN ⁽¹⁾
0x2EF3	12020	R	A	INT16U	0–32767	A/E	A/E/P/H	X	Maximum of I1, I2, I3, and IN
0x2EF4	12021	R	%Ig	INT16U	0–32767	A/E	A/E/P/H	X	Ground-fault current Ig ⁽²⁾
0x2EF5	12022	R	%IΔn	INT16U	0–32767	E	A/P/H	X	Earth-leakage current IΔn ⁽³⁾

(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral current transformer (ENCT).

(2) This value is only available:

- For MasterPacT MTZ MicroLogic 6.0 X control units, expressed as %Ig pick-up
- For MasterPacT NT/NW and ComPacT NS MicroLogic 6.0 trip units, expressed as %Ig pick-up
- For ComPacT NSX MicroLogic 6.2 and 6.3 trip units, expressed as %Ig pick-up

(3) This value is only available:

- For MasterPacT MTZ MicroLogic 7.0 X control units, expressed as %IΔn pick-up
- For MasterPacT NT/NW and ComPacT NS MicroLogic 7.0 trip units, expressed as %IΔn pick-up
- For ComPacT NSX MicroLogic 7.2 and 7.3 trip units, expressed as %IΔn pick-up

Maximum Current Values

Maximum current values can be reset with the reset minimum/maximum command.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2EF6	12023	R	A	INT16U	0–32767	A/E	A/E/P/H	X	Maximum RMS current on phase 1: I1
0x2EF7	12024	R	A	INT16U	0–32767	A/E	A/E/P/H	X	Maximum RMS current on phase 2: I2
0x2EF8	12025	R	A	INT16U	0–32767	A/E	A/E/P/H	X	Maximum RMS current on phase 3: I3
0x2EF9	12026	R	A	INT16U	0–32767	A/E	A/E/P/H	X	Maximum RMS current on the neutral: IN ⁽¹⁾
0x2EFA	12027	R	A	INT16U	0–32767	A/E	A/E/P/H	X	Maximum RMS current out of the 4 previous registers
0x2EFB	12028	R	%Ig	INT16U	0–32767	A/E	A/E/P/H	X	Maximum ground-fault current Ig ⁽²⁾
0x2EFC	12029	R	%IΔn	INT16U	0–32767	E	A/P/H	X	Maximum earth-leakage current ⁽³⁾

(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral current transformer (ENCT).

(2) This value is only available:

- For MasterPacT MTZ MicroLogic 6.0 X control units, expressed as %Ig pick-up
- For MasterPacT NT/NW and ComPacT NS MicroLogic 6.0 trip units, expressed as %Ig pick-up
- For ComPacT NSX MicroLogic 6.2 and 6.3 trip units, expressed as %Ig pick-up

(3) This value is only available:

- For MasterPacT MTZ MicroLogic 7.0 X control units, expressed as %IΔn pick-up
- For MasterPacT NT/NW and ComPacT NS MicroLogic 7.0 trip units, expressed as %IΔn pick-up
- For ComPacT NSX MicroLogic 7.2 and 7.3 trip units, expressed as %IΔn pick-up

Voltage

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2EFD	12030	R	V	INT16U	0–1200	E	E/P/H	X	RMS phase-to-phase voltage V12
0x2EFE	12031	R	V	INT16U	0–1200	E	E/P/H	X	RMS phase-to-phase voltage V23
0x2EFF	12032	R	V	INT16U	0–1200	E	E/P/H	X	RMS phase-to-phase voltage V31
0x2F00	12033	R	V	INT16U	0–1200	E	E/P/H	X	RMS phase-to-neutral voltage V1N ⁽¹⁾
0x2F01	12034	R	V	INT16U	0–1200	E	E/P/H	X	RMS phase-to-neutral voltage V2N ⁽¹⁾
0x2F02	12035	R	V	INT16U	0–1200	E	E/P/H	X	RMS phase-to-neutral voltage V3N ⁽¹⁾

(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral voltage transformer (ENVT).

Frequency

When the MicroLogic trip unit cannot calculate the frequency, it returns Not applicable = 32768 (0x8000).

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F03	12036	R	0.1 Hz	INT16U	400–600	E	P/H	X	Frequency
0x2F04	12037	R	0.1 Hz	INT16U	400–600	E	P/H	X	Maximum frequency ⁽¹⁾

(1) This value can be reset with the reset minimum/maximum command.

Power

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F05	12038	R	0.1 kW	INT16	-32767–+32767	E	E/P/H	X	Active power on phase 1: P1 ⁽¹⁾ ⁽²⁾
0x2F06	12039	R	0.1 kW	INT16	-32767–+32767	E	E/P/H	X	Active power on phase 2: P2 ⁽¹⁾ ⁽²⁾
0x2F07	12040	R	0.1 kW	INT16	-32767–+32767	E	E/P/H	X	Active power on phase 3: P3 ⁽¹⁾ ⁽²⁾
0x2F08	12041	R	0.1 kW	INT16	-32767–+32767	E	E/P/H	X	Total active power: Ptot ⁽²⁾
0x2F09	12042	R	0.1 kVAR	INT16	-32767–+32767	E	E/P/H	X	Reactive power on phase 1: Q1 ⁽¹⁾ ⁽²⁾
0x2F0A	12043	R	0.1 kVAR	INT16	-32767–+32767	E	E/P/H	X	Reactive power on phase 2: Q2 ⁽¹⁾ ⁽²⁾
0x2F0B	12044	R	0.1 kVAR	INT16	-32767–+32767	E	E/P/H	X	Reactive power on phase 3: Q3 ⁽¹⁾ ⁽²⁾
0x2F0C	12045	R	0.1 kVAR	INT16	-32767–+32767	E	E/P/H	X	Total reactive power: Qtot ⁽²⁾
0x2F0D	12046	R	0.1 kVA	INT16U	0–32767	E	E/P/H	X	Apparent power on phase 1: S1 ⁽¹⁾
0x2F0E	12047	R	0.1 kVA	INT16U	0–32767	E	E/P/H	X	Apparent power on phase 2: S2 ⁽¹⁾
0x2F0F	12048	R	0.1 kVA	INT16U	0–32767	E	E/P/H	X	Apparent power on phase 3: S3 ⁽¹⁾
0x2F10	12049	R	0.1 kVA	INT16U	0–32767	E	E/P/H	X	Total apparent power: Stot

(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral current transformer (ENCT).

(2) The sign for the active and reactive power depends on the configuration of:

- register 3316 for ComPacT NSX, ComPacT NS and MasterPacT NT/NW circuit breakers.
- register 8405 for MasterPacT MTZ circuit breakers.

Energy

Energy is stored in big-endian format: the most significant register is transmitted first, the least significant second.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F11– 0x2F12	12050– 12051	R	kWh	INT32	-1 999 999 999– +1 999 999 999	E	E/P/H	X	Active energy: Ep ⁽¹⁾
0x2F13– 0x2F14	12052– 12053	R	kVARh	INT32	-1 999 999 999– +1 999 999 999	E	E/P/H	X	Reactive energy: Eq ⁽¹⁾
0x2F15– 0x2F16	12054– 12055	R	kWh	INT32U	0–1 999 999 999	E	P/H	X	Active energy counted positively: Epln

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F17– 0x2F18	12056– 12057	R	kWh	INT32U	0–1 999 999 999	E	P/H	X	Active energy counted negatively: EpOut
0x2F19– 0x2F1A	12058– 12059	R	kVARh	INT32U	0–1 999 999 999	E	P/H	X	Reactive energy counted positively: EqIn
0x2F1B– 0x2F1C	12060– 12061	R	kVARh	INT32U	0–1 999 999 999	E	P/H	X	Reactive energy counted negatively: EqOut
0x2F1D– 0x2F1E	12062– 12063	R	kVAh	INT32U	0–1 999 999 999	E	E/P/H	X	Total apparent energy: Es
0x2F1F– 0x2F20	12064– 12065	R	kWh	INT32U	0–1 999 999 999	E	–	X	Active energy counted positively (non-resettable): Epln
0x2F21– 0x2F22	12066– 12067	R	kWh	INT32U	0–1 999 999 999	E	–	X	Active energy counted negatively (non-resettable): EpOut
0x2F23– 0x2F2E	12068– 12079	–	–	–	–	–	–	–	Reserved

(1) This value is always positive with MicroLogic E trip units for MasterPacT NT/NW and ComPacT NS circuit breakers.

Current Demand Values

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F2F	12080	R	A	INT16U	0–32767	E	E/P/H	X	Current demand value on phase 1: I1 Dmd
0x2F30	12081	R	A	INT16U	0–32767	E	E/P/H	X	Current demand value on phase 2: I2 Dmd
0x2F31	12082	R	A	INT16U	0–32767	E	E/P/H	X	Current demand value on phase 3: I3 Dmd
0x2F32	12083	R	A	INT16U	0–32767	E	E/P/H	X	Current demand value on the neutral: IN Dmd ⁽¹⁾

(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral current transformer (ENCT).

Power Demand Values

- For the block window, the demand value is updated at the end of the window.
- For the sliding window,
 - If window duration is configured for less or equal to 15 minutes, the demand value is updated every 15 seconds
 - If window duration is configured for more than 15 minutes, the demand value is updated every 1 minute.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F33	12084	R	0.1 kW	INT16U	0–32767	E	E/P/H	X	Total active power demand: P Dmd
0x2F34	12085	R	0.1 kVAR	INT16U	0–32767	E	P/H	X	Total reactive power demand: Q Dmd
0x2F35	12086	R	0.1 kVA	INT16U	0–32767	E	P/H	X	Total apparent power demand: S Dmd
0x2F36– 0x2F38	12087– 12089	–	–	–	–	–	–	–	Reserved

Maximum Voltage Values

Maximum voltage values can be reset with the reset minimum/maximum command.

Register = 0 if the voltage < 25 V.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F39	12090	R	V	INT16U	0–1200	E	E/P/H	X	Maximum RMS phase-to-phase voltage V12
0x2F3A	12091	R	V	INT16U	0–1200	E	E/P/H	X	Maximum RMS phase-to-phase voltage V23
0x2F3B	12092	R	V	INT16U	0–1200	E	E/P/H	X	Maximum RMS phase-to-phase voltage V31
0x2F3C	12093	R	V	INT16U	0–1200	E	E/P/H	X	Maximum RMS phase-to-neutral voltage V1N ⁽¹⁾
0x2F3D	12094	R	V	INT16U	0–1200	E	E/P/H	X	Maximum RMS phase-to-neutral voltage V2N ⁽¹⁾
0x2F3E	12095	R	V	INT16U	0–1200	E	E/P/H	X	Maximum RMS phase-to-neutral voltage V3N ⁽¹⁾
(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral voltage transformer (ENVT).									

Power Factor

The sign for the fundamental power factor ($\cos\phi$) depends on the MicroLogic configuration.

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F3F	12096	R	0.01	INT16	-100– +100	E	E/P/H	X	Power factor on phase 1: PF1 ⁽¹⁾
0x2F40	12097	R	0.01	INT16	-100– +100	E	E/P/H	X	Power factor on phase 2: PF2 ⁽¹⁾
0x2F41	12098	R	0.01	INT16	-100– +100	E	E/P/H	X	Power factor on phase 3: PF3 ⁽¹⁾
0x2F42	12099	R	0.01	INT16	-100– +100	E	E/P/H	X	Total power factor: PF
0x2F43	12100	R	0.01	INT16	-100– +100	E	H	X	Fundamental power factor on phase 1: $\cos\phi_1$ ⁽¹⁾
0x2F44	12101	R	0.01	INT16	-100– +100	E	H	X	Fundamental power factor on phase 2: $\cos\phi_2$ ⁽¹⁾
0x2F45	12102	R	0.01	INT16	-100– +100	E	H	X	Fundamental power factor on phase 3: $\cos\phi_3$ ⁽¹⁾
0x2F46	12103	R	0.01	INT16	-100– +100	E	H	X	Total fundamental power factor: $\cos\phi$
(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral voltage transformer (ENVT).									

Total Harmonic Distortion (THD)

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F47	12104	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of V12 compared to the fundamental
0x2F48	12105	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of V23 compared to the fundamental
0x2F49	12106	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of V31 compared to the fundamental
0x2F4A	12107	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of V1N compared to the fundamental ⁽¹⁾
0x2F4B	12108	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of V2N compared to the fundamental ⁽¹⁾
0x2F4C	12109	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of V3N compared to the fundamental ⁽¹⁾
0x2F4D	12110	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of I1 compared to the fundamental
0x2F4E	12111	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of I2 compared to the fundamental
0x2F4F	12112	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of I3 compared to the fundamental
0x2F50	12113	R	0.1 %	INT16U	0–5000	E	H	X	Total harmonic distortion of total current compared to the fundamental
(1) This value cannot be accessed for motor applications and in cases of three-pole circuit breakers without external neutral voltage transformer (ENVV).									

Counters

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Description
0x2F7F	12160	R	–	INT16U	0–32766	A/E	A/E/P/H	X	Trip counter
0x2F80	12161	R	–	INT16U	0–32766	A/E	A/E/P/H	X	Counter of alarms with priority level = 3 (high)
0x2F81	12162	R	–	INT16U	0–32766	A/E	A/E/P/H	X	Counter of alarms with priority level = 2 (medium)
0x2F82	12163	R	–	INT16U	0–32766	A/E	A/E/P/H	X	Counter of alarms with priority level = 1 (low)

Miscellaneous

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
0x2F83	12164	R	–	INT16U	–	A/E	A/E/P/H	X	–	Validity of the breaker close inhibit
									0	Validity of the breaker close inhibit by IO module
									1	Validity of the breaker close inhibit by remote controller
									2–15	Reserved
0x2F84	12165	R	–	INT16U	–	A/E	A/E/P/H	X	–	Status of the breaker close inhibit
									0	Status of the breaker close inhibit by IO module

Address	Register	RW	Unit	Type	Range	A/E	A/E/P/H	X	Bit	Description
									1	Status of the breaker close inhibit by remote controller
									2–15	Reserved
0x2F85– 0x2FA7	12166– 12200	–	–	–	–	–	–	–	–	Reserved

MicroLogic 5A/E, 6A/E, and 7E Trip Unit Data for ComPacT NSX Circuit Breakers

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MicroLogic 5A/E, 6A/E, and 7E Trip Unit Registers

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Real-Time Measurements

General Description

The real-time measurements are refreshed every second. Real-time measurements include:

- voltage and voltage unbalance
- current and current unbalance
- active, reactive, apparent, and distortion power
- reactive power with harmonic
- power factor and fundamental power factor
- frequency
- THD (total harmonic distortion)

Voltage

Register = 0 if voltage < 25 V.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x03E7	1000	R	1	V	INT16U	0–850	E	RMS phase-to-phase voltage V12
0x03E8	1001	R	1	V	INT16U	0–850	E	RMS phase-to-phase voltage V23
0x03E9	1002	R	1	V	INT16U	0–850	E	RMS phase-to-phase voltage V31
0x03EA	1003	R	1	V	INT16U	0–850	E	RMS phase-to-neutral voltage V1N ⁽¹⁾
0x03EB	1004	R	1	V	INT16U	0–850	E	RMS phase-to-neutral voltage V2N ⁽¹⁾
0x03EC	1005	R	1	V	INT16U	0–850	E	RMS phase-to-neutral voltage V3N ⁽¹⁾
0x03ED	1006	R	1	V	INT16U	0–850	E	Arithmetic mean of V12, V23, and V31: $(V12 + V23 + V31) / 3 = V_{avg} \text{ L-L}$
0x03EE	1007	R	1	V	INT16U	0–850	E	Arithmetic mean of V1N, V2N, and V3N: $(V1N + V2N + V3N) / 3 = V_{avg} \text{ L-N}^{(1)}$
0x0478	1145	R	1	V	INT16U	0–850	E	Vmax: maximum of V12, V23, and V31 ⁽²⁾
0x0479	1146	R	1	V	INT16U	0–850	E	Vmin: minimum of V12, V23, and V31 ⁽²⁾

(1) This value is not available for motor application and not accessible when the system type in register 3314 is 30 or 31. Refer to System Type, page 139.

(2) This value can be reset with the reset minimum/maximum command. Refer to Reset Minimum/Maximum, page 164.

Voltage Unbalance

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x03EF	1008	R	10	%	INT16U	-1000–+1000	E	V12 phase-to-phase voltage unbalance in relation to the arithmetic mean of phase-to-phase voltages
0x03F0	1009	R	10	%	INT16U	-1000–+1000	E	V23 phase-to-phase voltage unbalance in relation to the arithmetic mean of phase-to-phase voltages
0x03F1	1010	R	10	%	INT16U	-1000–+1000	E	V31 phase-to-phase voltage unbalance in relation to the arithmetic mean of phase-to-phase voltages
0x03F2	1011	R	10	%	INT16U	-1000–+1000	E	V1N phase-to-neutral voltage unbalance in relation to the arithmetic mean of phase-to-neutral voltages ⁽¹⁾

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x03F3	1012	R	10	%	INT16U	-1000—+1000	E	V2N phase-to-neutral voltage unbalance in relation to the arithmetic mean of phase-to-neutral voltages ⁽¹⁾
0x03F4	1013	R	10	%	INT16U	-1000—+1000	E	V3N phase-to-neutral voltage unbalance in relation to the arithmetic mean of phase-to-neutral voltages ⁽¹⁾
0x03F5	1014	R	10	%	INT16U	-1000—+1000	E	Maximum phase-to-phase voltage unbalance value of registers 1008, 1009, and 1010 ⁽²⁾
0x03F6	1015	R	10	%	INT16U	-1000—+1000	E	Maximum phase-to-neutral voltage unbalance value of registers 1011, 1012, and 1013 ⁽¹⁾⁽²⁾
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to <i>System Type</i> , page 139.								
(2) This value can be reset with the reset minimum/maximum command. Refer to <i>Reset Minimum/Maximum</i> , page 164.								

Current

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x03F7	1016	R	1	A	INT16U	0–20xIn	A/E	RMS current on phase 1: I1
0x03F8	1017	R	1	A	INT16U	0–20xIn	A/E	RMS current on phase 2: I2
0x03F9	1018	R	1	A	INT16U	0–20xIn	A/E	RMS current on phase 3: I3
0x03FA	1019	R	1	A	INT16U	0–20xIn	A/E	RMS current on the neutral: IN ⁽¹⁾
0x03FB	1020	R	1	A	INT16U	0–20xIn	A/E	Maximum of I1, I2, I3, and IN ⁽²⁾
0x03FC	1021	R	1	%Ig	INT16U	0–20xIn	A/E	Ground-fault current
0x03FD	1022	R	1	mA	INT16U	0–20xIn	E	Earth-leakage current
0x0401	1026	R	1	A	INT16U	0–20xIn	A/E	Minimum of I1, I2, and I3 ⁽²⁾
0x0402	1027	R	1	A	INT16U	0–20xIn	A/E	Arithmetic mean of I1, I2, and I3: $(I1 + I2 + I3) / 3 = I_{avg}$
(1) This value is not available for motor application and when the system type in register 3314 is 31 or 40. Refer to <i>System Type</i> , page 139.								
(2) This value can be reset with the reset minimum/maximum command. Refer to <i>Reset Minimum/Maximum</i> , page 164.								

Current Unbalance

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0403	1028	R	10	%	INT16	-1000—+1000	E	I1 current unbalance in relation to the arithmetic mean of the phase currents
0x0404	1029	R	10	%	INT16	-1000—+1000	E	I2 current unbalance in relation to the arithmetic mean of the phase currents
0x0405	1030	R	10	%	INT16	-1000—+1000	E	I3 current unbalance in relation to the arithmetic mean of the phase currents
0x0406	1031	R	10	%	INT16	-1000—+1000	E	IN current unbalance in relation to the arithmetic mean of the phase current ⁽¹⁾
0x0407	1032	R	10	%	INT16	-1000—+1000	E	Maximum current unbalance of registers 1028, 1029, and 1030 ⁽²⁾
(1) This value is not available for motor application and when the system type in register 3314 is 31 or 40. Refer to <i>System Type</i> , page 139.								
(2) This value can be reset with the reset minimum/maximum command. Refer to <i>Reset Minimum/Maximum</i> , page 164.								

Active Power

The flow sign of the active power depends on the configuration of register 3316. Refer to [Power Flow Sign](#), page 139.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0409	1034	R	10	kW	INT16	-10000—+10000	E	Active power on phase 1: P1 ⁽¹⁾
0x040A	1035	R	10	kW	INT16	-10000—+10000	E	Active power on phase 2: P2 ⁽¹⁾
0x040B	1036	R	10	kW	INT16	-10000—+10000	E	Active power on phase 3: P3 ⁽¹⁾
0x040C	1037	R	10	kW	INT16	-30000—+30000	E	Total active power: Ptot
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to System Type , page 139.								

Reactive Power

The flow sign of the reactive power depends on the configuration of register 3316 , page 139.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x040D	1038	R	10	kVAR	INT16	-10000—+10000	E	Reactive power on phase 1: Q1 ⁽¹⁾
0x040E	1039	R	10	kVAR	INT16	-10000—+10000	E	Reactive power on phase 2: Q2 ⁽¹⁾
0x040F	1040	R	10	kVAR	INT16	-10000—+10000	E	Reactive power on phase 3: Q3 ⁽¹⁾
0x0410	1041	R	10	kVAR	INT16	-30000—+30000	E	Total reactive power: Qtot
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to System Type , page 139.								

Apparent Power

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0411	1042	R	10	kVAR	INT16	0—10000	E	Apparent power on phase 1: S1 ⁽¹⁾
0x0412	1043	R	10	kVAR	INT16	0—10000	E	Apparent power on phase 2: S2 ⁽¹⁾
0x0413	1044	R	10	kVAR	INT16	0—10000	E	Apparent power on phase 3: S3 ⁽¹⁾
0x0414	1045	R	10	kVAR	INT16	0—30000	E	Total apparent power: Stot
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to System Type , page 139.								

Power Factor

The sign of the power factor depends on the configuration of register 3318 , page 140.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0415	1046	R	10-0	—	INT16	-100—+100	E	Power factor on phase 1: PF1 ⁽¹⁾
0x0416	1047	R	10-0	—	INT16	-100—+100	E	Power factor on phase 2: PF2 ⁽¹⁾
0x0417	1048	R	10-0	—	INT16	-100—+100	E	Power factor on phase 3: PF3 ⁽¹⁾
0x0418	1049	R	10-0	—	INT16	-100—+100	E	Total power factor: PF
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to System Type , page 139.								

Fundamental Power Factor ($\cos\phi$)

The sign of the fundamental power factor ($\cos\phi$) depends on the configuration of register 3318, page 140.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0419	1050	R	100	–	INT16	-100–+100	E	Fundamental power factor on phase 1: $\cos\phi$ 1 ⁽¹⁾
0x041A	1051	R	100	–	INT16	-100–+100	E	Fundamental power factor on phase 2: $\cos\phi$ 2 ⁽¹⁾
0x041B	1052	R	100	–	INT16	-100–+100	E	Fundamental power factor on phase 3: $\cos\phi$ 3 ⁽¹⁾
0x041C	1053	R	100	–	INT16	-100–+100	E	Total fundamental power factor: $\cos\phi$
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to System Type, page 139.								

Frequency

When the MicroLogic trip unit cannot calculate the frequency, it returns Not Evaluated = 32768 (0x8000).

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x041D	1054	R	10	Hz	INT16U	150–4400	E	System frequency: F

Distortion Power

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x043F	1088	R	10	kVAR	INT16U	0–10000	E	Distortion power on phase 1: D1 ⁽¹⁾
0x0440	1089	R	10	kVAR	INT16U	0–10000	E	Distortion power on phase 2: D2 ⁽¹⁾
0x0441	1090	R	10	kVAR	INT16U	0–10000	E	Distortion power on phase 3: D3 ⁽¹⁾
0x0442	1091	R	10	kVAR	INT16U	0–10000	E	Total distortion power: Dtot
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to System Type, page 139.								

Total Harmonic Distortion (THD)

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0443	1092	R	10	%	INT16U	0–32766	E	Total harmonic distortion of V12 in relation to the fundamental
0x0444	1093	R	10	%	INT16U	0–32766	E	Total harmonic distortion of V23 in relation to the fundamental
0x0445	1094	R	10	%	INT16U	0–32766	E	Total harmonic distortion of V31 in relation to the fundamental
0x0446	1095	R	10	%	INT16U	0–32766	E	Total harmonic distortion of V1N in relation to the fundamental ⁽¹⁾
0x0447	1096	R	10	%	INT16U	0–32766	E	Total harmonic distortion of V2N in relation to the fundamental ⁽¹⁾
0x0448	1097	R	10	%	INT16U	0–32766	E	Total harmonic distortion of V3N in relation to the fundamental ⁽¹⁾
0x0449	1098	R	10	%	INT16U	0–32766	E	Total harmonic distortion of I1 in relation to the fundamental
0x044A	1099	R	10	%	INT16U	0–32766	E	Total harmonic distortion of I2 in relation to the fundamental

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x044B	1100	R	10	%	INT16U	0–32766	E	Total harmonic distortion of I3 in relation to the fundamental
(1) This value is not available for motor application and when the system type in register 3314 is 30 or 31. Refer to <i>System Type</i> , page 139.								

Minimum/Maximum Values of Real-Time Measurements

Minimum/Maximum Measurements Rule

Minimum and maximum measurements take into account the absolute value of real-time measurements. Therefore the following rule applies:

$0 < 10 < 200 < -400 < 600 < -3800$.

In this case:

- the minimum value = 0
- the maximum value = -3800

NOTE: This rule does not apply for the power factor (PF) and for the fundamental power factor ($\cos\phi$):

- PFmax (or $\cos\phi$ max.) is obtained for the smallest positive value of PF (or $\cos\phi$).
- PFmin (or $\cos\phi$ min.) is obtained for the highest negative value of PF (or $\cos\phi$).

The reset minimum/maximum command (command code = 46728) can reset the content of the minimum/maximum real-time measurement registers.

Minimum of Real-Time Measurements

Registers 1300 to 1599 hold the minimum values of real-time metering parameters:

- The register of the minimum value of a real-time metering parameter is equal to the register of the real-time metering parameter plus 300.

Examples

- Register 1300 holds the minimum value of the phase-to-phase voltage V12 (register 1000).
- Register 1316 holds the minimum value of the current on phase 1 (register 1016).
- The order of the registers is the same as that of the real-time metering variables.
- The scale factors of the minimum values are the same as those of the real-time metering parameters.
- The minimum values of unbalance current and unbalance voltage are not available.
- The minimum values of Imin (register 1026), Vmax (register 1145), and Vmin (register 1146) are not available.

Maximum of Real-Time Measurements

Registers 1600 to 1899 hold the maximum values of real-time metering parameters:

- The register of the maximum value of a real-time metering parameter is equal to the register of the real-time metering parameter plus 600.

Examples

- Register 1600 holds the maximum value of the phase-to-phase voltage V12 (register 1000).
- Register 1616 holds the maximum value of the current on phase 1 (register 1016).
- The order of the registers is the same as that of the real-time metering variables.

- The scale factors of the maximum values are the same as those of the real-time metering parameters.
- The maximum values of Imin (register 1026), Vmax (register 1145), and Vmin (register 1146) are not available.

Energy Measurements

General Description

The energy measurements are refreshed every second. Energy measurements are saved every hour in the non-volatile memory of the MicroLogic trip unit.

Energy measurements include:

- active energy E_p
- reactive energy E_q
- apparent energy E_s
- active energy counted positively (E_{pIn}) or negatively (E_{pOut}), according to the configuration of register 3316 *Power Flow Sign*, page 139.
- reactive energy counted positively (E_{qIn}) or negatively (E_{qOut}), according to the configuration of register 3316 *Power Flow Sign*, page 139.
- active energy and reactive energy are accumulated according to the configuration of register 3324 (absolute mode by factory setting) *Energy Accumulation Mode*, page 140.

The reset minimum/maximum command (command code = 46728) can reset the content of the energy measurement registers, cumulative energy measurements excepted.

NOTE: The power flow sign configuration command (command code = 47240) can reset the content of the energy measurement registers excluding the cumulative energy measurements.

Energy Registers

Energies are stored in big-endian format: the most significant word is transmitted first and the least significant second.

Examples

If $E_p = 7589$ kWh, then:

- register 2000 = 0 (0x0000)
- register 2001 = 7589 (0x1DA5)

If $E_p = 4,589,625$ kWh, then:

- register 2000 = 70 (0x0046)
- register 2001 = 2105 (0x0839)

$$4589625 = 70 \times 65536 + 2105$$

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x07CF– 0x07D0	2000– 2001	R	1	kWh	INT32	-1 999 999 999–+1 999 999 999	E	Active energy: E_p
0x07D3– 0x07D4	2004– 2005	R	1	kVARh	INT32	-1 999 999 999–+1 999 999 999	E	Reactive energy: E_q
0x07D7– 0x07D8	2008– 2009	R	1	kWh	INT32	0–1 999 999 999	E	Active energy counted positively: E_{pIn}
0x07DB– 0x07DC	2012– 2013	R	1	kWh	INT32	0–1 999 999 999	E	Active energy counted negatively: E_{pOut}
0x07DF– 0x07E0	2016– 2017	R	1	kVARh	INT32	0–1 999 999 999	E	Reactive energy counted positively: E_{qIn}
0x07E3– 0x07E4	2020– 2021	R	1	kVARh	INT32	0–1 999 999 999	E	Reactive energy counted negatively: E_{qOut}

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x07E7– 0x07E8	2024– 2025	R	1	kVAh	INT32	0–1 999 999 999	E	Apparent energy: Es
0x07EB– 0x07EC	2028– 2029	R	1	kWh	INT32	0–1 999 999 999	E	Cumulative active energy counted positively (cannot be reset): EpIn
0x07ED– 0x07EE	2030– 2031	R	1	kWh	INT32	0–1 999 999 999	E	Cumulative active energy counted negatively (cannot be reset): EpOut

Demand Measurements

General Description

Demand registers include:

- current demand
- active, reactive, and apparent power demand

The window duration of current demand depends on the configuration of register 3352. Refer to [Demand Time](#), page 140.

The window duration and the window type of power demand depend on the configuration of registers 3354 and 3355. Refer to [Demand Time](#), page 140.

The demand measurements are refreshed every minute with the sliding window type.

The demand measurements are refreshed at the end of the window interval with the block window type.

Current Demand

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0897	2200	R	1	A	INT16U	0–20xIn	E	Current demand on phase 1: I1 Dmd
0x0898	2201	R	1	A	INT16U	0–20xIn	E	Current demand on phase 2: I2 Dmd
0x0899	2202	R	1	A	INT16U	0–20xIn	E	Current demand on phase 3: I3 Dmd
0x089A	2203	R	1	A	INT16U	0–20xIn	E	Current demand on the neutral: IN Dmd ⁽¹⁾
0x089B	2204	R	1	A	INT16U	0–20xIn	E	Maximum of current demand on phase 1: I1 Peak Dmd ⁽²⁾
0x089C	2205	R	1	A	INT16U	0–20xIn	E	Maximum of current demand on phase 2: I2 Peak Dmd ⁽²⁾
0x089D	2206	R	1	A	INT16U	0–20xIn	E	Maximum of current demand on phase 3: I3 Peak Dmd ⁽²⁾
0x089E	2207	R	1	A	INT16U	0–20xIn	E	Maximum of current demand on the neutral: IN Peak Dmd ^{(1) (2)}
(1) This value is not available when the system type in register 3314 is 31 or 40. Refer to System Type , page 139.								
(2) This value can be reset with the reset minimum/maximum command. Refer to Reset Minimum/Maximum , page 164.								

Active Power Demand

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x08AF	2224	R	10	kW	INT16	-30000–+30000	E	Total active power demand: P Dmd ⁽¹⁾
0x08B0	2225	R	10	kW	INT16	-30000–+30000	E	Maximum of total active power demand: P Peak Dmd ⁽²⁾
(1) For the block window type, this value is updated at the end of the window interval. For the sliding window type, the value is updated every minute.								
(2) This value can be reset with the reset minimum/maximum command. Refer to Reset Minimum/Maximum , page 164.								

Reactive Power Demand

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x08B5	2230	R	10	kVAR	INT16	-30000– +30000	E	Total reactive power demand: Q Dmd ⁽¹⁾
0x08B6	2231	R	10	kVAR	INT16	-30000– +30000	E	Maximum of total reactive power demand: Q Peak Dmd ⁽²⁾

(1) For the block window type, this value is updated at the end of the window interval. For the sliding window type, the value is updated every minute.

(2) This value can be reset with the reset minimum/maximum command. Refer to [Reset Minimum/Maximum](#), page 164.

Apparent Power Demand

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x08BB	2236	R	10	kVA	INT16	0–30000	E	Total apparent power demand: S Dmd ⁽¹⁾
0x08BC	2237	R	10	kVA	INT16	0–30000	E	Maximum of total apparent power demand: S Peak Dmd ⁽²⁾

(1) For the block window type, this value is updated at the end of the window interval. For the sliding window type, the value is updated every minute.

(2) This value can be reset with the reset minimum/maximum command. Refer to [Reset Minimum/Maximum](#), page 164.

Minimum/Maximum Measurements Reset Time

Overview

The minimum/maximum measurements reset time registers enable the user to know all the dates relative to the last reset minimum/maximum command.

The reset minimum/maximum command (command code 46728) can reset the content of the minimum/maximum measurements registers.

A read request of 30 registers is necessary to read the minimum/maximum measurements reset time.

Minimum/Maximum Measurements Reset Time

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0B53– 0x0B55	2900– 2902	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum current , page 54
0x0B56– 0x0B58	2903– 2905	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum voltage.
0x0B59– 0x0B5B	2906– 2908	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum power (P, Q, S).
0x0B5C– 0x0B5E	2909– 2911	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum power factor and $\cos\phi$.
0x0B5F– 0x0B61	2912– 2914	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum total harmonic distortion.
0x0B62– 0x0B64	2915– 2917	R	–	–	ULP DATE	–	E	Date and time of reset of peak current demand.
0x0B65– 0x0B67	2918– 2920	R	–	–	ULP DATE	–	E	Date and time of reset of peak active, reactive, and apparent power demand.
0x0B68– 0x0B6A	2921– 2923	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum frequency.
0x0B6B– 0x0B6D	2924– 2926	R	–	–	ULP DATE	–	E	Date and time of reset of minimum/ maximum thermal image of motor.
0x0B6E– 0x0B70	2927– 2929	R	–	–	ULP DATE	–	E	Date and time of reset of energy (active, reactive, and apparent).

MicroLogic Trip Unit Identification

Serial Number

The MicroLogic trip unit serial number is composed of a maximum of 11 alphanumeric characters with the following format: PPYYWWDnnnn.

- PP = plant code
- YY = year of fabrication (05–99)
- WW = week of fabrication (01–53)
- D = day of fabrication (1–7)
- nnnn = sequence number (0001–9999)

A read request of 6 registers is necessary to read the MicroLogic trip unit serial number.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x21FB	8700	R	–	–	OCTET STRING	–	A/E	'PP'
0x21FC	8701	R	–	–	OCTET STRING	'05'–'99'	A/E	'YY'
0x21FD	8702	R	–	–	OCTET STRING	'01'–'52'	A/E	'WW'
0x21FE	8703	R	–	–	OCTET STRING	D: '1'–'7' n: '0'–'9'	A/E	'Dn'
0x21FF	8704	R	–	–	OCTET STRING	'00'–'99'	A/E	'nn'
0x2200	8705	R	–	–	OCTET STRING	'0'–'9'	A/E	'n' (the NULL character ends the serial number)

Hardware Revision

For MicroLogic trip units with firmware revision up to V1.2.1, the hardware revision is an integer.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2204	8709	R	–	–	INT16U	0–15	A/E	Hardware revision of the MicroLogic trip unit.

For MicroLogic trip units with firmware revision greater than or equal to V1.2.2, the hardware revision is an ASCII string using the format XXX.YYY.ZZZ with:

- XXX = major version (000–127)
- YYY = minor version (000–255)
- ZZZ = revision number (000–255)

The NULL character ends the revision number.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2212–0x2217	8723–8728	R	–	–	OCTET STRING	–	A/E	Hardware revision of the MicroLogic trip unit.

Product Identification

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x220B	8716	R	–	–	INT16U	15143–15145	A/E	Product identification: 15143 = distribution application, type A 15144 = distribution application, type E 15145 = motor application, type E

Protection Type

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2223	8740	R	–	–	OCTET STRING	52–73	A/E	MicroLogic trip unit protection type: - For ComPacT NSX100/250: '52' = LSI '62' = LSIG '72' = LSIV - For ComPacT NSX400/630: '53' = LSI '63' = LSIG '73' = LSIV

Metering Type

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2224	8741	R	–	–	OCTET STRING	A-E	A/E	MicroLogic trip unit metering type A/E

Application

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x222A	8747	R	–	–	INT16U	1–2	A/E	Application: 1 = distribution 2 = motor

Standard

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x222B	8748	R	–	–	INT16U	1–3	A/E	Standard: 1 = UL 2 = IEC 3 = JIS

Nominal Current

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x222D	8750	R	1	A	INT16U	0–8000	A/E	Circuit breaker nominal current In

Pole

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x222E	8751	R	–	–	INT16U	0–1	A/E	0 = 3-pole 1 = 4-pole

16 Hz 2/3

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x222F	8752	R	–	–	INT16U	0–1	A/E	0 = not a 16 Hz 2/3 MicroLogic trip unit application 1 = 16 Hz 2/3 MicroLogic trip unit application

Firmware Revision

For MicroLogic trip units with firmware revision up to V1.2.1, the firmware revision is an ASCII string using the format VXXX.YYY.ZZZ with:

- XXX = major version (001-999)
- YYY = minor version (001-999)
- ZZZ = revision number (001-999)

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x7529– 0x752D	29994– 29998	R	–	–	OCTET STRING	–	A/E	Firmware revision of the MicroLogic trip unit.

For MicroLogic trip units with firmware revision greater than or equal to V1.2.2, the firmware revision is an ASCII string using the format XXX.YYY.ZZZ with:

- XXX = major version (000-127)
- YYY = minor version (000-255)
- ZZZ = revision number (000-255)

The NULL character ends the revision number.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x220C– 0x2211	8717–8722	R	–	–	OCTET STRING	–	A/E	Firmware revision of the MicroLogic trip unit.

Part Number

The MicroLogic part number starts with the characters LV4 and has the format, LV4XYZTW.

A read request of 4 registers is necessary to read the MicroLogic trip unit part number.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x752F	30000	R	–	–	OCTET STRING	–	A/E	Example: 'LV'
0x7530	30001	R	–	–	OCTET STRING	–	A/E	Example: '4X'
0x7531	30002	R	–	–	OCTET STRING	–	A/E	Example: 'YZ'
0x7532	30003	R	–	–	OCTET STRING	–	A/E	Example: 'TW'

Status

Alarms Status

The alarms status register indicates the current status of the alarms:

- alarm bit = 0: alarm is not active.
- alarm bit = 1: alarm is active.

The following table describes the values for each bit of the alarm status register:

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x1647	5704	R	–	–	INT16U	–	A/E	–	Alarm status
							A/E	0	User-defined alarm 201
							A/E	1	User-defined alarm 202
							A/E	2	User-defined alarm 203
							A/E	3	User-defined alarm 204
							A/E	4	User-defined alarm 205
							A/E	5	User-defined alarm 206
							A/E	6	User-defined alarm 207
							A/E	7	User-defined alarm 208
							A/E	8	User-defined alarm 209
							A/E	9	User-defined alarm 210
							A/E	10	Long-time protection Ir pre-alarm (PAL Ir)
							E	11	Earth-leakage protection IΔn pre-alarm (PAL IΔn)
							A/E	12	Ground-fault protection Ig pre-alarm (PAL Ig)
							—	13–15	Reserved

SDx Module Status

The SDx module status register indicates the status and the validity of the SDx outputs (2 outputs maximum):

- status bit = 0: the output is open.
- status bit = 1: the output is closed.
- validity bit = 0: the output status is unknown.
- validity bit = 1: the output status is known.

The following table describes the values for each bit of the SDx module status register:

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x2298	8857	R	–	–	INT16U	–	A/E	–	SDx module status
							A/E	0	Status of output 1
							A/E	1	Status of output 2
							—	2–7	Reserved
							A/E	8	Validity of output 1
							A/E	9	Validity of output 2
							—	10–15	Reserved

Trip Status

The trip status register indicates the current status of the trip:

- trip bit = 0: trip is not active.
- trip bit = 1: trip is active.

The following table describes the values for each bit of the trip status register:

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x270F	10000	R	–	–	INT16U	–	A/E	–	Trip status
							A/E	0	Long-time protection I _r
							A/E	1	Short-time protection I _{sd}
							A/E	2	Instantaneous protection I _i
							A/E	3	Ground-fault protection I _g
							E	4	Earth-leakage protection I Δ _n
							A/E	5	Integrated instantaneous protection
							A/E	6	Trip unit internal failure (STOP)
							E	7	Instantaneous with earth-leakage protection
							A/E	8	Unbalance motor protection I _{unbal}
							A/E	9	Jam motor protection I _{jam}
							A/E	10	Underload motor protection I _{underload}
							A/E	11	Long-start motor protection I _{longstart}
							A/E	12	Reflex tripping protection
							—	13–15	Reserved

Alarm History

General Description

The alarm history registers describe the last 10 alarms. The alarm history format corresponds to a series of 10 records. Each record is composed of 5 registers describing one alarm.

The alarm history registers return 32768 (0x8000) when they are not used.

Record Number

A read request of $5 \times (n)$ registers is necessary to read the last n alarm records, where 5 is the number of registers for each alarm record.

For example, a read request of $5 \times 3 = 15$ registers is necessary to read the last 3 alarm records of the alarm history:

- The first 5 registers describe the first alarm record (most recent alarm).
- The next 5 registers describe the second alarm record.
- The last 5 registers describe the third alarm record.

Address	Register	Description
0x1663–0x1667	5732–5736	Alarm record 1 (most recent alarm)
0x1668–0x166C	5737–5741	Alarm record 2
0x166D–0x1671	5742–5746	Alarm record 3
0x1672–0x1676	5747–5751	Alarm record 4
0x1677–0x167B	5752–5756	Alarm record 5
0x167C–0x1680	5757–5761	Alarm record 6
0x1681–0x1685	5762–5766	Alarm record 7
0x1686–0x168A	5767–5771	Alarm record 8
0x168B–0x168F	5772–5776	Alarm record 9
0x1690–0x1694	5777–5781	Alarm record 10 (oldest alarm)

Alarm Record

A read request of 5 registers is necessary to read an alarm record.

The order and the description of the alarm records registers are the same as that of the alarm record 1:

Alarm Record 1 (Most Recent Alarm)								
Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x1663	5732	R	1	–	INT16U	0–65535	A/E	Alarm code (see next paragraph)
0x1664– 0x1666	5733– 5735	R	–	–	ULP DATE	–	A/E	Date and time of alarm , page 54
0x1667	5736	R	1	–	INT16U	1–2	A/E	Event type MSB = 0 (reserved) Event occurrence: LSB = 1 Event completion: LSB = 2

Alarm Codes

Alarm Code	Description
201 (0x00C8)	User-defined alarm 201
202 (0x00C9)	User-defined alarm 202
203 (0x00CA)	User-defined alarm 203
204 (0x00CB)	User-defined alarm 204
205 (0x00CC)	User-defined alarm 205
206 (0x00CD)	User-defined alarm 206
207 (0x00CE)	User-defined alarm 207
208 (0x00CF)	User-defined alarm 208
209 (0x00D0)	User-defined alarm 209
210 (0x00D1)	User-defined alarm 210
1013 (0x03F4)	Long-time protection I_r pre-alarm (PAL I_r)
1014 (0x03F5)	Ground-fault protection I_g pre-alarm (PAL I_g)
1015 (0x03F6)	Earth-leakage protection $I_{\Delta n}$ pre-alarm (PAL $I_{\Delta n}$)
The list of the predefined alarms from which the user can choose the 10 user-defined alarms is available at User-Defined Alarms , page 130.	

Trip History

General Description

The trip history registers describe the last 17 trips. The trip history format corresponds to a series of 17 records. Each record is composed of 7 registers describing one trip.

The trip history registers return 32768 (0x8000) when they are not used.

Trip Record Number

A read request of $7 \times (n)$ registers is necessary to read the last n trip records, where 7 is the number of registers for each trip record.

For example, a read request of $7 \times 4 = 28$ registers is necessary to read the last 4 trip records of the trip history:

- The first 7 registers describe the first trip record (most recent trip).
- The next 7 registers describe the second trip record.
- The last 7 registers describe the third trip record.
- The last 7 registers describe the fourth trip record.

Address	Register	Description
0x238B–0x2391	9100–9106	Trip record 1 (most recent trip)
0x2392–0x2398	9107–9113	Trip record 2
0x2399–0x239F	9114–9120	Trip record 3
0x23A0–0x23A6	9121–9127	Trip record 4
0x23A7–0x23AD	9128–9134	Trip record 5
0x23AE–0x23B4	9135–9141	Trip record 6
0x23B5–0x23BB	9142–9148	Trip record 7
0x23BC–0x23C2	9149–9155	Trip record 8
0x23C3–0x23C9	9156–9162	Trip record 9
0x23CA–0x23D0	9163–9169	Trip record 10
0x23D1–0x23D7	9170–9176	Trip record 11
0x23D8–0x23DE	9177–9183	Trip record 12
0x23DF–0x23E5	9184–9190	Trip record 13
0x23E6–0x23EC	9191–9197	Trip record 14
0x23ED–0x23F3	9198–9204	Trip record 15
0x23F4–0x23FA	9205–9211	Trip record 16
0x23FB–0x2401	9212–9218	Trip record 17 (oldest trip)

Trip Record

A read request of 7 registers is necessary to read a trip record.

The order and the description of the trip record registers are the same as that of the trip record 1:

Trip Record 1 (Most Recent Trip)								
Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x238B	9100	R	1	–	INT16U	0–65535	A/E	Trip code (see next paragraph)
0x238C– 0x238E	9101– 9103	R	–	–	ULP DATE	–	A/E	Date and time of event (trip or acknowledge , page 54)
0x238F	9104	R	1	–	INT16U	1–2	A/E	Event type MSB = 0 (reserved) Event occurrence: LSB = 1 Event completion: LSB = 2
0x2390	9105	R	1	–	INT16U	0–5	A/E	Faulty phase 0 = failure (no faulty phase) 1 = phase 1 2 = phase 2 3 = phase 3 4 = phase N 5 = phase 1 2 3 (motor application, ground-fault, earth-leakage)
0x2391	9106	R	1	A	INT16U	0–65535	A/E	Interrupted current (peak) ⁽¹⁾
(1) Measurement depends on the application: - For electrical distribution application, the interrupted current is measured when a trip occurs, due to the long-time protection, the short-time protection, or the instantaneous protection. - For motor-feeder application, the interrupted current is measured when a trip occurs, due to the short-time protection. - In case of other trip types, the interrupted current is not measured and the value recorded is 65535 (0xFFFF).								

Trip Codes

Trip Code	Description
1000 (0x03E8)	Long-time protection I _r
1001 (0x03E9)	Short-time protection I _{sd}
1002 (0x03EA)	Instantaneous protection I _i
1003 (0x03EB)	Ground-fault protection I _g
1004 (0x03EC)	Earth-leakage protection IΔn
1005–1009 (0x03ED–0x03F1)	Reserved
1010 (0x03F2)	Integrated instantaneous protection
1011 (0x03F3)	STOP (trip unit internal failure)
1012 (0x03F4)	Instantaneous with earth-leakage protection
1013–1031 (0x03F5–0x0407)	Reserved
1032 (0x0408)	Unbalance motor protection
1033 (0x0409)	Jam motor protection
1034 (0x040A)	Underload motor protection

Trip Code	Description
1035 (0x040B)	Long-start motor protection
1036 (0x040C)	Reflex tripping protection

Earth-Leakage Test History

General Description

The earth-leakage test history registers describe the last 10 earth-leakage tests performed on a MicroLogic 7E trip unit. The earth-leakage test history format corresponds to a series of 10 records. Each record is composed of 5 registers describing one earth-leakage test.

The earth-leakage test history registers return 32768 (0x8000) when they are not used.

Earth-Leakage Test Number

A read request of 5 x n registers is necessary to read the last n earth-leakage test records, where 5 is the number of registers for each earth-leakage test record.

For example, a read request of 5 x 2 = 10 registers is necessary to read the last 2 earth-leakage test records of the earth-leakage test history:

- The first 5 registers describe the first earth-leakage test record (most recent earth-leakage test).
- The last 5 registers describe the second earth-leakage test record.

Address	Register	Description
0x7593–0x7597	30100–30104	Earth-leakage test record (most recent earth-leakage test)
0x7598–0x759C	30105–30109	Earth-leakage test record 2
0x759D–0x75A1	30110–30114	Earth-leakage test record 3
0x75A2–0x75A6	30115–30119	Earth-leakage test record 4
0x75A7–0x75AB	30120–30124	Earth-leakage test record 5
0x75AC–0x75B0	30125–30129	Earth-leakage test record 6
0x75B1–0x75B5	30130–30134	Earth-leakage test record 7
0x75B6–0x75BA	30135–30139	Earth-leakage test record 8
0x75BB–0x75BF	30140–30144	Earth-leakage test record 9
0x75C0–0x75C4	30145–30149	Earth-leakage test record 10 (oldest earth-leakage test)

Earth-Leakage Test Record

A read request of 5 registers is necessary to read a earth-leakage test record.

The order and the description of the earth-leakage test record registers are the same as that of the earth-leakage test record 1:

Earth-Leakage Test Record 1 (Most Recent Earth-Leakage Test)								
Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x7593	30100	R	1	–	INT16U	0–65535	E	Earth-leakage test code (see next section.)
0x7594– 0x7596	30101– 30103	R	–	–	ULP DATE	–	E	Date and time of earth-leakage test. , page 54
0x7597	30104	R	–	–	INT16U	0–1	E	Test status • 0: test succeeded • 1: test failed

Earth-Leakage Test Codes

Earth-Leakage Test Code	Description
5162 (0x142A)	Earth-leakage test by button
5163 (0x142B)	Earth-leakage test by HMI

Maintenance Operation History

General Description

The maintenance operation history registers describe the last 10 maintenance operations. The maintenance operation history format corresponds to a series of 10 records. Each record is composed of 5 registers describing one maintenance operation.

The maintenance operation history registers return 32768 (0x8000) when they are not used.

Maintenance Operation Number

A read request of 5 x n registers is necessary to read the last n maintenance operation records, where 5 is the number of registers for each maintenance operation record.

For example, a read request of 5 x 2 = 10 registers is necessary to read the last 2 maintenance operation records of the maintenance operation history:

- The first 5 registers describe the first maintenance operation record (most recent maintenance operation).
- The last 5 registers describe the second maintenance operation record.

Address	Register	Description
0x733B–0x733F	29500–29504	Maintenance operation record 1 (most recent maintenance operation)
0x7340–0x7344	29505–29509	Maintenance operation record 2
0x7345–0x7349	29510–29514	Maintenance operation record 3
0x734A–0x734E	29515–29519	Maintenance operation record 4
0x734F–0x7353	29520–29524	Maintenance operation record 5
0x7354–0x7358	29525–29529	Maintenance operation record 6
0x7359–0x735D	29530–29534	Maintenance operation record 7
0x735E–0x7362	29535–29539	Maintenance operation record 8
0x7363–0x7367	29540–29544	Maintenance operation record 9
0x7368–0x736C	29545–29549	Maintenance operation record 10 (oldest maintenance operation)

Maintenance Operation Record

A read request of 5 registers is necessary to read a maintenance operation record.

The order and the description of the maintenance operation record registers are the same as that of the maintenance operation record 1:

Maintenance Operation Record 1 (Most Recent Maintenance Operation)								
Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x733B	29500	R	1	–	INT16U	0–65535	A/E	Maintenance operation code (see next paragraph)
0x733C– 0x733E	29501– 29503	R	–	–	ULP DATE	–	A/E	Date and time of maintenance operation , page 54
0x733F	29504	–	–	–	–	–	–	Reserved

Maintenance Operation Codes

Maintenance Operation Code	Description
2000 (0x07D0)	Push to trip test (with USB maintenance interface)
2001 (0x07D1)	Ground-fault inhibition
2002 (0x07D2)	Reserved
2003 (0x07D3)	Start numerical injection test
2004 (0x07D4)	End numerical injection test
2005 (0x07D5)	Ground-fault test
2006 (0x07D6)	Earth-leakage test
2007 (0x07D7)	Start alarm test
2008 (0x07D8)	End alarm test
2009 (0x07D9)	Start long-time protection
2010 (0x07DA)	End long-time protection
2011 (0x07DB)	Start short-time protection
2012 (0x07DC)	End short-time protection
2013 (0x07DD)	Start instantaneous protection
2014 (0x07DE)	Stop instantaneous protection
2015 (0x07DF)	Start integrated instant protection
2016 (0x07E0)	Stop integrated instant protection
2017 (0x07E1)	Start unbalance protection
2018 (0x07E2)	Stop unbalance protection
2019 (0x07E3)	Start ground-fault protection
2020 (0x07E4)	Stop ground-fault protection
2021 (0x07E5)	Start earth-leakage protection
2022 (0x07E6)	Stop earth-leakage protection
2023 (0x07E7)	Start thermal memory
2024 (0x07E8)	Stop thermal memory
2025 (0x07E9)	Start connection with USB maintenance interface
2026 (0x07EA)	Stop connection with USB maintenance interface
2027 (0x07EB)	Turn rotary wheel 1
2028 (0x07EC)	Turn rotary wheel 2
2029 (0x07ED)	Locking pad open
2030 (0x07EE)	Locking pad closed
2031 (0x07EF)	ZSI test
2032 (0x07F0)	Reserved
2033 (0x07F1)	Reset software
2034 (0x07F2)	Reset minimum/maximum of current measurements
2035 (0x07F3)	Reset minimum/maximum of voltage measurements
2036 (0x07F4)	Reset minimum/maximum of power measurements
2037 (0x07F5)	Reset minimum/maximum of power factor measurements
2038 (0x07F6)	Reset minimum/maximum of total harmonic distortion measurements
2039 (0x07F7)	Reset maximum of current demand measurement

Maintenance Operation Code	Description
2040 (0x07F8)	Reset maximum of power demand (active, reactive, and apparent)
2041 (0x07F9)	Reset minimum/maximum of frequency measurement
2042 (0x07FA)	Reset minimum/maximum of thermal image measurements
2043 (0x07FB)	Reset energy measurements
2044 (0x07FC)	Reset energy counter

Pre-Alarms

General Description

EcoStruxure Power Commission software, page 13 enables the configuration of the following 3 pre-alarms:

- long-time protection pre-alarm (PAL Ir)
- ground-fault protection pre-alarm (PAL Ig)
- earth-leakage protection pre-alarm (PAL IΔn)

Each alarm has a corresponding alarm code:

- PAL Ir = 1013
- PAL Ig = 1014
- PAL IΔn = 1015

Each alarm has a priority level that manages the alarm display on the FDM121 display:

- no priority = N/A (not affected)
- low priority = 1.
- medium priority = 2.
- high priority = 3.

For more information, refer to DOCA0188** *ComPacT NSX MicroLogic 5/6/7 Electronic Trip Units – User Guide*, page 8.

The pre-alarms registers describe the settings of the pre-alarms:

Address	Register	Description
0x19F9–0x1A02	6650–6659	Long-time protection pre-alarm (PAL Ir)
0x1A03–0x1A0C	6660–6669	Ground-fault protection pre-alarm (PAL Ig)
0x1A0D–0x1A16	6670–6679	Earth-leakage protection pre-alarm (PAL IΔn)

Long-Time Protection Pre-Alarm (PAL Ir)

A read request of 10 registers is necessary to read the long-time protection pre-alarm parameters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x19F9	6650	R	–	–	INT16U	–	A/E	The MSB gives the activity of the alarm: 0 = On, 1 = Off. The factory setting is 0 (On). The LSB gives the priority of the alarm: N/A, 1, 2, or 3. The factory setting is 2 (medium priority).
0x19FA	6651	–	–	–	–	–	–	Reserved
0x19FB	6652	R	1	%	INT16	⁽¹⁾	A/E	% of Ir pick-up value. The factory setting is 90.
0x19FC	6653	–	–	–	–	–	–	Reserved
0x19FD	6654	R	1	s	INT16U	1	A/E	Pick-up time delay (fixed to 1 s)
0x19FE	6655	R	1	%	INT16	⁽¹⁾	A/E	% of Ir drop-out value. The factory setting is 85.
0x19FF	6656	–	–	–	–	–	–	Reserved
0x1A00	6657	R	1	s	INT16U	1	A/E	Drop-out time delay (fixed to 1 s)

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x1A01–0x1A02	6658–6659	–	–	–	–	–	–	Reserved
(1) For distribution application, the range is 40–100. For motor application, the range is 10–95.								

Ground-Fault Protection Pre-Alarm (PAL Ig)

A read request of 10 registers is necessary to read the ground-fault protection pre-alarm parameters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x1A03	6660	R	–	–	INT16U	–	A/E	The MSB gives the activity of the alarm: 0 = On, 1 = Off. The factory setting is 0 (On). The LSB gives the priority of the alarm: N/A, 1, 2, or 3. The factory setting is 2 (medium priority).
0x1A04	6661	–	–	–	–	–	–	Reserved
0x1A05	6662	R	1	%	INT16	40–100	A/E	% of Ig pick-up value. The factory setting is 90.
0x1A06	6663	–	–	–	–	–	–	Reserved
0x1A07	6664	R	1	s	INT16U	1	A/E	Pick-up time delay (fixed to 1 s)
0x1A08	6665	R	1	%	INT16	40–100	A/E	% of Ig drop-out value. The factory setting is 85.
0x1A09	6666	–	–	–	–	–	–	Reserved
0x1A0A	6667	R	1	s	INT16U	1	A/E	Drop-out time delay (fixed to 1 s)
0x1A0B–0x1A0C	6668–6669	–	–	–	–	–	–	Reserved

Earth-Leakage Protection Pre-Alarm (PAL IΔn)

A read request of 10 registers is necessary to read the earth-leakage protection pre-alarm parameters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x1A0D	6670	R	–	–	INT16U	–	E	The MSB gives the activity of the alarm: 0 = On, 1 = Off. The factory setting is 0 (On). The LSB gives the priority of the alarm: N/A, 1, 2, or 3. The factory setting is 2 (medium priority).
0x1A0E	6671	–	–	–	–	–	–	Reserved
0x1A0F	6672	R	1	%	INT16	50–80	E	% of IΔn pick-up value. The factory setting is 90.
0x1A10	6673	–	–	–	–	–	–	Reserved
0x1A11	6674	R	1	s	INT16U	1	E	Pick-up time delay (fixed to 1 s)
0x1A12	6675	R	1	%	INT16	50–80	E	% of IΔn drop-out value. The factory setting is 85.
0x1A13	6676	–	–	–	–	–	–	Reserved
0x1A14	6677	R	1	s	INT16U	1	E	Drop-out time delay (fixed to 1 s)
0x1A15–0x1A16	6678–6679	–	–	–	–	–	–	Reserved

User-Defined Alarms

General Description

EcoStruxure Power Commission software, page 13 enables the configuration of 10 user-defined alarms that can be chosen from a list of 150 predefined alarms.

Each user-defined alarm has a corresponding user-defined alarm number (201–210) and a corresponding alarm code (see next paragraph).

Each alarm has a priority level that manages the alarm display on the FDM121 display:

- no priority = N/A (not affected)
- low priority = 1
- medium priority = 2
- high priority = 3

For more information, refer to DOCA0188** *ComPacT NSX MicroLogic 5/6/7 Electronic Trip Units – User Guide*, page 8.

The settings of the 10 user-defined alarms are in the user-defined alarms registers:

Address	Register	Description
0x1A71–0x1A7C	6770–6781	User-defined alarm 201
0x1A7D–0x1A88	6782–6793	User-defined alarm 202
0x1A89–0x1A94	6794–6805	User-defined alarm 203
0x1A95–0x1AA0	6806–6817	User-defined alarm 204
0x1AA1–0x1AAC	6818–6829	User-defined alarm 205
0x1AAD–0x1AB8	6830–6841	User-defined alarm 206
0x1AB9–0x1AC4	6842–6853	User-defined alarm 207
0x1AC5–0x1AD0	6854–6865	User-defined alarm 208
0x1AD1–0x1ADC	6866–6877	User-defined alarm 209
0x1ADD–0x1AE8	6878–6889	User-defined alarm 210

User-Defined Alarm Record

A read request of 12 registers is necessary to read a user-defined alarm record.

The order and the description of the user-defined alarms records are the same as that of user-defined alarm record 1:

User-Defined Alarm 201								
Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x1A71	6770	R	–	–	INT16U	–	A/E	The MSB gives the activity of the alarm: 0 = On, 1 = Off. The factory setting is 1 (Off). The LSB gives the priority of the alarm: N/A, 1, 2, or 3. The factory setting is N/A (no priority).
0x1A72	6771	R	–	–	INT16U	–	A/E	Measurement identifier ⁽¹⁾
0x1A73	6772	–	–	–	–	–	–	Reserved
0x1A74	6773	R	1	⁽²⁾	INT16	-32767– +32767	A/E	Threshold pick-up value. The factory setting is 0.
0x1A75	6774	–	–	–	–	–	–	Reserved
0x1A76	6775	R	1	s	INT16U	0–3000	A/E	Pick-up time delay. The factory setting is 0.

User-Defined Alarm 201								
Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x1A77	6776	R	1	⁽²⁾	INT16	-32767– +32767	A/E	Threshold drop-out value. The factory setting is 0.
0x1A78	6777	–	–	–	–	–	–	Reserved
0x1A79	6778	R	1	s	INT16	0–3000	A/E	Drop-out time delay. The factory setting is 0.
0x1A7A	6779	R	–	–	INT16U	0–3	A/E	Operator: 0: ≥, 1: ≤, 2: =, 3: ≠
0x1A7B	6780	R	–	–	INT16U	1–1919	–	Alarm code, page 119
0x1A7C	6781	–	–	–	–	–	–	Reserved
(1) The value of the measurement identifier is the register number of the measurement. For example, the measurement identifier of current on phase 1 (I1) is 1016.								
(2) The unit of the threshold depends on the measurement identifier. For example, if the measurement identifier is I1, then the unit is A.								

Predefined Alarms Codes

The following table describes the list of predefined alarms and corresponding codes from which the user can choose the 10 user-defined alarms and configure them with EcoStruxure Power Commission software.

Alarm Code	Alarm Description
1 (0x0000)	Overcurrent instantaneous phase 1
2 (0x0001)	Overcurrent instantaneous phase 2
3 (0x0002)	Overcurrent instantaneous phase 3
4 (0x0003)	Overcurrent instantaneous neutral
5 (0x0004)	Ground-fault current
6 (0x0005)	Undercurrent instantaneous phase 1
7 (0x0006)	Undercurrent instantaneous phase 2
8 (0x0007)	Undercurrent instantaneous phase 3
9 (0x0008)	Overcurrent unbalance phase 1
10 (0x0009)	Overcurrent unbalance phase 2
11 (0x000A)	Overcurrent unbalance phase 3
12 (0x000B)	Overvoltage (phase 1 to neutral)
13 (0x000C)	Overvoltage (phase 2 to neutral)
14 (0x000D)	Overvoltage (phase 3 to neutral)
15 (0x000E)	Undervoltage (phase 1 to neutral)
16 (0x000F)	Undervoltage (phase 2 to neutral)
17 (0x0010)	Undervoltage (phase 3 to neutral)
18 (0x0011)	Overvoltage unbalance (phase 1 to neutral)
19 (0x0012)	Overvoltage unbalance (phase 2 to neutral)
20 (0x0013)	Overvoltage unbalance (phase 3 to neutral)
21 (0x0014)	Over total apparent power
22 (0x0015)	Over total active power
23 (0x0016)	Over total active reverse power
24 (0x0017)	Over total reactive power
25 (0x0018)	Over total reactive reverse power

Alarm Code	Alarm Description
26 (0x0019)	Under total apparent power
27 (0x001A)	Under total active power
28 (0x001B)	Reserved
29 (0x001C)	Under total reactive power
30 (0x001D)	Reserved
31 (0x001E)	Leading power factor (IEEE)
32 (0x001F)	Reserved
33 (0x001E)	Leading or lagging power factor (IEC)
34 (0x0021)	Lagging power factor (IEEE)
35 (0x0022)	Over total harmonic distortion current phase 1
36 (0x0023)	Over total harmonic distortion current phase 2
37 (0x0024)	Over total harmonic distortion current phase 3
38 (0x0025)	Over total harmonic distortion voltage (phase 1 to neutral)
39 (0x0026)	Over total harmonic distortion voltage (phase 2 to neutral)
40 (0x0027)	Over total harmonic distortion voltage (phase 3 to neutral)
41 (0x0028)	Over total harmonic distortion voltage (phase 1– 2)
42 (0x0029)	Over total harmonic distortion voltage (phase 2– 3)
43 (0x002A)	Over total harmonic distortion voltage (phase 3– 1)
44–53 (0x002B–0x0034)	Reserved
54 (0x0035)	Earth-leakage current
55 (0x0036)	Overcurrent (average)
56 (0x0037)	Over maximum current (I1, I2, I3, or neutral)
57 (0x0038)	Undercurrent instantaneous neutral
58–59 (0x0039–0x003A)	Reserved
60 (0x003B)	Undercurrent (average)
61 (0x003C)	Overcurrent demand phase 1
62 (0x003D)	Overcurrent demand phase 2
63 (0x003E)	Overcurrent demand phase 3
64 (0x003F)	Overcurrent demand neutral
65 (0x0040)	Under minimum current (I1, I2, or I3)
66 (0x0041)	Undercurrent demand phase 1
67 (0x0042)	Undercurrent demand phase 2
68 (0x0043)	Undercurrent demand phase 3
69 (0x0044)	Undercurrent demand neutral
70 (0x0045)	Over maximum current unbalance (I1, I2, or I3)
71 (0x0046)	Overvoltage (phase 1– 2)
72 (0x0047)	Overvoltage (phase 2– 3)
73 (0x0048)	Overvoltage (phase 3– 1)
74 (0x0049)	Reserved
75 (0x004A)	Overvoltage (average)
76 (0x004B)	Undervoltage (phase 1– 2)
77 (0x004C)	Undervoltage (phase 2– 3)

Alarm Code	Alarm Description
78 (0x004D)	Undervoltage (phase 3– 1)
79 (0x004E)	Over maximum voltage
80 (0x004F)	Undervoltage (average)
81 (0x0050)	Under minimum voltage
82 (0x0051)	Over maximum voltage unbalance (phases to neutral)
83–85 (0x0052–0x0054)	Reserved
86 (0x0055)	Overvoltage unbalance (phase 1– 2)
87 (0x0056)	Overvoltage unbalance (phase 2– 3)
88 (0x0057)	Overvoltage unbalance (phase 3– 1)
89 (0x0058)	Over maximum voltage unbalance
90 (0x0059)	Phase sequence
91 (0x005A)	Reserved
92 (0x005B)	Under frequency
93 (0x005C)	Over frequency
94–98 (0x005D–0x0061)	Reserved
99 (0x0062)	Over active power demand
100–120 (0x0063–0x0077)	Reserved
121 (0x0078)	Leading $\cos\phi$ (IEEE)
122 (0x0079)	Reserved
123 (0x007A)	Leading or lagging $\cos\phi$ (IEC)
124 (0x007B)	Lagging $\cos\phi$ (IEEE)
125 (0x007C)	Overcurrent thermal image motor
126 (0x007D)	Undercurrent thermal image motor
127–140 (0x007E–0x008B)	Reserved
141 (0x008C)	Overcurrent maximum demand phase 1
142 (0x008D)	Overcurrent maximum demand phase 2
143 (0x008E)	Overcurrent maximum demand phase 3
144 (0x008F)	Overcurrent maximum demand neutral
145 (0x0090)	Lead
146 (0x0091)	Lag
147 (0x0092)	Quadrant 1
148 (0x0093)	Quadrant 2
149 (0x0094)	Quadrant 3
150 (0x0095)	Quadrant 4
151–255 (0x0096–0x00FE)	Reserved
256 (0x00FF)	Contact wear

Protection Parameters

Long-Time Protection Parameters

A read request of 10 registers is necessary to read the long-time protection parameters.

The long-time protection command (command code = 45192) configures the content of the long-time protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2231	8754	R	–	–	INT16U	1–2	A/E	Status: 1 = On, 2 = Inhibit
0x2232	8755	–	–	–	–	–	–	Reserved
0x2233	8756	R-WC	1	A	INT16U	–	A/E	I _r pick-up value. The I _r range depends on the nominal current I _n .
0x2234	8757	–	–	–	–	–	–	Reserved
0x2235	8758	R-WC	1	ms	INT16U	500– 16000	A/E	t _r time delay (distribution application) t _r = 500, 1000, 2000, 4000, 8000, 16000 ms
0x2236	8759	R-WC	1	s	INT16U	5–30	E	Motor class (motor application only) Possible values = 5, 10, 20, 30 s
0x2237	8760	–	–	–	–	–	–	Reserved
0x2238	8761	R-WC	1	–	INT16U	1–2	E	Cool fan (motor application only) 1 = auto, 2 = motor
0x2239– 0x223A	8762– 8763	–	–	–	–	–	–	Reserved

Short-Time Protection Parameters

A read request of 10 registers is necessary to read the short-time protection parameters.

The short-time protection command (command code = 45193) configures the content of the short-time protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x223B	8764	R	–	–	INT16U	1–2	A/E	Status: 1 = On, 2 = Inhibit
0x223C	8765	R-WC	–	–	INT16U	0–1	A/E	Type of protection: 0 = I ² t On, 1 = I ² t Off. For motor application, t _{sd} = 30 ms and I ² t is Off (fixed values).
0x223D	8766	R-WC	10	–	INT16U	⁽¹⁾	A/E	I _{sd} coefficient, adjustable in steps of 5
0x223E	8767	R	1	A	INT16U	–	A/E	I _{sd} pick-up value = (I _r) x (I _{sd} coefficient) / 10
0x223F	8768	R-WC	1	ms	INT16U	0–400	A/E	t _{sd} time delay t _{sd} = 0, 30, 100, 200, 300, 400 ms If t _{sd} = 0 ms, then I ² t must be Off.
0x2240– 0x2244	8769– 8773	–	–	–	–	–	–	Reserved

(1) For distribution application, the range is 15–100. For motor application, the range is 50–130.

Instantaneous Protection Parameters

A read request of 10 registers is necessary to read the instantaneous protection parameters.

The instantaneous protection command (command code = 45194) configures the content of the instantaneous protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2245	8774	R	–	–	INT16U	1–2	A/E	Status: 1 = On, 2 = Inhibit
0x2246	8775	–	–	–	–	–	–	Reserved
0x2247	8776	R-WC	10	–	INT16U	(1)	A/E	li coefficient, adjustable in steps of 5
0x2248	8777	R	1	A	INT16U	–	A/E	li pick-up value = (ln) x (li coefficient) / 10
0x2249– 0x224E	8778– 8783	–	–	–	–	–	–	Reserved

(1) The li coefficient range depends on the circuit breaker size:

- for 100–160 A, the range is 15–150.
- for 250–400 A, the range is 15–120.
- for 630 A, the range is 15–110.

Ground-Fault Protection Parameters

A read request of 10 registers is necessary to read the ground-fault protection parameters.

The ground-fault protection command (command code = 45195) configures the content of the ground-fault protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x224F	8784	R	–	–	INT16U	0–2	A/E	Status: 0 = Off, 1 = On, 2 = Inhibit
0x2250	8785	R-WC	–	–	INT16U	0–1	A/E	Type of protection: 0 = I _{2t} On, 1 = I _{2t} Off For motor application, tg = 0 ms and I _{2t} is Off (fixed values).
0x2251	8786	R-WC	100	–	INT16U	–	A/E	Ig coefficient, adjustable in steps of 5
0x2252	8787	R	1	A	INT16U	–	A/E	Ig pick-up value = (ln) x (Ig coefficient) / 100 If ground-fault protection is set to Off, Ig pick-up value = ln.
0x2253	8788	R-WC	1	ms	INT16U	0–400	A/E	tg time delay tg = 0, 100, 200, 300, 400 ms. If tg = 0 ms, then I _{2t} must be Off.
0x2254– 0x2258	8789– 8793	–	–	–	–	–	–	Reserved

Earth-Leakage Protection Parameters

A read request of 10 registers is necessary to read the earth-leakage protection parameters.

The earth-leakage protection command (command code = 45196) configures the content of the earth-leakage protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2259	8794	R	–	–	INT16U	0–2	E	Status: 0 = Off, 1 = On, 2 = Inhibit
0x225A	8795	–	–	–	–	–	–	Reserved
0x225B	8796	R	1	mA	INT16U	–	E	Earth-leakage current I Δ n. The I Δ n range depends on the nominal current I _n .
0x225C	8797	–	–	–	–	–	–	Reserved
0x225D	8798	R-WC	1	ms	INT16U	0–1000	E	t Δ n time delay t Δ n = 0, 60, 150, 500, 1000 ms If I Δ n = 0.03 mA, then t Δ n = 0 ms.
0x225E– 0x2262	8799– 8803	–	–	–	–	–	–	Reserved

Jam Protection Parameters

A read request of 4 registers is necessary to read the jam protection parameters.

The jam protection is available for motor application only. The jam protection command (command code = 45448) configures the content of the jam protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22C3	8900	R-WC	–	–	INT16U	0–1	E	Status: 0 = Off, 1 = On
0x22C4	8901	R-WC	1–0	–	INT16U	10–80	E	I _{jam} coefficient, adjustable in steps of 1
0x22C5	8902	R	1	A	INT16U	–	E	I _{jam} pick-up value = (I _r) x (I _{jam} coefficient) / 10
0x22C6	8903	R-WC	1	s	INT16U	1–30	E	t _{jam} time delay

Unbalance Protection Parameters

A read request of 4 registers is necessary to read the unbalance protection parameters.

The unbalance protection is available for motor application only. The unbalance protection command (command code = 45450) configures the content of the unbalance protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22C7	8904	R	–	–	INT16U	0–2	E	Status: 0 = Off, 1 = On, 2 = Inhibit
0x22C8	8905	R-WC	1	%	INT16U	10–40	E	I _{unbal} coefficient
0x22C9	8906	R-WC	1	s	INT16U	1–10	E	t _{unbal} time delay
0x22CA	8907	–	–	–	–	–	–	Reserved

Underload Protection Parameters

A read request of 4 registers is necessary to read the underload protection parameters.

The underload protection is available for motor application only. The underload protection command (command code = 45449) configures the content of the underload protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22CB	8908	R-WC	–	–	INT16U	0–1	E	Status: 0 = Off, 1 = On
0x22CC	8909	R-WC	100	–	INT16U	30–90	E	lunderload coefficient, adjustable in steps of 1
0x22CD	8910	R	1	A	INT16U	–	E	lunderload pick-up value = (Ir) x (lunderload) / 100
0x22CE	8911	R-WC	1	s	INT16U	1–200	E	tunderload time delay

Long-Start Protection Parameters

A read request of 4 registers is necessary to read the long-start protection parameters.

The long-start protection is available for motor application only. The long-start protection command (command code = 45451) configures the content of the long-start protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22CF	8912	R-WC	–	–	INT16U	0–1	E	Status: 0 = Off, 1 = On
0x22D0	8913	R-WC	10	–	INT16U	10–80	E	llongstart coefficient, adjustable in steps of 1
0x22D1	8914	R	1	A	INT16U	–	E	llongstart pick-up value = (Ir) x (llongstart coefficient) / 10
0x22D2	8915	R-WC	1	s	INT16U	1–200	E	tlongstart time delay

Neutral Protection Parameters

The neutral protection is only available when system type in register 3314 is 30 or 41, page 139.

A read request of 4 registers is necessary to read the neutral protection parameters.

The neutral protection command (command code = 45197) configures the content of the neutral protection registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22D3	8916	R	–	–	INT16U	0–2	A/E	Status: 0 = Off, 1 = On, 2 = Inhibit ⁽¹⁾
0x22D4	8917	R-WC	–	–	INT16U	0–3	A/E	Neutral coefficient pickup value 0 = Off 1 = 0.5 2 = 1.0 3 = OSN
0x22D5	8918	R	1	A	INT16U	0–32766	A/E	Ir pick-up value
0x22D6	8919	R	1	A	INT16U	0–32766	A/E	Isd pick-up value
(1) For 40 A IEC and 60 A UL circuit breakers, the user cannot set up the neutral coefficient pickup value to 0.5.								

Thermal Memory Inhibit Parameter

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22E1	8930	R	–	–	INT16U	1–2	A/E	Status: 1 = On, 2 = Inhibit

Configuration of the SDx Module

Output 1

A read request of 3 registers is necessary to read the output 1 parameters.

The user can check the status and the validity of output 1 at register 8857, page 116.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2648	9801	R	1	–	INT16U	0–4	A/E	Output mode 0 = normal mode 1 = latched mode 2 = time delayed mode 3 = closed forced mode 4 = open forced mode
0x2649	9802	R	1	s	INT16U	1–360	A/E	Delay (if the output mode is set to 2). The factory setting is 1 s.
0x264A	9803	R	1	–	INT16U	0–65535	A/E	Alarm identifier (201–210, 1013, 1014, 1015). The alarm identifier is set to 0 if there is no alarm.

Output 2

A read request of 3 registers is necessary to read the output 2 parameters.

The user can check the status and the validity of output 2 at register 8857, page 116.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x264F	9808	R	1	–	INT16U	0–4	A/E	Output mode 0 = normal mode 1 = latched mode 2 = time delayed mode 3 = closed forced mode 4 = open forced mode
0x2650	9809	R	1	s	INT16U	1–360	A/E	Delay (if the output mode is set to 2). The factory setting is 1 s.
0x2651	9810	R	1	–	INT16U	0–65535	A/E	Alarm identifier (201–210, 1013, 1014, 1015). The alarm identifier is set to 0 if there is no alarm.

Measurement Parameters

System Type

The setup ENVT (External Neutral Voltage Tap) presence command (command code = 46472) configures the content of the system type register.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0CF1	3314	R-WC	–	–	INT16U	30–41	A/E	System type

Determining system type:

If...	Then...	Result
the system type is 3-pole circuit breaker with external neutral current transformer and without external neutral voltage tap	system type = 30	- Measurements of the phase-to-phase voltages are available. - Measurements of the phase-to-neutral voltages are not available. - Measurement of the neutral current is available. 3 wattmeters method is not possible.
the system type is 3-pole circuit breaker without external neutral current transformer and without external neutral voltage tap	system type = 31	- Measurements of the phase-to-phase voltages are available. - Measurements of the phase-to-neutral voltages are not available. - Measurement of the neutral current is not available. 3 wattmeters method is not possible.
the system type is 3-pole circuit breaker without external neutral current transformer and with external neutral voltage tap	system type = 40	- Measurements of the phase-to-phase voltages are available. - Measurements of the phase-to-neutral voltages are available. - Measurement of the neutral current is not available. 3 wattmeters method is possible.
the system type is 3-pole circuit breaker with external neutral current transformer and external neutral voltage tap, or if the system type is 4-pole circuit breaker	system type = 41	- Measurements of the phase-to-phase voltages are available. - Measurements of the phase-to-neutral voltages are available. - Measurement of the neutral current is available. 3 wattmeters method is possible.

Quadrant Total

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x08C1	2242	R	–	–	INT16U	1–4	E	Quadrant total
0x08C2	2243	R	–	–	INT16U	0–1	E	0 = lead 1 = lag

Power Flow Sign

The power flow sign command (command code = 47240) configures the content of the power flow sign register.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0CF3	3316	R-WC	–	–	INT16U	0–1	E	Power flow sign: 0 = the active power flows from upstream (top) to downstream (bottom) (factory setting). 1 = the active power flows from downstream (bottom) to upstream (top).

Power Factor Sign

The power factor sign configuration command (command code = 47241) configures the content of the power factor sign register.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0CF5	3318	R-WC	–	–	INT16U	0–2	E	Sign convention for the power factor and the fundamental power factor ($\cos\phi$): 0 = IEC convention 2 = IEEE convention (factory setting)

Energy Accumulation Mode

The energy accumulation mode configuration command (command code = 47242) configures the content of the energy accumulation mode register.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0CFB	3324	R-WC	–	–	INT16U	0–1	E	Energy accumulation mode: 0 = absolute accumulation (factory setting) $E_p = E_{pIn} + E_{pOut}$ $E_q = E_{qIn} + E_{qOut}$ 1 = signed accumulation $E_p = E_{pIn} - E_{pOut}$ $E_q = E_{qIn} - E_{qOut}$

Demand Time

The current demand configuration command (command code 47243) configures the content of register 3352.

The power demand configuration command (command code 47244) configures the content of registers 3354 and 3355.

For more information regarding the demand calculation method, refer to DOCA0188•• *ComPacT NSX MicroLogic 5/6/7 Electronic Trip Units – User Guide*, page 8.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0D17	3352	R-WC	–	min	INT16U	5–60	E	Duration of the current demand calculation window, adjustable in steps of 1 minute. The factory setting is 15 minutes.
0x0D19	3354	R-WC	–	–	INT16U	0–5	E	Power demand calculation method (window type): 0 = sliding (factory setting) 2 = block 5 = synchronized to communication
0x0D1A	3355	R-WC	–	min	INT16U	5–60	E	Duration of the power demand calculation window, adjustable in steps of 1 minute. The factory setting is 15 minutes.

Nominal Voltage

The setup nominal voltage Vn display command (command code = 47245) configures the content of the nominal voltage register.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x258F	9616	R-WC	1	V	INT16U	0–65535	A/E	Nominal voltage Vn. Factory setting = 400 V

Time-Stamped Information

General Description

The time-stamped information enables the user to know all the dates relative to important information such as previous protection settings and minimum/maximum values of currents, voltages, and system frequency.

The time-stamped information table describes:

- the previous protection setup parameters and corresponding dates
- the minimum and maximum values of voltage measurements and corresponding dates
- the maximum values of current measurements and corresponding dates
- the minimum and maximum system frequencies and corresponding dates
- the peak demand of current and power and corresponding dates

A read request of 100 registers is necessary to read the time-stamped previous protection registers (29600–29699).

Previous Short-Time Protection Setup

Address	Register	RW	Unit	Type	Range	A/E	X	Description
0x73AF	29616	R	–	INT16U	⁽¹⁾	A/E	10	Previous Isd coefficient pick-up value.
73x0B0– 73x0B2	29617– 29619	R	–	ULP DATE	–	A/E	–	Date and time of the previous setup.
73x0B3	29620	R	ms	INT16U	0– 400	A/E	1	Previous tsd time delay tsd = 0, 100, 200, 300, 400 ms If tsd = 0 ms, I2t must be off.
73x0B4– 73x0B6	29621– 29623	R	–	ULP DATE	–	A/E	–	Date and time of the previous setup.
73x0B7	29624	R	–	INT16U	0– 1	A/E	–	Previous type of protection: 0 = I2t On, 1 = I2t Off
73x0B8– 0x73BA	29625– 29627	R	–	ULP DATE	–	A/E	–	Date and time of the previous setup.

⁽¹⁾ The Isd coefficient range depends on the:

- the application:
 - for distribution application, the range is 15–100, adjustable in steps of 5.
 - for motor application, the range is 50–130, adjustable in steps of 5.
- the MicroLogic trip unit rotary switch if it exists.

Previous Instantaneous Protection Setup

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73BB	29628	R	10	–	INT16U	(1)	A/E	Previous I _i coefficient pick-up value
0x73BC– 0x73BE	29629– 29631	R	–	–	ULP DATE	–	A/E	Date and time of the previous setup.

(1) The I_i coefficient range depends on the circuit breaker size:

- for 100–160 A, the range is 15–150.
- for 250–400 A, the range is 15–120.
- for 630 A, the range is 15–110.

Previous Ground-Fault Protection Setup

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x73BF	29632	R	10-0	ms	INT16U	(1)	A/E	Previous I _g coefficient pick-up value.
0x73C0– 0x73C2	29633– 29635	R	–	–	ULP DATE	–	A/E	Date and time of the previous setup.
0x73C3	29636	R	1	ms	INT16U	0–400	A/E	Previous t _g time delay t _g = 0, 100, 200, 300, 400 ms
0x73C4– 0x73C6	29637– 29639	R	–	–	ULP DATE	–	A/E	Date and time of the previous setup.
0x73C7	29640	R	–	–	INT16U	0–1	A/E	Previous type of protection: 0 = I ₂ t On, 1 = I ₂ t Off
0x73C8– 0x73CA	29641– 29643	R	–	–	ULP DATE	–	A/E	Date and time of the previous setup.

(1) The I_g coefficient range depends on the nominal current I_n and the MicroLogic trip unit rotary switch if it exists.
For example, 0 (protection off) or 0.40 I_n to I_n in steps of 0.05.

Previous Earth-Leakage Protection Setup

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x73CB	29644	R	1	mA	INT16U	–	E	Previous I _{Δn} pick-up value. The I _{Δn} depends on the nominal current I _n .
0x73CC– 0x73CE	29645– 29647	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73CF	29648	R	1	ms	INT16U	0–1000	E	Previous t _{Δn} time delay t _{Δn} = 0, 60, 150, 500, 1000 ms If I _{Δn} = 0.03 mA, then T _{Δn} = 0 ms.
0x73D0– 0x73D2	29649– 29651	R	–	–	ULP DATE	–	E	Date and time of the previous setup.

Previous Jam Protection Setup

The jam protection is available for motor application only.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73D3	29652	R	–	–	INT16U	0–2	E	Previous setup status: 0 = Off, 1 = On
0x73D4– 0x73D6	29653– 29655	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73D7	29656	R	10	–	INT16U	10–80	E	Previous ljam coefficient pick-up value
0x73D8– 0x73DA	29657– 29659	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73DB	29660	R	1	s	INT16U	1–30	E	Previous tjam time delay
0x73DC– 0x73DE	29661– 29663	R	–	–	ULP DATE	–	E	Date and time of the previous setup.

Previous Unbalance Protection Setup

The unbalance protection is available for motor application only.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73DF	29664	R	1	%	INT16U	10–40	E	Previous unbalance coefficient pick-up value
0x73E0– 0x73E2	29665– 29667	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73E3	29668	R	1	s	INT16U	1–10	E	Previous tunbal time delay
0x73E4– 0x73E6	29669– 29671	R	–	–	ULP DATE	–	E	Date and time of the previous setup.

Previous Underload Protection Setup

The underload protection is available for motor application only.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73E7	29672	R	–	–	INT16U	0–2	E	Previous setup status: 0 = Off, 1 = On
0x73E8– 0x73EA	29673– 29675	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73EB	29676	R	100	–	INT16U	30–90	E	Previous lunderload coefficient pick-up value
0x73EC– 0x73EE	29677– 29679	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73EF	29680	R	1	s	INT16U	1–200	E	Previous tunderload time delay
0x73F0– 0x73F2	29681– 29683	R	–	–	ULP DATE	–	E	Date and time of the previous setup.

Previous Long-Start Protection Setup

The long-start protection is available for motor application only.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73F3	29684	R	–	–	INT16U	0–2	E	Previous setup status: 0 = Off, 1 = On
0x73F4–	29685–	R	–	–	ULP DATE	–	E	Date and time of the previous setup.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73F6	29687							
0x73F7	29688	R	10	–	INT16U	10–50	E	Previous Ilongstart coefficient pick-up value
0x73F8– 0x73FA	29689– 29691	R	–	–	ULP DATE	–	E	Date and time of the previous setup.
0x73FB	29692	R	1	s	INT16U	1–30	E	Previous tlongstart time delay
0x73FC– 0x73FE	29693– 29695	R	–	–	ULP DATE	–	E	Date and time of the previous setup.

Previous Neutral Protection Setup

The neutral protection is only available when system type in register 3314 is 30 or 41 Measurement Parameters, page 139.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x73FF	29696	R	–	–	INT16U	0–3	A/E	Previous neutral coefficient pick-up value: 0 = Off 1 = 0.5 2 = 1.0 3 = OSN
0x7400– 0x7402	29697– 29699	R	–	–	ULP DATE	–	A/E	Date and time of the previous setup.

Minimum/Maximum V12 Voltage Measurements

A read request of 48 registers is necessary to read the time-stamped minimum/maximum values of voltage, current, and frequency registers (29780–29827).

Register = 0 if voltage < 25 V.

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x7453	29780	R	1	V	INT16U	0–850	E	Minimum of RMS phase-to-phase voltage V12
0x7454– 0x7456	29781– 29783	R	–	–	ULP DATE	–	E	Date and time.
0x7457	29784	R	1	V	INT16U	0–850	E	Maximum of RMS phase-to-phase voltage V12
0x7458– 0x745A	29785– 29787	R	–	–	ULP DATE	–	E	Date and time.

Minimum/Maximum V23 Voltage Measurements

Register = 0 if voltage < 25 V.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x745B	29788	R	1	V	INT16U	0–850	E	Minimum of RMS phase-to-phase voltage V23
0x745C– 0x745E	29789– 29791	R	–	–	ULP DATE	–	E	Date and time.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x745F	29792	R	1	V	INT16U	0–850	E	Maximum of RMS phase-to-phase voltage V23
0x7460– 0x7462	29793– 29795	R	–	–	ULP DATE	–	E	Date and time.

Minimum/Maximum V31 Voltage Measurements

Register = 0 if voltage < 25 V.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x7463	29796	R	1	V	INT16U	0–850	E	Minimum of RMS phase-to-phase voltage V31
0x7464– 0x7466	29797– 29799	R	–	–	ULP DATE	–	E	Date and time.
0x7467	29800	R	1	V	INT16U	0–850	E	Maximum of RMS phase-to-phase voltage V31
0x7468– 0x746A	29801– 29803	R	–	–	ULP DATE	–	E	Date and time.

Maximum I1 Current Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x746B	29804	R	1	A	INT16U	0–20xIn	A/E	Maximum of RMS current on phase 1: I1
0x746C– 0x746E	29805– 29807	R	–	–	ULP DATE	–	A/E	Date and time.

Maximum I2 Current Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x746F	29808	R	1	A	INT16U	0–20xIn	A/E	Maximum of RMS current on phase 2: I2
0x7470– 0x7472	29809– 29811	R	–	–	ULP DATE	–	A/E	Date and time.

Maximum I3 Current Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x7473	29812	R	1	A	INT16U	0–20xIn	A/E	Maximum of RMS current on phase 3: I3
0x7474– 0x7476	29813– 29815	R	–	–	ULP DATE	–	A/E	Date and time.

Maximum IN Current Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x7477	29816	R	1	A	INT16U	0–20xIn	A/E	Maximum of RMS current on neutral: IN
0x7478– 0x747A	29817– 29819	R	–	–	ULP DATE	–	A/E	Date and time.

Minimum System Frequency

When the MicroLogic trip unit cannot calculate the frequency, it returns Not Evaluated = 32768 (0x8000).

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x747B	29820	R	10	Hz	INT16U	150–4400	A/E	Minimum of system frequency
0x747C– 0x747E	29821– 29823	R	–	–	ULP DATE	–	A/E	Date and time.

Maximum System Frequency

When the MicroLogic trip unit cannot calculate the frequency, it returns Not Evaluated = 32768 (0x8000).

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x747F	29824	R	10	Hz	INT16U	150–4400	E	Maximum of system frequency
0x7480– 0x7482	29825– 29827	R	–	–	ULP DATE	–	E	Date and time.

I1 Peak Demand Measurement

A read request of 20 registers is necessary to read the time-stamped peak demand of current and power (29828–29847).

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x7483	29828	R	1	A	INT16U	0–20xIn	E	I1 peak demand
0x7484– 0x7486	29829– 29831	R	–	–	ULP DATE	–	E	Date and time.

I2 Peak Demand Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x7487	29832	R	1	A	INT16U	0–20xIn	E	I2 peak demand
0x7488– 0x748A	29833– 29835	R	–	–	ULP DATE	–	E	Date and time.

I3 Peak Demand Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x748B	29836	R	1	A	INT16U	0–20xIn	E	I3 peak demand
0x748C– 0x748E	29837– 29839	R	–	–	ULP DATE	–	E	Date and time.

IN Peak Demand Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x748F	29840	R	1	A	INT16U	0–20xIn	E	IN peak demand
0x7490– 0x7492	29841– 29843	R	–	–	ULP DATE	–	E	Date and time.

P Peak Demand Measurement

Address	Register	R-W	X	Unit	Type	Range	A/E	Description
0x7493	29844	R	10	kW	INT16	-30000– +30000	E	P peak demand
0x7494– 0x7496	29845– 29847	R	–	–	ULP DATE	–	E	Date and time.

Maintenance Indicators

Time of Use Counter

The time of use counter reports the time of use of the circuit breaker. The time of use is written in the EEPROM every hour. If the time of use counter reaches the maximum value 4 294 967 295 and a new time of use event occurs, the counter is reset to 0.

A read request of 2 registers is necessary to read the time of use counter.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x749A– 0x749B	29851– 29852	R	1	Hour	INT32U	0–4 294 967 295	A/E	Time of use counter

Rate of Wear Counter

The rate of wear counter reports the percentage of the circuit breaker contact use.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x749C	29853	R	1	%	INT16U	0–32766	A/E	Rate of wear contact 0% = The circuit breaker contact is new. > 100% = The circuit breaker contact must be changed.

Boot Counter

The boot counter reports the number of cold starts (power-up) and the number of warm starts (software reset of the MicroLogic trip unit).

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x749D	29854	R	1	–	INT16U	0–32766	A/E	Boot counter

EEPROM Writing Counter

The EEPROM writing counter reports the number of storages of the energy measurement in the EEPROM. The energy measurement is written in the EEPROM every hour. If the EEPROM writing counter reaches the maximum value 4 294 967 295 and a new EEPROM writing event occurs, the EEPROM writing counter is reset to 0.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x749E– 0x749F	29855– 29856	R	1	–	INT32U	0–4 294 967 295	A/E	Counter of EEPROM writing

Load Profile Counters

The load profile counters report the number of hours for each range of current in the MicroLogic trip unit. If the load profile counters reach the maximum value 4 294 967 295 and a new load profile event occurs, the load profile counters are reset to 0.

A read request of 8 registers is necessary to read the load profile counters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x74B7– 0x74B8	29880– 29881	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours current was 0-49% of nominal range
0x74B9– 0x74BA	29882– 29883	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours current was 50-79% of nominal range
0x74BB– 0x74BC	29884– 29885	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours current was 80-89% of nominal range
0x74BD– 0x74BE	29886– 29887	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours current was 90-100% of nominal range

Temperature Profile Counters

The temperature profile counters report the number of hours for each range of temperature in the MicroLogic trip unit. If the temperature profile counters reach the maximum value 4 294 967 295 and a new temperature profile event occurs, the temperature profile counters are reset to 0.

A read request of 12 registers is necessary to read the temperature profile counters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x74C1– 0x74C2	29890– 29891	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours temperature was < -30 °C (-22 °F)
0x74C3– 0x74C4	29892– 29893	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours temperature was in the -30–+59 °C range (22–138.2 °F)
0x74C5– 0x74C6	29894– 29895	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours temperature was in the +60–+74 °C range (140–165.2 °F)
0x74C7– 0x74C8	29896– 29897	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours temperature was in the +75–+89 °C range (167–192.2 °F)
0x74C9– 0x74CA	29898– 29899	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours temperature was in the +90–+99 °C range (194–210.2 °F)
0x74CB– 0x74CC	29900– 29901	R	1	Hour	INT32U	0–4 294 967 295	A/E	Hours temperature was > +100 °C (212 °F)

Protection Trips Counters

The protection trips counters report the number of protection trips for each type of protection: long-time, short-time, instantaneous, ground-fault, earth-leakage jam, unbalance, long-start, and underload protections.

The protection trips counters stop incrementing when they reach the maximum value 10000.

A read request of 9 registers is necessary to read the protection trips counters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x74D5	29910	R	1	–	INT16U	0–10000	A/E	Number of long-time protection trips
0x74D6	29911	R	1	–	INT16U	0–10000	A/E	Number of short-time protection trips
0x74D7	29912	R	1	–	INT16U	0–10000	A/E	Number of instantaneous protection trips

Address	Register	RW	X	Unit	Type	Range	A/E	Description
								(including integrated instantaneous protection, instantaneous with earth-leakage protection, and reflex protection)
0x74D8	29913	R	1	–	INT16U	0–10000	A/E	Number of ground-fault protection trips
0x74D9	29914	R	1	–	INT16U	0–10000	E	Number of earth-leakage protection trips
0x74DA	29915	R	1	–	INT16U	0–10000	A/E	Number of jam protection trips
0x74DB	29916	R	1	–	INT16U	0–10000	A/E	Number of unbalance protection trips
0x74DC	29917	R	1	–	INT16U	0–10000	A/E	Number of long-start protection trips
0x74DD	29918	R	1	–	INT16U	0–10000	A/E	Number of underload protection trips

Alarms Counters

The alarms counters report the number of occurrences of the alarms. When an alarm is configured, the associated counter is set to 0. The alarm counters stop incrementing when they reach the maximum value 10000.

A read request of 13 registers is necessary to read the alarms counters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x74F3	29940	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 201
0x74F4	29941	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 202
0x74F5	29942	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 203
0x74F6	29943	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 204
0x74F7	29944	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 205
0x74F8	29945	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 206
0x74F9	29946	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 207
0x74FA	29947	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 208
0x74FB	29948	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 209
0x74FC	29949	R	1	–	INT16U	0–10000	A/E	Counter of user-defined alarm 210
0x74FD	29950	R	1	–	INT16U	0–10000	A/E	Counter of pre-alarm Ir
0x74FE	29951	R	1	–	INT16U	0–10000	A/E	Counter of pre-alarm Ig
0x74FF	29952	R	1	–	INT16U	0–10000	A/E	Counter of pre-alarm IΔn

Maintenance Operation Counters

The maintenance operation counters report the number of some maintenance operations. The maintenance operation counters stop incrementing when they reach the maximum value 10000.

A read request of 7 registers is necessary to read the maintenance operation counters.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x751B	29980	R	1	–	INT16U	0–10000	A/E	Counter of MicroLogic trip unit locking pad activation
0x751C	29981	R	1	–	INT16U	0–10000	A/E	Counter of connections of the USB maintenance interface
0x751D	29982	R	1	–	INT16U	0–10000	A/E	Counter of ground-fault test operation (using MicroLogic keypad only)
0x751E	29983	R	1	–	INT16U	0–10000	E	Counter of earth-leakage test operation

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x751F	29984	R	1	–	INT16U	0–10000	A/E	Counter of ZSI (Zone Selective Interlocking) test operation
0x7520	29985	R	1	–	INT16U	0–10000	A/E	Counter of numerical injection test operation
0x7521	29986	R	1	–	INT16U	0–10000	A/E	Counter of reset Firmware command

Miscellaneous

Current Date and Time

A read request of 3 registers is necessary to read the current date.

The set absolute time command (command code 769) configures the content of the current date registers.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0BB7– 0x0BB9	3000– 3002	R-WC	–	–	ULP DATE	–	A/E	Current date and time

Temperature

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x2292	8851	R	1	°C	INT16	-30– +120	A/E	Temperature of the MicroLogic trip unit

Time Remaining Until Long-Time Tripping

Time remaining until long-time tripping is evaluated every second. If another protection is tripped, then time remaining until long-time tripping continues to be evaluated.

Time remaining until long-time tripping = 32768 (0x8000) if

- long-time protection is already tripped, or
- time remaining until long-time tripping is below 1 s, or
- no default is detected by long-time protection.

If time remaining until long-time tripping is > 7200 s, time remaining until long-time tripping = 7200 s.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22A0	8865	R	1	s	INT16U	1–7200	A/E	Time remaining until long-time tripping

Phase Rotation

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x22A7	8872	R	1	–	INT16U	0–1	E	0 = 123 phase sequence 1 = 132 phase sequence

Failure Status

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x72CD	29390	R	–	–	INT16U	–	A/E	–	Failure status
							A/E	0	Reserved
							A/E	1	STOP (internal failure)

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
									0 = No internal failure 1 = Internal failure
							A/E	2	ERROR (internal failure) 0 = No internal failure 1 = Internal failure
							A/E	3–15	Reserved

NOTE: In the case of a STOP event, it is mandatory to replace the MicroLogic trip unit. In the case of an ERROR event, it is advised to replace the MicroLogic trip unit (the core protection functions still work but it is preferable to replace the MicroLogic trip unit).

MicroLogic Trip Unit Rotary Switches

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x7525	29990	R	1	–	INT16U	1–9	A/E	Position of MicroLogic trip unit rotary switch 1 (I _r)
0x7526	29991	R	1	–	INT16U	1–9	A/E	Position of MicroLogic trip unit rotary switch 2 (I _{sd} , I _g /I _{Δn})

MicroLogic Trip Unit Locking Pad Status

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x7527	29992	R	1	–	INT16U	0–1	A/E	0 = MicroLogic trip unit locking pad open 1 = MicroLogic trip unit locking pad closed

Auxiliary 24 Vdc Power Supply

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x7528	29993	R	1	–	INT16U	0–1	A/E	0 = auxiliary 24 Vdc power supply is not present. 1 = auxiliary 24 Vdc power supply is present.

MicroLogic Trip Unit LED

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x7534	30005	R	–	–	INT16U	–	A/E	–	MicroLogic trip unit LED
							A/E	0	Ready LED 0 = not ready (LED is not blinking). 1 = ready (LED is blinking).
							A/E	1	Pre-alarm LED (distribution application only)

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
									0 = pre-alarm is not active (LED is steady off). 1 = pre-alarm is active (LED is steady on).
							A/E	2	Overload LED 0 = overload is not active (LED is steady off). 1 = overload is active (LED is steady on).
							A/E	3–15	Reserved

MicroLogic 5A/E, 6A/E, and 7E Trip Unit Commands

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List of MicroLogic Trip Unit Commands and Error Codes

List of Commands

The following table lists the available MicroLogic trip unit commands, their corresponding command codes and user profiles. Follow the command execution procedures, page 38.

Command	Command code	User profile
Long-time protection, page 158	45192	Administrator
Short-time protection, page 158	45193	Administrator
Instantaneous protection, page 159	45194	Administrator
Ground-fault protection, page 159	45195	Administrator
Earth-leakage protection, page 160	45196	Administrator
Neutral protection, page 160	45197	Administrator
Jam protection, page 160	45448	Administrator
Underload protection, page 161	45449	Administrator
Unbalance protection, page 161	45450	Administrator
Long-start protection, page 162	45451	Administrator
Acknowledge a latched output, page 163	46216	Administrator or Operator
Acknowledge a trip, page 163	46217	Administrator
Set up ENVT presence, page 164	46472	Administrator
Reset minimum/maximum, page 164	46728	Administrator or Operator
Start/stop synchronization, page 165	46729	Administrator or Operator
Power flow sign configuration, page 165	47240	Administrator
Power factor sign configuration, page 166	47241	Administrator
Energy accumulation mode configuration, page 166	47242	Administrator
Current demand configuration, page 167	47243	Administrator
Power demand configuration, page 167	47244	Administrator
Set up nominal voltage Vn display, page 167	47245	Administrator

Error Codes

Error codes generated by MicroLogic trip units are the generic error codesResult of the Command, page 40.

MicroLogic Trip Unit Protection Commands

Long-Time Protection

The user can read the long-time protection parameters from registers 8754 to 8763 Protection Parameters, page 134.

To set the long-time protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45192	A/E	Command code = 45192
0x1F40	8001	–	–	INT16U	18	A/E	Number of parameters (bytes) = 18
0x1F41	8002	–	–	INT16U	5121	A/E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET- STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	1	A	INT16U	–	A/E	I _r pick-up value. The I _r range depends on the nominal current I _n and on the position of the MicroLogic trip unit rotary switch 1 (I _r).
0x1F46	8007	1	ms	INT16U	500– 16000	A/E	t _r time delay (distribution application only) t _r = 500, 1000, 2000, 4000, 8000, 16000 ms
0x1F47	8008	–	–	INT16U	5–30	A/E	Motor class (motor application only) Possible values = 5, 10, 20, 30 s
0x1F48	8009	–	–	INT16U	1–2	A/E	Cool fan (motor application only) 1 = auto, 2 = motor

Short-Time Protection

The user can read the short-time protection parameters from registers 8764 to 8773 Short-Time Protection Parameters, page 134.

To set the short-time protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45193	A/E	Command code = 45193
0x1F40	8001	–	–	INT16U	16	A/E	Number of parameters (bytes) = 16
0x1F41	8002	–	–	INT16U	5121	A/E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET- STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	1– 0	–	INT16U	⁽¹⁾	A/E	I _{sd} coefficient, adjustable in steps of 5. I _{sd} pick-up value = (I _r) x (I _{sd} coefficient) / 10
0x1F46	8007	1	ms	INT16U	0–400	A/E	t _{sd} time delay t _{sd} = 0, 100, 200, 300, 400 ms If t _{sd} = 0 ms, then I ₂ t must be Off.
0x1F47	8008	–	–	INT16U	0–1	A/E	Type of protection: 0 = I ₂ t On, 1 = I ₂ t Off For motor application, t _{sd} = 0 ms and I ₂ t is Off (fixed values).

(1) For distribution application, the range is 15–100. For motor application, the range is 50–130.

Instantaneous Protection

The user can read the instantaneous protection parameters from registers 8774 to 8783 *Instantaneous Protection Parameters*, page 135.

To set the instantaneous protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45194	A/E	Command code = 45194
0x1F40	8001	–	–	INT16U	12	A/E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	A/E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	10	–	INT16U	⁽¹⁾	A/E	li coefficient, adjustable in steps of 5. li pick-up value = (ln) x (li coefficient) / 10
(1) The li coefficient range depends on the circuit breaker size: - for 100–160 A, the range is 15–150. - for 250–400 A, the range is 15–120. - for 630 A, the range is 15–110.							

Ground-Fault Protection

The user can read the ground-fault protection parameters from registers 8784 to 8793 *Ground-Fault Protection Parameters*, page 135.

To set the ground-fault protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45195	A/E	Command code = 45195
0x1F40	8001	–	–	INT16U	16	A/E	Number of parameters (bytes) = 16
0x1F41	8002	–	–	INT16U	5121	A/E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	10- 0	–	INT16U	⁽¹⁾	A/E	Ig coefficient, adjustable in steps of 5. The value is defined by the position of the MicroLogic trip unit rotary switch 2 (Ig). Ig pick-up value = (ln) x (Ig coefficient) / 100
0x1F46	8007	1	ms	INT16U	0–400	A/E	tg time delay tg = 0, 100, 200, 300, 400 ms If tg = 0 ms, then I ² t must be Off.
0x1F47	8008	–	–	INT16U	0–1	A/E	Type of protection: 0 = I ² t On, 1 = I ² t Off For motor application, tg = 0 ms and I ² t is Off (fixed values).
(1) The Ig coefficient depends on the sensor rating ln and the MicroLogic trip unit rotary switch if it exists. For example, 0 (protection Off) or 0.40 ln to ln in steps of 0.05. The Ig coefficient cannot be set using the command interface when the MicroLogic rotary switch position Ig is Off.							

Earth-Leakage Protection

The user can read the earth-leakage protection parameters from registers 8794 to 8803 Earth-Leakage Protection Parameters, page 135.

To set the earth-leakage protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45196	E	Command code = 45196
0x1F40	8001	–	–	INT16U	14	E	Number of parameters (bytes) = 14
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	1	mA	INT16U	0–FFFF	E	Reserved
0x1F46	8007	1	ms	INT16U	0–1000	E	t _{Δn} time delay t _{Δn} = 0, 60, 150, 500, 1000 ms If I _{Δn} = 0.03 mA, then t _{Δn} = 0 ms

Neutral Protection

The neutral protection is only available when system type in register 3314 is 30 or 41 Measurement Parameters, page 139.

The user can read the neutral protection parameters from registers 8916 to 8919 Neutral Protection Parameters, page 137.

To set the neutral protection registers, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45197	E	Command code = 45197
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–3	A/E	Neutral coefficient pickup value 0 = Off 1 = 0.5 2 = 1.0 3 = OSN

Jam Protection

The jam protection is available for motor application only.

The user can read the jam protection parameters from registers 8900 to 8903 Jam Protection Parameters, page 136.

To set the jam protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45448	E	Command code = 45448
0x1F40	8001	–	–	INT16U	16	E	Number of parameters (bytes) = 16
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	E	Activation: 0 = Off, 1 = On
0x1F46	8007	1- 0	–	INT16U	10–80	E	ljam coefficient, adjustable in steps of 1. ljam pick-up value = (Ir) x (ljam coefficient) / 10
0x1F47	8008	1	s	INT16U	1–30	E	tjam time delay

Underload Protection

The underload protection is available for motor application only.

The user can read the underload protection parameters from registers 8908 to 8911 [Underload Protection Parameters](#), page 136.

To set the underload protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45449	E	Command code = 45449
0x1F40	8001	–	–	INT16U	16	E	Number of parameters (bytes) = 16
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	E	Activation: 0 = Off, 1 = On
0x1F46	8007	10- 0	–	INT16U	30–90	E	lunderload coefficient, adjustable in steps of 1. lunderload pick-up value = (Ir) x (lunderload) / 100
0x1F47	8008	1	s	INT16U	1–200	E	tunderload time delay

Unbalance Protection

The unbalance protection is available for motor application only.

The user can read the unbalance protection parameters from registers 8904 to 8907 [Unbalance Protection Parameters](#), page 136.

To set the unbalance protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45450	E	Command code = 45450
0x1F40	8001	–	–	INT16U	14	E	Number of parameters (bytes) = 14
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	1	%	INT16U	10–40	E	lunbal coefficient
0x1F46	8007	1	s	INT16U	1–10	E	tunbal time delay

Long-Start Protection

The long-start protection is available for motor application only.

The user can read the underload protection parameters from registers 8912 to 8915 Long-Start Protection Parameters, page 137.

To set the long-start protection parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	45451	E	Command code = 45451
0x1F40	8001	–	–	INT16U	16	E	Number of parameters (bytes) = 16
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	E	Activation: 0 = Off, 1 = On
0x1F46	8007	1- 0	–	INT16U	10–80	E	llongstart coefficient, adjustable in steps of 1. llongstart pick-up value = (Ir) x (llongstart coefficient) / 10
0x1F47	8008	1	s	INT16U	1–200	E	llongstart time delay

Event Commands

Acknowledge a Latched Output

The user can read the SDx module outputs parameters from registers 9801 to 9810 Configuration of the SDx Module, page 138.

To acknowledge a latched output, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	46216	A/E	Command code = 46216
0x1F40	8001	–	–	INT16U	12	A/E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	A/E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator or Operator user profile password
0x1F45	8006	–	–	INT16U	1–2	A/E	1 = relay 1, 2 = relay 2

Acknowledge a Trip

To acknowledge a trip, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	46217	A/E	Command code = 46217
0x1F40	8001	–	–	INT16U	10	A/E	Number of parameters (bytes) = 10
0x1F41	8002	–	–	INT16U	5121	A/E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password

Measurement Configuration Commands

Set Up ENVT Presence

The user can read the ENVT (External Neutral Voltage Tap) presence parameters at register 3314. See [System Type](#), page 139.

To set up ENVT presence, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	46472	E	Command code = 46472
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	E	0 = ENVT is not present. 1 = ENVT is present.

Reset Minimum/Maximum

The reset minimum/maximum command resets the minimum values of real-time measurements (registers 1300 to 1599) and the maximum values of real-time measurements (registers 1600 to 1899). See [Minimum/Maximum Values of Real-Time Measurements](#), page 105.

The reset minimum/maximum command resets the energy measurements (registers 2000 to 2025). See [Energy Measurements](#), page 107.

The reset minimum/maximum command resets the peak demand measurements (registers 2200 to 2237). See [Demand Measurements](#), page 109.

The user can read the minimum and maximum values of current, voltage, and frequency measurements and the corresponding dates from registers 29780 to 29827. See [Minimum/Maximum V12 Voltage Measurements](#), page 145.

The user can read the dates of the reset minimum/maximum command from registers 2900 to 2929. See [Minimum/Maximum Measurements Reset Time](#), page 111.

To reset minimum/maximum values of measurements, the user must set up the command registers as follows:

Address	Register	Type	Range	A/E	Bit	Description
0x1F3F	8000	INT16U	46728	–	–	Command code = 46728
0x1F40	8001	INT16U	12	–	–	Number of parameters (bytes) = 12
0x1F41	8002	INT16U	5121	–	–	Destination = 5121 (0x1401)
0x1F42	8003	INT16U	1	–	–	Security type of the command
0x1F43– 0x1F44	8004– 8005	OCTET STRING	–	–	–	Password of the command: Administrator or Operator user profile password
0x1F45	8006	INT16U	–	–	–	Reset minimum/maximum of metering variables: - To reset the metering variable, set the bit to 1. - To keep the current values, set the bit to 0.
				A/E	0	Reset minimum/maximum current (I1, I2, I3, IN, I _{max} , I _g , I _{Δn} , I _{avg} , and I _{unbal})

Address	Register	Type	Range	A/E	Bit	Description
				E	1	Reset minimum/maximum voltage (V12, V13, V23, V1N, V2N, V3N, VavgL-L, VavgL-N, and Vunbal)
				E	2	Reset minimum/maximum power (active power, reactive power, apparent power, and distortion power)
				E	3	Reset minimum/maximum power factor and $\cos\phi$
				E	4	Reset minimum/maximum total harmonic distortion (THD)
				E	5	Reset peak of current demand
				E	6	Reset peak of active power, reactive power, and apparent power demand
				E	7	Reset minimum/maximum frequency
				E	8	Reset minimum/maximum thermal image (motor application only)
				E	9	Reset energy (active, reactive, apparent)
				–	10–15	Reserved

Start/Stop Synchronization

The start/stop synchronization command is used to start or stop the calculation of the current or power demand. The first command starts the calculation, the next command updates the value of current or power demand, and then restarts the calculation. The time period between 2 commands must be less than 1 hour.

To start/stop synchronization, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	46729	E	Command code = 46729
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator or Operator user profile password
0x1F45	8006	–	–	INT16U	–	E	Start/stop synchronization = 1

Power Flow Sign Configuration

The user can read the power flow sign configuration at register 3316 Power Flow Sign, page 139.

To set the power flow sign parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	47240	E	Command code = 47240
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	E	Power flow sign: 0 = the active power flows from upstream (top) to downstream (bottom) (factory setting). 1 = the active power flows from downstream (bottom) to upstream (top).

NOTE: The setting of the power flow sign can reset the content of the energy measurement registers excluding the cumulative energy measurements.

Power Factor Sign Configuration

The user can read the power factor sign configuration at register 3318 *Power Factor Sign*, page 140.

To set the power factor sign parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	47241	E	Command code = 47241
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–2	E	Sign convention for the power factor and the fundamental power factor ($\cos\phi$): 0 = IEC convention 2 = IEEE convention (factory setting)

Energy Accumulation Mode Configuration

The user can read the energy accumulation mode configuration at register 3324 *Energy Accumulation Mode*, page 140.

To set the energy accumulation mode parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	47242	E	Command code = 47242
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	E	Energy accumulation mode: 0 = absolute accumulation (factory setting) 1 = signed accumulation

Current Demand Configuration

The user can read the duration of the current demand calculation window at register 3352 Demand Time, page 140.

The user can read the current demand parameters from registers 2200 to 2207 Current Demand, page 109.

To start current demand, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	47243	E	Command code = 47243
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	Min	INT16U	5–60	E	Duration of the current demand calculation window, adjustable in steps of 1. The factory setting is 15 minutes (sliding).

Power Demand Configuration

The user can read the power demand calculation method from registers 3354 to 3355 Demand Time, page 140.

The user can read the power demand parameters from registers 2224 to 2237 Active Power Demand, page 109.

To start power demand, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	47244	E	Command code = 47244
0x1F40	8001	–	–	INT16U	14	E	Number of parameters (bytes) = 14
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	Min	INT16U	0–5	E	Power demand calculation method (window type): 0 = sliding 2 = fixed 5 = synchronized to communication The factory setting is 0 (sliding).
0x1F46	8007	–	Min	INT16U	5–60	E	Duration of the power demand calculation window, adjustable in steps of 1. The factory setting is 15 minutes.

Set Up Nominal Voltage Vn Display

The user can read the nominal voltage at register 9616 Nominal Voltage, page 141.

To set the nominal voltage Vn display parameters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	47245	E	Command code = 47245
0x1F40	8001	–	–	INT16U	12	E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	5121	E	Destination = 5121 (0x1401)
0x1F42	8003	–	–	INT16U	1	E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	E	Password of the command: Administrator user profile password
0x1F45	8006	–	V	INT16U	0–65535	E	Nominal voltage Vn (factory setting = 400 V)

DCM Modbus SL Module Data for ComPacT NSX Circuit Breakers

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DCM Modbus SL Module Registers

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DCM Modbus SL Module Identification

Product Identification

Address	Register	RW	Unit	Type	Range	Description
0x2E7C	11901	R	–	INT16U	–	Product identification = 16200 for the DCM Modbus SL module

DCM Modbus SL Module Firmware Revision

DCM firmware revision starts at register 11776 and has a maximum length of eight registers.

The firmware revision is an ASCII string using the format VXXX.YYY.ZZZ with:

- XXX = major version (000–127)
- YYY = minor version (000–255)
- ZZZ = revision number (000–255)

The NULL character ends the revision number.

Address	Register	RW	Unit	Type	Range	Description
0x2DFF– 0x2E04	11776– 11781	R	–	OCTET STRING	–	Firmware revision

DCM Modbus SL Module Hardware Revision

DCM hardware revision starts at register 11922 and has a maximum length of ten registers.

The firmware revision is an ASCII string using the format VXXX.YYY.ZZZ with:

- XXX = major version (000–127)
- YYY = minor version (000–255)
- ZZZ = revision number (000–255)

The NULL character ends the revision number

Address	Register	RW	Unit	Type	Range	Description
0x2E91– 0x2E9A	11922– 11931	R	–	OCTET STRING	–	Hardware revision

DCM Modbus SL Module Serial Number

The DCM Modbus SL module (commercial reference: LV434311) serial number is composed of a maximum of 15 alphanumeric characters with the following format: PPYYWWDLLNNNNN.

- PP = plant code (example, plant code of WPF is WX)
- YY = year of fabrication (05–99)
- WW = week of fabrication (01–53)
- D = day of fabrication (1–7)
- LLL = unique final assembly manufacturing line number or machine number within the plant (a–z or 0–9).

- NNNNN= unique product number generated on the day of manufacture by the plant manufacturing line or the plant machine (00001–99999).

A read request of 6 registers is necessary to read the DCM module serial number.

Address	Register	RW	X	Unit	Type	Range	Description
0x02E5C	11869	R	–	–	OCTET STRING	–	'PP'
0x02E5D	11870	R	–	–	OCTET STRING	05–99	'YY'
0x02E5E	11871	R	–	–	OCTET STRING	01–53	'WW'
0x02E5F	11872	R	–	–	OCTET STRING	1–7	'D'
0x02E60	11873	R	–	–	OCTET STRING	a–z, 0–9	'LLL'
0x02E61	11874	R	–	–	OCTET STRING	00–99	'NN'
0x02E62	11875	R	–	–	OCTET STRING	01–99	'NN'
0x02E63	11876	R	–	–	OCTET STRING	01–99	'N' (the NULL character ends the serial number)

DCM Modbus SL Module Identification

Address	Register	RW	Unit	Type	Range	Description
0x2D9E–0x2DDD	11679–11742	R	–	OCTET STRING	–	Vendor URL = https://www.se.com/
0x2DDF–0x2DEE	11744–11759	R	–	OCTET STRING	–	Product family = Gateway and Server
0x2DEF–0x2DF6	11760–11767	R	–	OCTET STRING	–	Product range = ComPacT NSX
0x2DF7–0x2DFE	11768–11775	R	–	OCTET STRING	–	Product model = DCM Modbus SL
0x2E2F–0x2E38	11824–11833	R	–	OCTET STRING	–	Vendor name = Schneider Electric
0x2E39–0x2E4B	11843–11852	R	–	OCTET STRING	–	Product code = LV434311

User Application Name

Address	Register	RW	Unit	Type	Range	Description
0x2D76–0x2D95	11639–11670	R-WC	–	OCTETSTRING	–	User application name

Current Date and Time

Address	Register	RW	Unit	Type	Range	Description
0x2E73–0x2E	11892–11895	R-WC	–	DATETIME	–	Current date and time in DATETIME format
0x2E77–0x2E78	11896–11897	R	Sec- onds	INT32U	0x00–0xFFFF- FFF	Number of seconds counted since last start

Circuit Breaker Status

Circuit Breaker Status

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x0232	563	R	–	–	INT16U	–	A/E	–	Circuit breaker status register
							A/E	0	OF input status 0 = The circuit breaker is open. 1 = The circuit breaker is closed.
							A/E	1	SD input status 0 = The circuit breaker is not tripped. 1 = The circuit breaker is tripped due to electrical fault, shunt trip or push-to-trip.
							A/E	2	SDE input status 0 = The circuit breaker is not tripped. 1 = The circuit breaker is tripped due to electrical fault (including ground-fault test and earth-leakage test).
							–	3–15	Reserved (forced to 0)

Communicating Motor Mechanism Status

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
0x0233	564	R	–	–	INT16U	–	A/E	–	Communicating motor mechanism status register
							A/E	0	Motor mechanism 0 = not available 1 = available
							A/E	1	Manu/auto mode 0 = manu 1 = auto
							A/E	2	Last command 0 = last command succeeded 1 = last command failed
							A/E	3	Enable automatic reset 0 = automatic reset is not enabled. 1 = automatic reset is enabled.
							A/E	4	Enable reset even if SDE 0 = reset is not enabled if the circuit breaker is tripped on electrical fault. 1 = reset is enabled even if the circuit breaker is tripped on electrical fault.

Address	Register	RW	X	Unit	Type	Range	A/E	Bit	Description
							A/E	5	Local/Remote mode selected in the FDM12B/ FDM12H display menu. 0 = Remote mode (default value), the communicating motor mechanism is controlled through a Modbus command only. 1 = Local mode, the communicating motor mechanism is controlled through the FDM12B/ FDM12H display menu only.
							–	6–15	Reserved (forced to 0)

Maintenance Indicators

General Description

The DCM Modbus SL module has seven counters that help to manage the ComPacT NSX circuit breaker.

The DCM Modbus SL module counters have the following properties:

- All the counters are saved in non-volatile memory to prevent data loss in case of power loss.
- The cumulating OF counter is read only. It stops incrementing when it reaches the maximum value 4 294 967 295.
- The user can preset all counters (except the cumulating OF counter) to any value from 0 to 65535. The counters stop incrementing when they reach the maximum value 65535.
- The OF counter and close circuit breaker command counter have a threshold which can be set by the user, to any value from 0 to 65534.

The factory setting is 5000. An alarm is generated when a counter reaches the threshold.

DCM Modbus SL Module Counters

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x023A–0x023B	571–572	R	1	–	INT32U	0–4 294 967 295	A/E	Cumulating OF counter (non-resettable open to close counter)
0x023C	573	R-WC	1	–	INT16U	0–65535	A/E	OF counter (resettable open to close counter)
0x023D	574	R-WC	1	–	INT16U	0–65535	A/E	SD counter (Close to SD position)
0x023E	575	R-WC	1	–	INT16U	0–65535	A/E	SDE counter (Close to SDE position)
0x023F	576	R-WC	1	–	INT16U	0–65535	A/E	Open circuit breaker command counter
0x0240	577	R-WC	1	–	INT16U	0–65535	A/E	Close circuit breaker command counter
0x0241	578	R-WC	1	–	INT16U	0–65535	A/E	Reset circuit breaker command counter
0x0242–0x0243	579–580	–	–	–	–	–	–	Reserved
0x0244	581	R-WC	1	–	INT16U	0–65535	A/E	OF counter threshold The factory setting is 5000.
0x0245	582	R-WC	1	–	INT16U	0–65535	A/E	Close circuit breaker command counter threshold The factory setting is 5000.

Event History

General Description

The DCM Modbus SL module event history registers describe the last 10 events. The DCM Modbus SL module event format corresponds to a series of 10 records. Each record is composed of 5 registers describing one DCM Modbus SL module event.

A read request of $5 \times (n)$ registers is necessary to read the last n DCM Modbus SL module event records, where 5 is the number of registers for each event record.

For example, a read request of $5 \times 3 = 15$ registers is necessary to read the last 3 DCM Modbus SL module event records of the DCM Modbus SL module event history:

- The first 5 registers describe the first DCM Modbus SL module event record (most recent event).
- The next 5 registers describe the second DCM Modbus SL module event record.
- The last 5 registers describe the third DCM Modbus SL module event record.

Address	Register	Description
0x0259	602	Event counter
0x025A–0x025E	603–607	Event record 1 (most recent event)
0x025F–0x0263	608–612	Event record 2
0x0264–0x0268	613–617	Event record 3
0x0269–0x026D	618–622	Event record 4
0x026E–0x0272	623–627	Event record 5
0x0273–0x0277	628–632	Event record 6
0x0278–0x027C	633–637	Event record 7
0x027D–0x0281	638–642	Event record 8
0x0282–0x0286	643–647	Event record 9
0x0287–0x028B	648–652	Event record 10 (oldest event)

Event Counter

The event counter is incremented when a new event is logged. If the counter reaches the maximum value 65535 and a new event is logged, the counter is reset to 0.

Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x0259	602	R	1	–	INT16U	0–65535	A/E	DCM Modbus SL module event counter

Event Record

The order and the description of the event record registers are the same as that of event record 1:

Event 1 (Most Recent Event)								
Address	Register	RW	X	Unit	Type	Range	A/E	Description
0x025A	603	R	1	–	INT16U	0–65535	A/E	DCM Modbus SL module event identifier (see next paragraph)
0x025B– 0x025D	604–606	R	–	–	ULP DATE	–	A/E	Date and time of event , page 54
0x025E	607	R	1	–	INT16U	1–2	A/E	Event status 1 = event occurrence 2 = event completion

Event Identifier

Event Identifier	Event
1024	SD contact change (occurrence = close to SD position)
1025	Threshold of OF counter is reached
1026	Threshold of close command counter is reached
1027	STOP (internal failure)
1028	ERROR (internal failure)
1029	OF contact change (occurrence = open to close position)
1030	SDE contact change (occurrence = close to SDE position)
1031	Manu/auto mode (occurrence = manu to auto position)
1040	Open command
1041	Close command
1042	Reset command

NOTE: In the case of a STOP event, it is mandatory to replace the DCM Modbus SL module. In the case of an ERROR event, it is advised to replace the DCM Modbus SL module (the core protection functions still work but it is preferable to replace the DCM Modbus SL module).

Modbus Network Parameters

Modbus Remote Control Commands (Intrusive Commands) Writing Mode Status

Address	Register	RW	Unit	Type	Range	Description
0x2E72	11891	R	–	INT16U	1–3	Modbus remote control command (intrusive command) writing Mode: 1 = Modbus remote control command (intrusive command) writing is not allowed. 3 = Modbus remote control command (intrusive command) writing is allowed.

Validity Duration of Data

Address	Register	RW	Unit	Type	Range	Description
0x306A	12395	R	s	INT16U	5–300 (in steps of 5 s)	Duration of validity of data in the dataset

Auto-Speed Sensing State

Address	Register	RW	Unit	Type	Range	Description
0x306E	12399	R	–	INT16U	0–1	Auto-Speed sensing state 0 = Auto-Speed sensing is disabled 1 = Auto-Speed sensing is enabled (factory setting)

DCM Modbus SL Module Modbus Address

Address	Register	RW	Unit	Type	Range	Description
0x306F	12400	R	–	INT16U	1–99	DCM Modbus address

Modbus Parity

Address	Register	RW	Unit	Type	Range	Description
0x3070	12401	R	–	INT16U	1–3	Modbus parity 1 = no parity (none) 2 = even parity (factory setting) 3 = odd parity

Modbus Baud Rate

Address	Register	RW	Unit	Type	Range	Description
0x3071	12402	R	–	INT16U	5–8	Modbus Baud rate • 5 = 4800 Baud • 6 = 9600 Baud • 7 = 19200 Baud (factory setting) • 8 = 38400 Baud

Number of Stop Bits

Address	Register	RW	Unit	Type	Range	Description
0x3072	12403	R	–	INT16U	0–5	Number of stop bits • 0 = no change • 1 = standard Modbus • 2 = 1/2 stop bit • 3 = 1 stop bit • 4 = 1 and 1/2 stop bit • 5 = 2 stop bits

DCM Modbus SL Module Commands

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List of DCM Module Commands and Error Codes

List of Commands

The following table lists the available DCM module commands, their corresponding command codes and user profiles. Follow the command execution procedures , page 38.

Command	Command code	User profile
Open Circuit Breaker, page 183	904	Administrator or Operator
Close Circuit Breaker, page 183	905	Administrator or Operator
Reset Circuit Breaker, page 183	906	Administrator or Operator
Enable/Inhibit a Circuit Breaker Close Order, page 184	910	Administrator or Operator
Enable/Disable Automatic Reset, page 184	42636	Administrator
Enable/Disable Reset even if SDE, page 184	42637	Administrator
Preset Counters, page 186	42638	Administrator
Set Up Thresholds, page 186	42639	Administrator
Get Current Time , page 188	768	No password required
Set Absolute Time, page 188	769	No password required
Read IMU Name and Location, page 189	1024	No password required
Write IMU Name and Location, page 190	1032	Administrator
Set Validity Duration of Data, page 190	41868	Administrator, Services, Engineer, or Operator
Write User Application Name, page 190	1032	Administrator
Set Remote Control Command (Intrusive Command) Mode, page 192	41867	Administrator
Set DCM Modbus Baud Rate, page 193	38664	Administrator
Set Modbus Address Using DCM Serial Number, page 193	38667	Administrator
Read DCM Modbus SL Module Location, page 194	2	No password required

Error Codes

Error codes generated by the DCM module commands are returned in register 8021. In addition to [generic error codes](#), page 40, DCM commands generate the following error codes:

Error Code	Description
4363 (0x110B)	DCM module is out of order.
4503 (0x1197)	Circuit breaker is tripped. It must be reset before the command.
4504 (0x1198)	Circuit breaker is already closed.
4505 (0x1199)	Circuit breaker is already open.
4506 (0x119A)	Circuit breaker is already reset.
4507 (0x119B)	Actuator is in manual mode. Remote commands are not allowed.
4508 (0x119C)	Actuator is not present.
4510 (0x119E)	A previous command is still in progress.
4511 (0x119F)	Reset command is forbidden when SDE is set.
4512 (0x11A0)	Inhibit close order is enabled.

Any other positive error code indicates an internal error.

Circuit Breaker Control Commands

Open Circuit Breaker

To open the circuit breaker with the communicating motor mechanism, set the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	904	A/E	Command code = 904
0x1F40	8001	–	–	INT16U	10	A/E	Number of parameters (bytes) = 10
0x1F41	8002	–	–	INT16U	10241	A/E	Destination = 10241 (0x2801)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004–8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator or Operator user profile password

Close Circuit Breaker

To close the circuit breaker with the communicating motor mechanism, set the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	905	A/E	Command code = 905
0x1F40	8001	–	–	INT16U	10	A/E	Number of parameters (bytes) = 10
0x1F41	8002	–	–	INT16U	10241	A/E	Destination = 10241 (0x2801)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004–8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator or Operator user profile password

Reset Circuit Breaker

To reset the circuit breaker with the communicating motor mechanism, set the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	906	A/E	Command code = 906
0x1F40	8001	–	–	INT16U	10	A/E	Number of parameters (bytes) = 10
0x1F41	8002	–	–	INT16U	10241	A/E	Destination = 10241 (0x2801)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004–8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator or Operator user profile password

Enable/Inhibit a Circuit Breaker Close Order

To enable or inhibit a close order of the circuit breaker with the communicating motor mechanism, set the command register as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	910	A/E	Command code = 910
0x1F40	8001	–	–	INT16U	13	A/E	Number of parameters (bytes) = 13
0x1F41	8002	–	–	INT16U	10241	A/E	Destination = 10241 (0x2801)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004–8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator or Operator user profile password
0x1F45	8006	–	–	INT16U	1	A/E	0 = Enable close order 1 = Inhibit close order
0x1F46	8007	–	–	INT16U	–	A/E	Origin of the command: 256 = command sent via communication interface (IFM or IFE)

Enable/Disable Automatic Reset

The user can read the automatic reset parameters at register 564 (bit 3), page 173.

To enable/disable automatic reset of the circuit breaker with the communicating motor mechanism, set the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	42636	A/E	Command code = 42636
0x1F40	8001	–	–	INT16U	12	A/E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	10241	A/E	Destination = 10241
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004–8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	A/E	0 = automatic reset is not enabled. 1 = automatic reset is enabled.

Enable/Disable Reset even if SDE

The user can read the reset parameters at register 564 (bit 4), page 173.

To enable/disable reset of the circuit breaker with the communicating motor mechanism even if SDE, set the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	42637	A/E	Command code = 42637
0x1F40	8001	–	–	INT16U	12	A/E	Number of parameters (bytes) = 12
0x1F41	8002	–	–	INT16U	10241	A/E	Destination = 10241 (0x2801)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F43– 0x1F44	8004–8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	–	–	INT16U	0–1	A/E	0 = reset is not enabled if SDE = 1. 1 = reset is enabled even if SDE = 1.

Counter Commands

Preset Counters

The user can read the values of the counters from registers 571 to 578, page 175.

To preset the counters, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	42638	A/E	Command code = 42638
0x1F40	8001	–	–	INT16U	22	A/E	Number of parameters (bytes) = 22
0x1F41	8002	–	–	INT16U	10240	A/E	Destination = 10240 (0x2800)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	1	–	INT16U	0–65535	A/E	0–65534 = preset value of the OF counter 65535 = do not preset the OF counter
0x1F46	8007	1	–	INT16U	0–65535	A/E	0–65534 = preset value of the SD counter 65535 = do not preset the SD counter
0x1F47	8008	1	–	INT16U	0–65535	A/E	0–65534 = preset value of the SDE counter 65535 = do not preset the SDE counter
0x1F48	8009	1	–	INT16U	0–65535	A/E	0–65534 = preset value of the open circuit breaker command counter 65535 = do not preset the open circuit breaker command counter
0x1F49	8010	1	–	INT16U	0–65535	A/E	0–65534 = preset value of the close circuit breaker command counter 65535 = do not preset the close circuit breaker command counter
0x1F4A	8011	1	–	INT16U	0–65535	A/E	0–65534 = preset value of the reset circuit breaker command counter 65535 = do not preset the reset circuit breaker command counter

Set Up Thresholds

The user can read the values of the thresholds from registers 581 to 582, page 175.

To set up the thresholds, the user must set up the command registers as follows:

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F3F	8000	–	–	INT16U	42639	A/E	Command code = 42639
0x1F40	8001	–	–	INT16U	22	A/E	Number of parameters (bytes) = 22
0x1F41	8002	–	–	INT16U	10240	A/E	Destination = 10240 (0x2800)
0x1F42	8003	–	–	INT16U	1	A/E	Security type of the command
0x1F43– 0x1F44	8004– 8005	–	–	OCTET STRING	–	A/E	Password of the command: Administrator user profile password
0x1F45	8006	1	–	INT16U	0–65535	A/E	0–65534 = value of the OF counter threshold 65535 = do not change the OF counter threshold
0x1F46	8007	1	–	INT16U	65535	A/E	65535 (no threshold is set up for the SD counter)

Address	Register	X	Unit	Type	Range	A/E	Description
0x1F47	8008	1	–	INT16U	65535	A/E	65535 (no threshold is set up for the SDE counter)
0x1F48	8009	1	–	INT16U	65535	A/E	65535 (no threshold is set up for the open circuit breaker command counter)
0x1F49	8010	1	–	INT16U	0–65535	A/E	0–65534 = value of the close circuit breaker command counter threshold 65535 = do not change close circuit breaker command counter threshold
0x1F4A	8011	1	–	INT16U	65535	A/E	65535 (no threshold is set up for the reset circuit breaker command counter)

Configuration Commands

Get Current Time

The get current time command is a non-intrusive command. The command is always enabled, even if the remote padlock parameter is set to **Disabled**.

To get the current time for all modules, set the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	768	Command code = 768
0x1F40	8001	–	INT16U	10	Number of parameters (bytes) = 10
0x1F41	8002	–	INT16U	10240	Destination = 10240 (0x2800)
0x1F42	8003	–	INT16U	0	Security type of the command
0x1F43– 0x1F44	8004–8005	–	OCTET STRING	0	Password of the command = 0 (no password required)

The following registers contain the time data:

- Register 8023 holds the month in the MSB, the day in the LSB.
- Register 8024 holds the year offset in the MSB (add 2000 to get the year) and the hour in the LSB.
- Register 8025 holds the minutes in the MSB, the seconds in the LSB.
- Register 8026 holds the milliseconds.

Set Absolute Time

The set absolute time command is a non-intrusive command. The command is always enabled, even if the remote padlock parameter is set to **Disabled**.

To set the absolute time for all the IMU modules, set the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	769	Command code = 769
0x1F40	8001	–	INT16U	18	Number of parameters (bytes) = 18
0x1F41	8002	–	INT16U	10240	Destination = 10240 (0x2800)
0x1F42	8003	–	INT16U	0	Security type of the command
0x1F43– 0x1F44	8004–8005	–	OCTET STRING	0	Password of the command = 0 (no password required)
0x1F45	8006	–	INT16U	–	MSB = month (1–12) LSB = day in the month (1–31)
0x1F46	8007	–	INT16U	–	MSB = year (0–99, 0 meaning year 2000) LSB = hour (0–23)
0x1F47	8008	–	INT16U	–	MSB = minutes (0–59) LSB = seconds (0–59)
0x1F48	8009	ms	INT16U	0–999	Milliseconds (0–999)

In case of 24 Vdc power loss, the date and time counters are reset to January 1 2000. It is therefore mandatory to set absolute time for all the IMU modules after recovering the 24 Vdc power supply.

Furthermore, due to the clock drift of each IMU module, it is mandatory to set absolute time for all the IMU modules periodically. Recommended period is at least every 15 minutes.

Read IMU Name and Location

The FDM121 displays the first 14 characters of the IMU name.

To read the IMU name and location, set the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	1024	Command code = 1024
0x1F40	8001	–	INT16U	16	Number of parameters (bytes) = 16
0x1F41	8002	–	INT16U	10240	Destination = 10240 (0x2800)
0x1F42	8003	–	INT16U	0	Security type of the command
0x1F43– 0x1F44	8004–8005	–	OCTET STRING	0	Password of the command = 0 (no password required)
0x1F45– 0x1F46	8006–8007	–	INT32U	–	17039489 = read IMU name (load 0x0104 into register 8006, 0x0081 into 8007) 17039490 = read IMU location (load 0x0104 into register 8006, 0x0082 into 8007)
0x1F47	8008	–	INT16U	2048	2048

The IMU name and location are returned to command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F53	8020	–	INT16U	1024	Last Command Code
0x1F54	8021	–	INT16U	–	Command status 0 = command succeeded Otherwise, command failed
0x1F55	8022	–	INT16U	–	Number of bytes returned (0 if command failed)
0x1F56	8023	–	OCTET STRING	–	If command succeeded MSB = first character of IMU name or location LSB = second character of IMU name or location
0x1F57– 0x1F6D	8024–8046	–	OCTET STRING	–	Depends on the length of the IMU name or location and ends with the NULL character 0x00

Write IMU Name and Location

The IMU name and location can be read from registers 11801 to 11868.

To write the IMU name and location, the user must set up the command registers as follows. It is mandatory to set date and time for all Modbus servers periodically, due to the clock drift of each Modbus server. Recommended period is at least every 15 minutes.

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	1032	Command code = 1032
0x1F40	8001	–	INT16U	16–62	Number of parameters (bytes) = depends on the length of the IMU name or location (up to 46 characters)
0x1F41	8002	–	INT16U	0	Destination = 0 (0x0000)
0x1F42	8003	–	INT16U	1	Security type of the command
0x1F43– 0x1F44	8004–8005	–	OCTET STRING	–	Password of the command = Administrator user profile password
0x1F45– 0x1F46	8006–8007	–	INT32U	–	17039489 = write IMU name (load 0x0104 into register 8006, 0x0081 into 8007) 17039490 = write IMU location (load 0x0104 into register 8006, 0x0082 into 8007)
0x1F47	8008	–	INT16U	2048	2048
0x1F48	8009	–	OCTET STRING	–	MSB = first character of the IMU name or location LSB = second character of the IMU name or location
0x1F49– 0x1F5F	8010–8038	–	OCTET STRING	–	Depends on the length of the IMU name or location and ends with the NULL character 0x00

Set Validity Duration of Data

This command is used to set the validity duration of data in the standard and legacy datasets.

The validity duration of data can be read in a register [Validity Duration of Data](#), page 178.

To set the validity duration of data, set the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	41868	Command code = 41868
0x1F40	8001	–	INT16U	12	Number of parameters (bytes) = 12
0x1F41	8002	–	INT16U	10241	Destination = 10240 (0x2801)
0x1F42	8003	–	INT16U	1	Security type of the command
0x1F43– 0x1F44	8004–8005	–	OCTET STRING	–	Password of the command: Administrator, Services, Engineer, or Operator user profile password
0x1F45	8006	s	INT16U	5–300 (in steps of 5 s)	Validity duration of data Factory setting: 10 s

Write User Application Name

The user application name can be read from registers 11639 to 11670.

To write the user application name, set the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	41868	Command code = 41868
0x1F40	8001	–	INT16U	-	Number of parameters (bytes) = 16 + the length of the user application name + 1 (NULL character 0x00)
0x1F41	8002	–	INT16U	10241	Destination = 10241 (0x2801)
0x1F42	8003	–	INT16U	1	Security type of the command
0x1F43- 0x1F44	8004-8005	–	OCTET STRING	–	Password of the command: Administrator user profile password
0x1F45 - 0x1F46	8006–8007	–	INT32U	–	17039378 = User application name (load 0x0104 into register 8006, 0x0012 into register 8007)
0x1F47	8008	–	INT16U	2048	2048
0x1F48	8009	–	OCTETSTRING		MSB = First character of the user application name LSB = Second character of the user application name Depends on the length of the user application name and ends with the NULL character 0x00

If the length of the user application name is more than 14 bytes, use the following command format.

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	41868	Command code = 41868
0x1F40	8001	–	INT16U		Number of parameters (bytes) = 16 + the length of the user application name + 1 (NULL character 0x00) NOTE: Number of parameters register 8016–8021 is not counted as command parameter.
0x1F41	8002	–	INT16U	10241	Destination = 10241 (0x2801)
0x1F42	8003	–	INT16U	1	Security type of the command
0x1F43- 0x1F44	8004-8005	–	OCTET STRING	–	Password of the command : Administrator user profile password
0x1F45 - 0x1F46	8006–8007	–	INT32U	–	17039378 = User application name (load 0x0104 into register 8006,

Address	Register	Unit	Type	Range	Description
					0x0012 into register 8007)
0x1F47	8008	–	INT16U	2048	2048
0x1F48 - 0x1F4E	8009 - 8015	–	OCTETSTRING		First to fourteenth characters of the user application name
0x1F4F	8016	–	–	–	Forced to 0 (factory setting)
0x1F50	8017	–	–	–	Forced to 8019 (factory setting)
0x1F51	8018	–	–	–	Forced to 8020 (factory setting)
0x1F52	8019	–	–	–	Forced to 8021 (factory setting)
0x1F53	8020	–	–	–	Forced to 0
0x1F54	8021	–	–	–	Forced to 0
0x1F55	8022	–	OCTETSTRING	–	Fifteen and sixteenth characters of the user application name
					Depends on the length of the user application name and ends with the NULL character 0x00

Set Remote Control Command (Intrusive Command) Mode

The remote control command (intrusive command) mode can be read from register 11891. Default setting of the remote control command (intrusive command) mode is enabled.

To change the mode, set up the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	41867	Command code = 41867
0x1F40	8001	–	INT16U	12	Number of parameters (bytes) = 12
0x1F41	8002	–	INT16U	10241	Destination = 10241 (0x2801)
0x1F42	8003	–	INT16U	1	1
0x1F43-0x1F44	8004-8005	–	OCTETSTRING	–	Level 4 password (factory setting = 0000 = 0x30303030)
0x1F45	8006	–	INT16U		Locking states: • 1 = Intrusive command mode is locked • 3 = Intrusive command mode is not locked

Set DCM Modbus Baud Rate

When the DCM Modbus Baud rate and Modbus address are provided, set up the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	38664	Command code = 38664
0x1F40	8001	–	INT16U	18	Number of parameters (bytes) = 18
0x1F41	8002	–	INT16U	10240	Destination = 10240 (0x2800)
0x1F42	8003	–	INT16U	1	1
0x1F43-0x1F44	8004-8005	–	OCTETSTRING	–	Level 4 password (factory setting = 0000 = 0x30303030)
0x1F45	8006	–	INT16U		Baud rate 0 = no change 5 = 4800 baud 6 = 9600 baud 7 = 19200 baud 8 = 38400 baud
0x1F46	8007	–	INT16U	-	Modbus parity 0 = no change 1 = None 2 = Even parity 3 = Odd parity
0x1F47	8008	–	INT16U	–	Number of stop bits 0 = no change 1 = standard Modbus (the default setting when baud rate self-adaption) 3 = 1 stop bit 5 = 2 stop bits
0x1F48	8009		INT16U		Auto Go function activation: 0 = Auto Go function not activated (set register 8009 to 0 when setting Baud rate, parity and stop bit) 1 = Auto Go function activated (when activated, set registers 8006-8008 to 0)

Set Modbus Address Using DCM Serial Number

To set the Modbus address using DCM serial number through broadcast address 00, set the command registers as follows:

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	38667	Command code = 38667
0x1F40	8001	–	INT16U	30	Number of parameters (bytes) = 30
0x1F41	8002	–	INT16U	0	Destination = 0 (0x0000)
0x1F42	8003	–	INT16U	1	1
0x1F43– 0x1F44	8004– 8005	–	OCTETSTRING	–	Level 4 password (factory setting = '0000' = 0x30303030)
0x1F45	8006	–	INT16U	1-247	New Modbus address. When using with EPS Hub, the address range is 1-99
0x1F46	8007	–	INT16U	1	1
0x1F47- 0x1F4E	8008- 8015	–	OCTETSTRING	–	Product serial number: Example, product serial number = WX2342200100031 = 0x5758323334323-230303130303033-3100

Read DCM Modbus SL Module Location

To read location of the DCM with specified Modbus address, set the command registers as follow. Then the LEDs on the correct DCM module blink for 15 seconds.

Address	Register	Unit	Type	Range	Description
0x1F3F	8000	–	INT16U	2	Command code = 2
0x1F40	8001	–	INT16U	10	Number of parameters (bytes) = 10
0x1F41	8002	–	INT16U	10240	Destination = 10240 (0x2800)
0x1F42	8003	–	INT16U	0	0
0x1F43– 0x1F44	8004– 8005	–	OCTETSTRING	0	0= 0x00000000

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Introduction

The appendix lists the Modbus registers by ascending order, with cross references to the corresponding pages of the manual.

Cross References to Modbus Registers for ComPacT NSX

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Cross References to Modbus Registers 197

Introduction

The cross references table gives the list of Modbus registers in ascending order, with cross references to the corresponding pages of the guide.

Cross References to Modbus Registers

General Description

The following table gives cross references to the Modbus registers used by the communication modules. The registers are listed in ascending order.

Cross References Table

Address	Register	Module	Description	Page
0x0232	563	DCM Modbus SL module	Circuit breaker status	Circuit Breaker Status, page 173
0x0233	564	DCM Modbus SL module	Communicating motor mechanism status	DCM Modbus SL Module Counters, page 175
0x023A–0x0245	571–582	DCM Modbus SL module	DCM Modbus SL module counters	Event Record, page 176
0x0259–0x028B	602–652	DCM Modbus SL module	DCM Modbus SL module event history	Event History, page 176
0x03E7–0x03EE	1000–1007	MicroLogic	Voltage (real-time measurements)	Voltage, page 100
0x03EF–0x03F6	1008–1015	MicroLogic	Voltage unbalance (real-time measurements)	Voltage Unbalance, page 100
0x03F7–0x0402	1016–1027	MicroLogic	Current (real-time measurements)	Current, page 101
0x0403–0x0407	1028–1032	MicroLogic	Current unbalance (real-time measurements)	Current Unbalance, page 101
0x0409–0x040C	1034–1037	MicroLogic	Active power (real-time measurements)	Active Power, page 102
0x040D–0x0410	1038–1041	MicroLogic	Reactive power (real-time measurements)	Reactive Power, page 102
0x0411–0x0414	1042–1045	MicroLogic	Apparent power (real-time measurements)	Apparent Power, page 102
0x0415–0x0418	1046–1049	MicroLogic	Power factor (real-time measurements)	Power Factor, page 102
0x0419–0x041C	1050–1053	MicroLogic	Fundamental power factor (real-time measurements)	Fundamental Power Factor, page 103
0x041D	1054	MicroLogic	Frequency (real-time measurements)	Frequency, page 103
0x0437–0x043A	1080–1083	MicroLogic	Fundamental reactive power (real-time measurements)	Frequency Reactive Power, page 103
0x043F–0x0442	1088–1091	MicroLogic	Distortion power (real-time measurements)	Distortion Power, page 103
0x0443–0x044B	1092–1100	MicroLogic	Total harmonic distortion (THD) (real-time measurements)	Total Harmonic Distortion, page 103
0x0478–0x0479	1145–1146	MicroLogic	Voltage (real-time measurements)	Voltage, page 100
0x0513–0x0522	1300–1315	MicroLogic	Voltage (minimum of real-time measurement)	Minimum of Real-Time Measurements, page 105
0x0523–0x0533	1316–1332	MicroLogic	Current (minimum of real-time measurement)	Minimum of Real-Time Measurements, page 105
0x07CF–0x07EE	2000–2031	MicroLogic	Energy measurements	Energy Measurements, page 107
0x0897–0x08BC	2200–2237	MicroLogic	Demand measurements	Current Demand, page 109
0x08C1–0x08C2	2242–2243	MicroLogic	Quadrant total	Quadrant Total, page 139

Address	Register	Module	Description	Page
0x0B53–0x0B70	2900–2929	MicroLogic	Minimum/maximum measurements reset time	Minimum/Maximum Measurements Reset Time, page 111
0x0BB7–0x0BB9	3000–3002	MicroLogic	Current date and time	Current Date and Time, page 153
0x0CF1	3314	MicroLogic	System type	System Type, page 139
0x0CF3	3316	MicroLogic	Power flow sign	Power Flow Sign, page 139
0x0CF5	3318	MicroLogic	Power factor sign	Power Factor Sign, page 140
0x0CFB	3324	MicroLogic	Energy accumulation mode	Energy Accumulation Mode, page 140
0x0D17–0x0D1A	3352–3355	MicroLogic	Demand time	Demand Time, page 140
0x1647	5704	MicroLogic	Alarms status	Alarms Status, page 116
0x1663–0x1694	5732–5781	MicroLogic	Alarm history	Alarms Record, page 118
0x19F9–0x1A02	6650–6659	MicroLogic	Long-time protection pre-alarm	Long-Time Protection Pre-Alarm (PAL Ir), page 128
0x1A03–0x1A0C	6660–6669	MicroLogic	Ground-fault protection pre-alarm	Ground-Fault Protection Pre-Alarm (PAL Ig), page 129
0x1A0D–0x1A16	6670–6679	MicroLogic	Earth-leakage protection pre-alarm	Earth-Leakage Protection Pre-Alarm (PAL Idn), page 129
0x1A71–0x1AE8	6770–6889	MicroLogic	User-defined alarms	User-Defined Alarms, page 130
0x1F3F–0x1FD4	8000–8149	MicroLogic	Command interface	Command Interface, page 38
0x21FB–0x2200	8700–8705	MicroLogic	Serial number	Serial Number, page 112
0x220B	8716	MicroLogic	Product identification	Product Identification, page 113
0x2212	8723	MicroLogic	Hardware revision	Hardware Revision, page 112
0x2223	8740	MicroLogic	Protection type	Protection Type, page 113
0x2224	8741	MicroLogic	Metering type (A, E)	Metering Type, page 113
0x222A	8747	MicroLogic	Application	Application, page 113
0x222B	8748	MicroLogic	Standard	Standard, page 113
0x222D	8750	MicroLogic	Nominal current	Nominal Current, page 114
0x222E	8751	MicroLogic	Pole	Pole, page 114
0x222F	8752	MicroLogic	16 Hz 2/3	16 Hz 2/3, page 114
0x2231–0x223A	8754–8763	MicroLogic	Long-time protection	Long-Time Protection Parameters, page 134
0x223B–0x2244	8764–8773	MicroLogic	Short-time protection	Short-Time Protection Parameters, page 134
0x2245–0x224E	8774–8783	MicroLogic	Instantaneous protection	Instantaneous Protection

Address	Register	Module	Description	Page
				Parameters, page 135
0x224F–0x2258	8784–8793	MicroLogic	Ground-fault protection	Ground-Fault Protection Parameters, page 135
0x2259–0x2262	8794–8803	MicroLogic	Earth leakage protection	Earth Leakage Protection Parameters, page 135
0x2292	8851	MicroLogic	Temperature	Temperature, page 153
0x2298	8857	MicroLogic	SDx module status	SDx Module Status, page 116
0x22A0	8865	MicroLogic	Time remaining until long-time tripping	Time Remaining Until Long-Time Tripping, page 153
0x22A7	8872	MicroLogic	Phase rotation	Phase Rotation, page 153
0x22C3–0x22C6	8900–8903	MicroLogic	Jam protection	Jam Protection Parameters, page 136
0x22C7–0x22CA	8904–8907	MicroLogic	Unbalance protection	Unbalance Protection Parameters, page 136
0x22CB–0x22CE	8908–8911	MicroLogic	Underload protection	Underload Protection Parameters, page 136
0x22CF–0x22D2	8912–8915	MicroLogic	Long-start protection	Long-Start Protection Parameters, page 137
0x22D3–0x22D6	8916–8919	MicroLogic	Neutral protection	Neutral Protection Parameters, page 137
0x22E1	8930	MicroLogic	Thermal memory inhibition	Thermal Memory Inhibition Parameters, page 137
0x238B–0x2401	9100–9218	MicroLogic	Trip history	Trip History, page 120
0x258F	9616	MicroLogic	Nominal voltage Vn	Nominal Voltage, page 141
0x2648–0x2651	9801–9810	MicroLogic	SDx module outputs	Configuration of the SDx Module, page 138
0x270F	10000	MicroLogic	Trip status	Trip Status, page 117
0x2D76–0x2D95	11639–11670	DCM Modbus SL module	User application name	User Application Name, page 172
0x2DFF–0x2E06	11776–11783	DCM Modbus SL module	Fireware revision	DCM Modbus SL Module Firmware Revision, page 171
0x2E91–0x2E9A	11922–11931	DCM Modbus SL module	Hardware revision	DCM Modbus SL Module Hardware Revision, page 171
0x2D9E–0x2E4B	11679–11843	DCM Modbus SL module	Module identification	DCM Modbus SL Module Identification, page 172
0x2E5C–0x2E63	11869–11878	DCM Modbus SL module	Serial number	DCM Modbus SL Module Serial Number, page 171
0x2E18–0x2E2E	11801–11823	DCM Modbus SL module	IMU name	Read IMU Name and Location, page 189

Address	Register	Module	Description	Page
0x2E45-0x2E5B	11846-11868	DCM Modbus SL module	IMU location	Read IMU Name and Location, page 189
0x2E72	11891	DCM Modbus SL module	Intrusive command mode status	Modbus Remote Control Commands (Intrusive Commands) Writing Mode Status, page 178
0x2E73-0x2E78	11896-11899	DCM Modbus SL module	Current date and time	Current Date and Time, page 172
0x2E7C	11901	DCM Modbus SL module	Product identification	Product Identification, page 171
0x2EDF-0x2F82	12000-12163	DCM Modbus SL module	Legacy dataset	Legacy Dataset Common Registers, page 82
0x306E	12399	DCM Modbus SL module	Auto-Speed Sensing State (baud rate self-adaptive detection)	Auto-Speed Sensing State, page 178
0x306F	12400	DCM Modbus SL module	Modbus address	DCM Modbus SL Module Modbus Address, page 178
0x3070	12401	DCM Modbus SL module	Modbus parity	Modbus Parity, page 178
0x3071	12402	DCM Modbus SL module	Modbus baud rate	Modbus Baud Rate, page 179
0x3072	12403	DCM Modbus SL module	Stop bit	Number of Stop Bits, page 179
0x72CD	29390	MicroLogic	Failure status	Failure Status, page 153
0x733B-0x733F	29500-29504	MicroLogic	Maintenance operation record	Maintenance Operation Record, page 125
0x733B-0x736C	29500-29549	MicroLogic	Maintenance operation number	Maintenance Operation Number, page 125
0x73BB-0x73BE	29628-29631	MicroLogic	Previous instantaneous protection setup	Previous Instantaneous Protection Setup, page 143
0x73BF-0x73CA	29632-29643	MicroLogic	Previous ground-fault protection setup	Previous Ground-Fault Protection Setup, page 143
0x73CB-0x73D2	29644-29651	MicroLogic	Previous earth-leakage protection setup	Previous Earth-Leakage Protection Setup, page 143
0x73D3-0x73DE	29652-29663	MicroLogic	Previous jam protection setup	Previous Jam Protection Setup, page 143
0x73DF-0x73E6	29664-29671	MicroLogic	Previous unbalance protection setup	Previous Unbalance Protection Setup, page 144
0x73E6-0x73F2	29672-29683	MicroLogic	Previous underload protection setup	Previous Underload Protection Setup, page 144
0x73F3-0x73FE	29684-29695	MicroLogic	Previous long-start protection setup	Previous Long-Start Protection setup, page 144
0x73FF-0x7402	29696-29699	MicroLogic	Previous neutral protection setup	Previous Neutral Protection Setup, page 145
0x7453-0x745A	29780-29787	MicroLogic	Minimum/maximum V12 voltage measurements	Minimum/maximum V12 voltage

Address	Register	Module	Description	Page
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0x745B–0x7462	29788–29795	MicroLogic	Minimum/maximum V23 voltage measurements	Minimum/maximum V23 voltage measurements, page 146
0x7463–0x746A	29796–29803	MicroLogic	Minimum/maximum V31 voltage measurements	Minimum/maximum V31 voltage measurements, page 146
0x746B–0x746E	29804–29807	MicroLogic	Maximum I1 current measurement	Maximum I1 Current Measurement, page 146
0x746F–0x7472	29808–29811	MicroLogic	Maximum I2 current measurement	Maximum I2 Current Measurement, page 146
0x7473–0x7476	29812–29815	MicroLogic	Maximum I3 current measurement	Maximum I3 Current Measurement, page 146
0x7477–0x747A	29816–29819	MicroLogic	Maximum IN current measurement	Maximum IN Current Measurement, page 147
0x747B–0x747E	29820–29823	MicroLogic	Minimum system frequency	Minimum System Frequency, page 147
0x747F–0x7482	29824–29827	MicroLogic	Maximum system frequency	Maximum System Frequency, page 147
0x7483–0x7486	29828–29831	MicroLogic	I1 peak demand measurement	I1 Peak Demand Measurement, page 147
0x7487–0x748A	29832–29835	MicroLogic	I2 peak demand measurement	I2 Peak Demand Measurement, page 147
0x748B–0x748E	29836–29839	MicroLogic	I3 peak demand measurement	I3 Peak Demand Measurement, page 148
0x748F–0x7492	29840–29843	MicroLogic	IN peak demand measurement	IN Peak Demand Measurement, page 148
0x7493–0x7496	29844–29847	MicroLogic	P peak demand measurement	P Peak Demand Measurement, page 148
0x749A–0x749B	29851–29852	MicroLogic	Time of use counter	Time of Use Counter, page 149
0x749C	29853	MicroLogic	Rate of wear counter	Rate of Wear Counter, page 149
0x749D	29854	MicroLogic	Boot counter	Boot Counter, page 149
0x749E–0x749F	29885–29886	MicroLogic	EEPROM writing counter	EEPROM Writing Counter, page 149
0x74B7–0x74BE	29880–29887	MicroLogic	Load profile counters	Load Profile Counters, page 149
0x74C1–0x74CC	29890–29901	MicroLogic	Temperature profile counters	Temperature Profile Counters, page 150
0x74D5–0x74DD	29910–29918	MicroLogic	Protection trips counters	Protection Trips Counters, page 150
0x74F3–0x74FF	29940–29952	MicroLogic	Alarms counters	Alarms Counters, page 151
0x751B–0x7520	29980–29985	MicroLogic	Maintenance operation counters	Maintenance Operation Counters, page 151
0x7525–0x7526	29990–29991	MicroLogic	MicroLogic trip unit rotary switches	MicroLogic Trip Unit Rotary Switches, page 154

Address	Register	Module	Description	Page
0x7527	29992	MicroLogic	MicroLogic trip unit locking pad status	MicroLogic Trip Unit Locking Pad Status, page 154
0x7528	29993	MicroLogic	Auxiliary 24 Vdc power supply	Auxiliary 24 Vdc Power Supply, page 154
0x7529–0x752D	29994–29998	MicroLogic	Firmware revision	Firmware Revision, page 114
0x752F–0x7532	30000–30003	MicroLogic	Part number	Part Number, page 114
0x7534	30005	MicroLogic	MicroLogic trip unit LED	MicroLogic Trip Unit LED, page 154
0x7CFF–0x7EFE	32000–32335	IFE communication interface	Standard dataset	Standard Dataset Common Registers, page 66

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